Test Suite ACAS sXu V4R0

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TEST SUITE ACAS SXU V4R0

This package includes the encounter set of the Test Suite for Airborne Collision Avoidance System sXu (ACAS sXu). The main purpose of this Test Suite is to validate the correct implementation of the Algorithm Design Description (ADD), specified in the Minimum Operational Performance Standards (MOPS) as Julia code. The development of the Test Suite encounters, along with the unit tests and functional requirements of the algorithms, helps to validate the correct implementation of the ADD. The Test Suite provides validation at the system-level, while unit-testing provides validation at the algorithm-level. The package contains encounters for testing specific functionalities (*functional encounters*) and additional encounters to cover branches of the code (*branch coverage encounters*) in the ADD.

Functional Coverage:

The coverage of specific functionality testing per the Test Suite can be seen in Table 1. The following areas are of particular interest.

Resolution Advisories:

The functional encounters of the Test Suite contain broad coverage of different resolution advisories (RA), in both the vertical and horizontal domains. Of the 30 vertical RA rows and 144 vertical RA combinations as defined by Table 2-49 in the MOPS, 19 rows and 46 combinations are covered in this Test Suite. Of the 5 horizontal RA rows and 21 horizontal RA combinations as defined by Table 2-50 in the MOPS, all 5 rows and 15 combinations are covered in this Test Suite.

Considering combinations of vertical and horizontal RAs, different timing combinations are possible where one dimension starts or ends before the other one, both occur at the same time, or where one dimension starts when the other one ends. Of the 8 total RA combinations possible, 6 are covered in the Test Suite. The 2 combinations not included are the cases where one dimension has a COC at the same time cycle the other dimension begins an RA. In future Test Suite releases, we intend to achieve full row coverage of both vertical and horizontal RAs and full coverage of all sequential RA combinations.

Coordination:

The Test Suite contains a set of encounters to test coordination functionality. In V4R0, a fully functional coordination scheme has been implemented, allowing ownship to coordinate with intruders.

Systematic tests of coordination combinations are exercised in this Test Suite, including: coordination messages with vertical and horizontal messages timing out / being cancelled, incoming coordination messages before the start of RAs to test vertical coordination delay, invalid coordination messages, coordination to exercise all possible combinations of ownship settings, and coordination with reachback.

Correlation / Decorrelation:

Both ownship and intruder track correlation / decorrelation are features of ACAS sXu, and the Test Suite contains encounters to test all possible track combinations in which correlation / decorrelation is allowed. Ownship track correlation / decorrelation is tested via spatial correlation and ID correlation with AGT tracks. Intruder spatial correlation / decorrelation is tested on all allowable combinations of intruder track sources, and the various methods of intruder ID correlation / decorrelation are tested on all allowable combinations of intruder track sources.

Track Source Selection:

To test the track source selection tables in the MOPS (Tables 2-5, 2-6, 2-7, & 2-8), this Test Suite contains an encounter for every possible combination of every row of these tables. These tables illustrate the STM track source selection rules for different combinations of available sources (ADS-B, V2V, pre-tracked inputs), validation status, and altitude validity.

Altitude Sources:

ACAS sXu accepts pressure, HAE, and AGL as measurements of altitude for both ownship and intruder. The document, "ACAS sXu V4R0: Altimetry Report"[1], describes the various altitude transformations that should occur given the intruder input type (ADS-B, UAT, V2V, AGT, ORNCT, point obstacle), the intruder altitude type, and the ownship altitude type. Specifically, its tables indicate the type of altitude reported to the TRM for the ownship and the intruder, along with any degraded surveillance flags or special conditions (e.g., assume co-altitude, set to NaN, do not send to TRM, etc.). The Test Suite contains an encounter to test each row of these tables. Additionally, the Test Suite contains encounters to exhaustively test the new altitude prioritization code that compares real time HAE accuracy to pressure altitude accuracy in determining altitude selection.

Intruder Filtering and Prioritization:

The ACAS sXu code contains several prioritizations and down-selects, e.g. when restricting the number of intruders passed by the STM to the TRM to the maximum (presently set to 30), or when selecting a maximum of 5 intruders to be used to calculate the horizontal RAs in the HTRM. Several encounters for these down-selections are available.

Low-Altitude Behaviors:

ACAS sXu focuses on operations at lower altitudes with manual mode switching and low-altitude point obstacle avoidance (POA) / ground point obstacle avoidance (GPOA). All automatic mode switching and descend inhibit thresholds have been removed in V4R0.

The Test Suite contains encounters to test the manual switching of modes via the requested_opmode flag (STANDBY, SURVEILLANCE_ONLY, RA). Encounters to test that all automatic mode switching and descend inhibit thresholds have been removed are also included in this Test Suite.

Additionally, the Test Suite includes encounters to test several terrain and obstacle scenarios, with and without airborne intruders. Depending on ownship's settings for enabling or disabling point obstacle and ground point obstacle avoidance, different protection is provided and different RAs are issued.

Ownship Parameter Settings:

The Test Suite contains encounters that test the correct processing of ownship values like the ownship operational flag, the requested operational mode flag, the prefer wind relative flag, the POA flag, the GPOA flag, and the turn & vert rate limits.

Structure of the Test Suite:

The encounters are divided into test groups as shown below in Table 1. The Test Suite consists of 791 total encounters. The package contains the encounters including Input, STM report, TRM report, and Cost files. In the header of the Input file, each of the encounters contains text describing the encounter and its purpose, the test group number, and a flag to indicate if the encounter is prescriptive.

In the ADD, most of the algorithms are prescriptive and must be implemented exactly the way they are described. Some of the algorithms are suggestive; those algorithms can be implemented as suggested, but any other implementations leading to the same results are permitted. Accordingly, some of the test encounters are flagged as prescriptive. They test only the prescriptive algorithms and must be executed for any ACAS sXu implementation.

The encounters flagged as non-prescriptive require suggestive algorithms to execute. These encounters can be used for testing if the implementer chooses to implement the suggestive algorithms of the ADD. Due to the structure of the code, the prescriptive and suggestive algorithms are not completely isolated. If an implementation differing from the suggestive algorithms of the ADD is chosen, running only the encounters marked as prescriptive may leave some of the branches in the prescriptive algorithms uncovered. In this case, it is up to the vendor to generate additional

[1] Wikle, J. "ACAS sXu V4R0: Altimetry Report." MIT-LL, FAA TCAS PO. August 2021.

tests covering these branches.

The ADD contains some branches that are unreachable. Those branches are considered useful as defensive or deactivated code and will not be removed. Full branch analysis has not been completed yet in this version of the Test Suite. The branches from prescriptive algorithms which this Test Suite does not yet reach are listed in Table 3, and the branches from suggestive algorithms which this Test Suite does not yet reach are listed in Table 4. Branch types fall into one of four categories: THEN - the *then* conditional branch, ELSE - the *else* conditional branch, ELSE_IF - the *elseif* conditional branch, and IMP_ELSE - an implicit else branch, where there is no explicit *else* conditional branch.

An overview table of all encounters is included in Tables 5 through 12, arranged by Test Group. It includes descriptions of the encounters and information such as input values (ORNCT, AGT, ADS-B, UAT, V2V, point obstacle, coordination), RAs issued, multithreat indication, test groups, and prescriptive flags as described above.

The list of all branches that are covered, and one encounter that covers each branch, are in Table 13. Branches may be covered by other encounters as well.

Table 1. Test Groups

Test Group Number	Functionality	Encounter Numbering
10	RA combinations	1000000-1999999
2x	V2V coordination	
21	V2V coordination during combinations of / transitions between vertical and horizontal RAs	2100000-2199999
22	Cycling through coordination combinations	2200000-2299999
23	Reachback encounters	2300000-2399999
3x	Correlation / decorrelation tests	
31x	Ownship correlation / decorrelation	
311	Spatial correlation	3110000-3119999
312	ID correlation	3120000-3129999
313	Spatial decorrelation	3130000-3139999
314	ID decorrelation	3140000-3149999
32x	Intruder correlation / decorrelation	
321	Spatial correlation	3210000-3219999
322	ID correlation	3220000-3229999
323	Spatial decorrelation	3230000-3239999
324	ID decorrelation	3240000-3249999
4x	Surveillance sources	
41x	Track source selection	
411	Nominal intruder surveillance inputs	4110000-4119999
412	NAR ADS-B intruder surveillance inputs	4120000-4129999
413	NAR V2V intruder surveillance inputs	4130000-4139999
414	NAR pre-tracked intruder surveillance inputs	4140000-4149999
42x	Altitude selection and prioritization	
421	ADS-B intruder	4210000-4219999
422	V2V intruder	4220000-4229999
423	UAT intruder	4230000-4239999
424	AGT intruder	4240000-4249999
425	ORNCT intruder	4250000-4259999
426	Point obstacle intruder	4260000-4269999
427	Ground point obstalce intruder	4270000-4279999
428	Altitude prioritization tests	4280000-4289999
43x	Validation and quality checks	
431	ADS-B validation checks	4310000-4319999
432	ADS-B quality checks	4320000-4329999
433	V2V validation checks	4330000-4339999
434	V2V quality checks	4340000-4349999
435	Ownship quality checks	4350000-4359999
44x	Intruder filtering	
441	Down-select from STM to TRM	4410000-4419999
442	Down-select for HTRM	4420000-4429999
5x	Low-altitude operations	
51	Manual transitions between modes	5100000-5199999
52	Terrain and obstacle scenarios	5200000-5299999
6x	sUAS behaviors and geometries	
61	Wind-relative	6100000-6199999
62	Turning	6200000-6299999
63	Ownship discretes	6300000-6399999
64	Improper inputs	6400000-6499999
65	Priority flags	6500000-6599999
70	Branch coverage	7000000-7999999
80	New functionality testing	8000000-8999999
99	Correlation bench testing (reserved)	9900000-9999999

Table 2. Branches of the prescriptive algorithms not exercised by executing prescriptive encounters

Algorithm	Branch Type	Line Number
UpdatePreventEarlyCOCCState	THEN	49
DetermineCrossing	ELSEIF	33
HorizontalAlertLookahead	IMP_ELSE	39
HorizontalAlertLookahead	ELSE	48
SelectHorizontalAdvisoryWindRelative	ELSE	24
AssociateV2VToV2VAndAGTTargets	IMP_ELSE	14
SetCoordination	THEN	33
SetCoordination	THEN	43
UpdateHorizontalVelocityMode	THEN	14
SelectFinalTrack	IMP_ELSE	10
GetPreviouslySelectedTrack	IMP_ELSE	20
RemoveStaleTracks	THEN	17
RemoveStaleTracks	IMP_ELSE	18
UpdateORNCTTrack	THEN	27
UpdateAGTTrack	IMP_ELSE	7
SetIntruderIDDirectory	IMP_ELSE	15
SetIntruderIDDirectory	IMP_ELSE	31
ReceiveStateVectorUatReport	IMP_ELSE	20

Table 3. Branches not yet reached in prescriptive algorithms

Algorithm	Branch Type	Line Number
DetermineVerticalScore	IMP_ELSE	26
GetIndexFromClassification	IMP_ELSE	18
VerticalTRMUpdatePrep	ELSEIF	16
SetHorizontalTRMSurvOnlyMode	THEN	16
SetVerticalTRMSurvOnlyMode	ELSEIF	20
TauEstimation	IMP_ELSE	42
IndividualCostEstimation	THEN	15
LookupOfflineCost	ELSE	39
GetModifiedGlobalRates	ELSE	16
MaintainRates	THEN	9
MaintainRates	THEN	17
DetermineLDI	THEN	15
DetermineLDI	THEN	16
DetermineLDI	THEN	17
DetermineLDI	ELSE	19
DetermineLDI	ELSEIF	22
DetermineLDI	IMP_ELSE	22
DetermineLDI	THEN	27
DetermineLDI	ELSEIF	29
DetermineLDI	ELSEIF	31
UpdateAltitudeInhibitCState	THEN	17
UpdateAltitudeInhibitCState	IMP_ELSE	19
AltitudeInhibitCost	THEN	18
UpdateMaxReversalCState	THEN	16
MaxReversalCost	THEN	16
MaxReversalCost	ELSEIF	20
MaxReversalCost	THEN	22
MaxReversalCost	IMP_ELSE	22
CompatibilityCost	ELSEIF	44
UpdateSA01HeuristicCState	THEN	14
SA01Heuristic	THEN	16
SA01Heuristic	THEN	17
SA01Heuristic	ELSEIF	19
SA01Heuristic	IMP_ELSE	19
BadMaintainTransitionCost	ELSEIF	23
UpdateCrossingNoAlertCState	THEN	6
ShouldReverse	THEN	15
ShouldReverse	IMP_ELSE	15
ShouldReverse	THEN	21
ShouldReverse	THEN	24

Algorithm	Branch Type	Line Number
ShouldReverse	IMP_ELSE	24
ShouldReverse	THEN	30
ShouldReverse	ELSEIF	32
ShouldReverse	ELSEIF	34
ShouldReverse	IMP_ELSE	34
ShouldReverse	THEN	42
ShouldReverse	ELSEIF	44
ShouldReverse	IMP_ELSE	44
ShouldReverse	THEN	49
ShouldReverse	ELSEIF	51
ShouldReverse	ELSEIF	53
ShouldReverse	ELSE	55
ShouldReverse	IMP_ELSE	21
UpdateCriticalIntervalProtectionCState	IMP_ELSE	34
CriticalIntervalProtectionCost	ELSEIF	23
DetermineOwnNonComplianceFactor	THEN	20
DetermineMultiIntruderAction	IMP_ELSE	22
DetermineMultiIntruderAction	THEN	38
DetermineMultiIntruderAction	THEN	49
DetermineMultiIntruderAction	THEN	50
DetermineMultiIntruderAction	ELSE	52
MTLODetermination	THEN	11
MTLODetermination	IMP_ELSE	17
ArbitrateConflictingSenses	ELSE	41
PersistMTLO	THEN	10
PersistMTLO	ELSEIF	12
RatesToAction	THEN	9
RatesToAction	IMP_ELSE	20
DetermineCrossing	THEN	25
EncodeTIDAltitude	ELSEIF	7
EncodeTIDAltitude	ELSEIF	9
EncodeTIDAltitude	THEN	13
EncodeTIDAltitude	ELSEIF	15
EncodeTIDRange	THEN	6
EncodeTIDRange	ELSEIF	10
EncodeTIDRange	THEN	14
EncodeTIDRange	ELSEIF	16
EncodeTIDBearing	THEN	10
EncodeTIDBearing	IMP_ELSE	4
DetermineTTIAndTID	IMP_ELSE	6
HRCToAdvisory	IMP_ELSE	12

Algorithm	Branch Type	Line Number
IsHorizontalWeakening	THEN	11
CalculateCrossingCostPsiFactor	THEN	18
CalculateCrossingCostPsiFactor	ELSEIF	20
CalculateCrossingCostThetaFactor	ELSEIF	21
CalculateCrossingCostThetaFactor	ELSEIF	23
CalculateCrossingCostVerticalTauFactor	IMP_ELSE	6
UpdatePolicySpeedBinIndividual	THEN	14
HorizontalStateEstimation	IMP_ELSE	14
HorizontalTRMUpdatePrep	ELSEIF	13
ResetHTRMState	IMP_ELSE	12
SelectHorizontalAlertLookaheadAction	IMP_ELSE	28
ApplyHorizontalCoordinationHysteresis	IMP_ELSE	12
DetermineHorizontalScore	IMP_ELSE	24
UpdateTargetTrackAngle	IMP_ELSE	8
SetReportRATerminationBits	ELSEIF	28
SetReportRATerminationBits	ELSEIF	44
SetBlendedCoordinationData	ELSE	13
AddADSBTrackToDB	ELSE	9
AddAltBiasAndSample	IMP_ELSE	32
AssociateV2VToV2VAndAGTTargets	THEN	14
ReceiveStateVectorV2VReport	THEN	5
AddPointObstacleTrackToReport	THEN	13
AddPointObstacleTrackToReport	IMP_ELSE	17
AddHorBias	THEN	9
AssociateICAOtoTarget	THEN	5
HAEAltAtToa	ELSE	16
FilterTracksForTRM	ELSEIF	20
FilterTracksForTRM	IMP_ELSE	14
HeadingAtToa	THEN	6
mahal	THEN	13
nanmin	ELSEIF	5
nanmax	THEN	3
PredictedAltitude	ELSE	29
PredictedAltitude	THEN	30
PredictedAltitude	ELSE	32
ReceivePresAltObservation	THEN	8
ReceivePresAltObservation	THEN	9
ReceivePresAltObservation	IMP_ELSE	9
ReceiveADSBOperationalStatusMessage	IMP_ELSE	4
ReceiveV2VOperationalStatusMessage	IMP_ELSE	5
ReceiveModeStatusReport	IMP_ELSE	5

Algorithm	Branch Type	Line Number
RemoveStaleORNCTTrack	IMP_ELSE	14
SetOwnV2VOperationalStatusMessage	IMP_ELSE	7
SetCoordination	IMP_ELSE	17
SetCoordination	IMP_ELSE	22
SetCoordination	IMP_ELSE	47
SetClassification	IMP_ELSE	26
UpdatePassiveQualityHistory	IMP_ELSE	12
RemoveStaleV2VTrack	IMP_ELSE	8
RemoveStaleV2VTrack	THEN	14
GetGroundSpeed	ELSE	8
UpdateIntruderVRC	THEN	6
ConvertECEFToWGS84	ELSE	26
ConvertECEFToWGS84	THEN	28
ConvertECEFToWGS84	ELSE	30
AdvancePassiveTrackPosition	IMP_ELSE	11
ConvertGVAToSigmaVEPU	THEN	4
ConvertGVAToSigmaVEPU	ELSEIF	8
ConvertGVAToSigmaVEPU	IMP_ELSE	8
SurroundingPoints	THEN	3
PassiveQualityCheck	IMP_ELSE	12
AddTracksToReport	IMP_ELSE	17
GetPreviouslySelectedTrack	IMP_ELSE	7
GetPreviouslySelectedTrack	IMP_ELSE	11
GetPreviouslySelectedTrack	IMP_ELSE	15
GetPreviouslySelectedTrack	IMP_ELSE	18
RemoveStaleTracks	THEN	9
RemoveStaleTracks	THEN	18
ReceiveV2VCoordination	IMP_ELSE	9
AddORNCTTrackToReport	THEN	15
ReceiveOwnRelNonCoopTrack	THEN	10
ReceiveOwnRelNonCoopTrack	IMP_ELSE	12
DetermineORNCTDeletion	THEN	5
DetermineORNCTDeletion	THEN	10
DetermineORNCTDeletion	ELSE	12
DetermineORNCTDeletion	THEN	14
DetermineORNCTDeletion	IMP_ELSE	14
DetermineORNCTDeletion	IMP_ELSE	5
ReceiveAbsoluteGeodeticTrack	THEN	4
ReceiveAbsoluteGeodeticTrack	IMP_ELSE	21
ReceiveAbsoluteGeodeticTrack	ELSEIF	24
ReceiveAbsoluteGeodeticTrack	IMP_ELSE	24

Algorithm	Branch Type	Line Number
DeleteAGTTrack	THEN	3
DeleteAGTTrack	ELSE	5
DeleteAGTTrack	THEN	8
DeleteAGTTrack	IMP_ELSE	8
UpdateAGTTrack	ELSEIF	41
UpdateAGTTrack	IMP_ELSE	56
ReceiveStateVectorUatReport	IMP_ELSE	22
AdvancePassiveTrackState	IMP_ELSE	18
UpdateIntruderHRC	THEN	6
Predict1DKalmanFilter	IMP_ELSE	12

Table 4. Branches not yet reached in suggestive algorithms

Algorithm	Branch Type	Line Number
AddDecorrelatedTrackToTarget	IMP_ELSE	7
ExtrapolateTrack	IMP_ELSE	10
MergeTargets	THEN	27
MergeTargets	THEN	32
CorrelateTargetsToOwnship	THEN	14
CorrelateIndividualTracksToOwnship	IMP_ELSE	20
RemoveDecorrelatedTrackFromOwnship	IMP_ELSE	4
CorrelateByType	THEN	9
CorrelateByType	ELSE	18
CorrelationTrackType	IMP_ELSE	21
GetCorrelationMechanism	IMP_ELSE	7
CorrelateIndividualTracks	IMP_ELSE	18
DecorrelateTargets	IMP_ELSE	39

Table 5. Test Group 10

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
1000001	10		X							1000 6020 4020			X	Different RA combinations. No horizontal RAs.
1000002	10						X		X	1000 6020 4110	1 3 4		X	Different RA combinations. Vert RA starting before Horiz RA. Vert RA ending before Horiz RA.
1000003	10						X		X	5001 6002	3		X	Different RA combinations. Vert RA starting before Horiz RA.
1000004	10						X			1000 4010			X	Different RA combinations. No horizontal RAs.
1000005	10				X				X	1000 4020 4010 6020			X	Different RA combinations. No horizontal RAs.
1000006	10						X			4020 4010 6020			X	Different RA combinations. No horizontal RAs.
1000007	10						X			1000 6020 4010 4020			X	Different RA combinations. No horizontal RAs.
1000008	10						X			4010 4020			X	Different RA combinations. No horizontal RAs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
1000009	10				X					1000 4410 6020	1 2		X	Different RA combinations. Vert and Horiz starting at same time Vert RA ending after Horiz RA
1000010	10				X					4010 4110 5201 4210 4020 5401	1 2	X	X	Different RA combinations. Vert RA starting before Horiz RA. Vert RA ending after Horiz RA
1000011	10				X					1000 5401 6002	1 2		X	Different RA combinations. Vert and Horiz starting at same time. Vert RA ending after Horiz RA.

Table 6. Test Group 20

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2100001	21						X		X	1000 5001 6002	1 3 4		X	V2V coordination #1. Coordination message from intruder, opposite to ownship desired RA. Testing timeout and cancellation. Base encounter, no coordination messages.)
2100002	21						X		X	1000 5001 6002	1 3 4		X	V2V coordination #2. Coordination message from intruder, opposite to ownship desired RA. Testing timeout and cancellation. Coordination message from intruder (opposite to ownship desired RA) received after begin RA. Coordination times out. Ownship is master.
2100003	21						X		X	1000 5001 4210 6020 4010	1 2 3 4		X	V2V coordination #3. Coordination message from intruder, opposite to ownship desired RA. Testing timeout and cancellation. Coordination message from intruder (opposite to ownship desired RA) received after begin RA. Coordination times out. Ownship is slave.
2100004	21						X		X	1000 5001 6002	1 3 4		X	V2V coordination #4. Coordination message from intruder, opposite to ownship desired RA. Testing timeout and cancellation. Coordination message from intruder (opposite to ownship desired RA) received after begin RA. Coordination then cancelled. Ownship is master.
2100005	21						X		X	1000 5001 4210 6020 4010	1 3 4		X	V2V coordination #5. Coordination message from intruder, opposite to ownship desired RA. Testing timeout and cancellation. Coordination message from intruder (opposite to ownship desired RA) received after begin RA. Coordination then cancelled. Ownship is slave.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2100006	21						X		X	1000 5001 6002	1 2 4		X	V2V coordination delay #1. Coordination message from intruder, opposite to ownship desired RA. Testing coordination delay (delay for vertical, no delay for horiontal). Coordination message from intruder (opposite to ownship desired RA) received 3s before begin RA. Ownship is master.
2100007	21						X		X	1000 6020 4110	1 2 4		X	V2V coordination delay #2. Coordination message from intruder, opposite to ownship desired RA. Testing coordination delay (delay for vertical, no delay for horiontal). Coordination message from intruder (opposite to ownship desired RA) received 3s before begin RA. Ownship is slave.
2100011	21						X			1000 5001 4210 4010	1 2 4		X	V2V Coordination, vertical and horizontal combination #1: vert and horiz separate, together in one message, and combined with valid message in one and invalid message in the other dimension. Base encounter without coordination message.
2100012	21						X		X	6020 1000 5001 4010 5201 4210	1 3 4		X	V2V Coordination encounters, vertical and horizontal combination #2: vert and horiz separate, together in one message, and combined with valid message in one and invalid message in the other dimension. Vertical and horizontal coordination in separate coordination messages. Ownship is slave.
2100013	21						X		X	6020 1000 5001 4010 5201 4210	1 3 4		X	V2V Coordination encounters, vertical and horizontal combination #3: vert and horiz separate, together in one message, and combined with valid message in one and invalid message in the other dimension. Vertical and horizontal coordination in same coordination messages. Ownship is slave.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2100014	21						X		X	6020 1000 5001 4010 5201 4210	1 3 4		X	V2V Coordination encounters, vertical and horizontal combination #4: vert and horiz separate, together in one message, and combined with valid message in one and invalid message in the other dimension. Vertical and horizontal coordination in separate coordination messages with other dimension having invalid values. Ownship is slave.
2100021	21						X			1000 5001 4010 4210 6002 6422	1 3 4	X	X	V2V Coordination encounters with multiple intruders #1: verify that vrc/hrc are correctly applied. Baseline encounter, no coordination messages.
2100022	21						X		X	1000 5001 4010 5201 4210 6422	1 3 4	X	X	V2V Coordination encounters with multiple intruders #2: verify that vrc/hrc are correctly applied. Coordination messages for main intruder. Ownship is slave.
2100023	21						X		X	1000 5001 4010 4210 6002 6422	1 3 4	X	X	V2V Coordination encounters with multiple intruders #3: verify that vrc/hrc are correctly applied. Coordination messages for secondary intruder. Ownship is slave.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2100024	21						X		X	1000 5001 4010 4210 6002 6422	1 3 4	X	X	V2V Coordination encounters with multiple intruders #4: verify that vrc/hrc are correctly applied. Coordination messages with invalid v2v_uid. Ownship is slave.
2200000	22				X		X			5001	3			V2V Coordination Combinations Encounter #1 of 25: Base encounter without any Operational Capability or Coordination Messages.
2200001	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #2 of 25: CA Operational status is 1, Type Capability is 1, Priority is 7, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200002	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #3 of 25: CA Operational status is 1, Type Capability is 1, Priority is 9, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200003	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #4 of 25: CA Operational status is 1, Type Capability is 1, Priority is 5, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200004	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #5 of 25: CA Operational status is 1, Type Capability is 1, Priority is 7, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200005	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #6 of 25: CA Operational status is 1, Type Capability is 1, Priority is 9, and Equipment is 31, V2V Coordination message received after 7 seconds.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2200006	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #7 of 25: CA Operational status is 1, Type Capability is 1, Priority is 5, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200007	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #8 of 25: CA Operational status is 1, Type Capability is 2, Priority is 7, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200008	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #9 of 25: CA Operational status is 1, Type Capability is 2, Priority is 9, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200009	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #10 of 25: CA Operational status is 1, Type Capability is 2, Priority is 5, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200010	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #11 of 25: CA Operational status is 1, Type Capability is 2, Priority is 7, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200011	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #12 of 25: CA Operational status is 1, Type Capability is 2, Priority is 9, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200012	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #13 of 25: CA Operational status is 1, Type Capability is 2, Priority is 5, and Equipment is 31, V2V Coordination message received after 7 seconds.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2200013	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #14 of 25: CA Operational status is 2, Type Capability is 1, Priority is 7, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200014	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #15 of 25: CA Operational status is 2, Type Capability is 1, Priority is 9, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200015	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #16 of 25: CA Operational status is 2, Type Capability is 1, Priority is 5, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200016	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #17 of 25: CA Operational status is 2, Type Capability is 1, Priority is 7, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200017	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #18 of 25: CA Operational status is 2, Type Capability is 1, Priority is 9, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200018	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #19 of 25: CA Operational status is 2, Type Capability is 1, Priority is 5, and Equipment is 31, V2V Coordination message received after 7 seconds.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2200019	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #20 of 25: CA Operational status is 2, Type Capability is 2, Priority is 7, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200020	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #21 of 25: CA Operational status is 2, Type Capability is 2, Priority is 9, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200021	22						X		X	5001	3		X	V2V Coordination Combinations Encounter #22 of 25: CA Operational status is 2, Type Capability is 2, Priority is 5, and Equipment is 15, V2V Coordination message received after 7 seconds.
2200022	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #23 of 25: CA Operational status is 2, Type Capability is 2, Priority is 7, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200023	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #24 of 25: CA Operational status is 2, Type Capability is 2, Priority is 9, and Equipment is 31, V2V Coordination message received after 7 seconds.
2200024	22						X		X	5001 4210 4010	3		X	V2V Coordination Combinations Encounter #25 of 25: CA Operational status is 2, Type Capability is 2, Priority is 5, and Equipment is 31, V2V Coordination message received after 7 seconds.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2200100	22				X		X			5001	3			ADSB Coordination Combinations Encounter #1 of 13: Base encounter without any Operational Capability or Coordination Messages.
2200101	22				X		X		X	5001	3			ADSB Coordination Combinations Encounter #2 of 13: V2V CA Status is 0, ADSB CA Operational status is 0, Type Capability is 0, V2V Coordination message received after 8 seconds.
2200102	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #3 of 13: V2V CA Status is 0, ADSB CA Operational status is 0, Type Capability is 1, V2V Coordination message received after 8 seconds.
2200103	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #4 of 13: V2V CA Status is 0, ADSB CA Operational status is 0, Type Capability is 2, V2V Coordination message received after 8 seconds.
2200104	22				X		X		X	5001	3			ADSB Coordination Combinations Encounter #5 of 13: V2V CA Status is 0, ADSB CA Operational status is 3, Type Capability is 0, V2V Coordination message received after 8 seconds.
2200105	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #6 of 13: V2V CA Status is 0, ADSB CA Operational status is 3, Type Capability is 1, V2V Coordination message received after 8 seconds.
2200106	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #7 of 13: V2V CA Status is 0, ADSB CA Operational status is 3, Type Capability is 2, V2V Coordination message received after 8 seconds.
2200107	22				X		X		X	5001	3			ADSB Coordination Combinations Encounter #8 of 13: V2V CA Status is 1, ADSB CA Operational status is 0, Type Capability is 0, V2V Coordination message received after 8 seconds.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2200108	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #9 of 13: V2V CA Status is 1, ADSB CA Operational status is 0, Type Capability is 1, V2V Coordination message received after 8 seconds.
2200109	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #10 of 13: V2V CA Status is 1, ADSB CA Operational status is 0, Type Capability is 2, V2V Coordination message received after 8 seconds.
2200110	22				X		X		X	5001	3			ADSB Coordination Combinations Encounter #11 of 13: V2V CA Status is 1, ADSB CA Operational status is 3, Type Capability is 0, V2V Coordination message received after 8 seconds.
2200111	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #12 of 13: V2V CA Status is 1, ADSB CA Operational status is 3, Type Capability is 1, V2V Coordination message received after 8 seconds.
2200112	22				X		X		X	5001 4210 4010	3			ADSB Coordination Combinations Encounter #13 of 13: V2V CA Status is 1, ADSB CA Operational status is 3, Type Capability is 2, V2V Coordination message received after 8 seconds.
2300001	23						X			5001 6002	3		X	Reachback encounter #1: Base encounter without coordination message. Ownship is slave.
2300002	23						X		X	1000 4010 6020	1 2 4		X	Reachback encounter #2: Coordination messages are received right before RAs (vert and horiz) occurr. Ownship is slave.
2300003	23						X		X	1000 5001 4210 6020 4010	1 3 4		X	Reachback encounter #3: Coordination messages are received right after RAs (vert and horiz) occurr. Ownship is slave.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
2300004	23						X		X	1000 4010 6020	1 2 4		X	Reachback encounter #4: Coordination messages are received 200ms after the cycle the RAs (vert and horiz) occurr. Run with 600ms reachback, coordination messages are effective for begin of RA (same as reachnack encounter #2). Ownship is slave.

Table 7. Test Group 30

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3110000	311			X						1000 5001 6002	2 3 4			Ownship correlation / decorrelation encounters (#1 of 4): Spatial correlation. Ownship has WGS84, intruder has AGT-ID. Intruder's AGT is separate from ownship and then begins trailling ownship close enough that its track correlates with ownship. Intruder track seems to disappear and alerting stops at t=24. [Note that the agt_id field is not populated for ownship - this will be fixed in V4R1.]
3120000	312		X										X	Ownship correlation encounters (#2 of 4): ID correlation. Ownship has WGS84, intruder has AGT-ID and V2V-UID. Because the intruder AGT has no valid Mode S and has valid V2V UID which matches ownship's V2V-UID, the AGT track is associated with ownship throughout the encounter. [Note that the agt_id field is not populated for ownship - this will be fixed in V4R1.]
3130000	311			X						1000 5001	1 2			Ownship correlation / decorrelation encounters (#3 of 4): Spatial decorrelation. Ownship has WGS84, intruder has AGT-ID. Intruder's AGT is trailing ownship close enough that their tracks correlate. Then the intruder's track begins to split off and decorrelates from ownship. Intruder track appears at t=37 and alerting begins at t=39. [Note that the agt_id field is not populated for ownship - this will be fixed in V4R1.]

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3140001	312		X										X	Ownship correlation encounters (#4 of 4): ID correlation. Ownship has WGS84, intruder has AGT-ID and V2V-UID. Because the intruder AGT has no valid Mode S and has valid V2V UID which does not match ownship's V2V-UID, the AGT track is not correlated with ownship during the encounter.
3211000	321				X		X				3			Intruder spatial correlation / decorrelation encounters (#1 of 34): Spatial correlation. T1: V2V with V2V-UID and ICAO Mode S. T2: ADS-B with anonymous Mode S. At t=33, the tracks spatially correlate and both IDs (V2V-UID of 111, anonymous Mode S of 222, ICAO Mode S of 333) populate the intruder ID dictionary.
3211001	321				X		X							Intruder spatial correlation / decorrelation encounters (#2 of 34): Spatial correlation. T1: V2V with V2V-UID and anonymous Mode S. T2: ADS-B with ICAO Mode S. At t=33, the tracks spatially correlate and both IDs (V2V-UID of 111, ICAO Mode S of 222, anonymous Mode S of 333) populate the intruder ID dictionary.
3212000	321	X			X						1 3			Intruder spatial correlation / decorrelation encounters (#3 of 34): Spatial correlation. T1: ORNCT with ORNCT-ID. T2: ADS-B with ICAO Mode S. At t=54, the tracks spatially correlate and both IDs (ORNCT-ID of 11 and ICAO Mode S of 222) populate the intruder ID dictionary.
3212001	321	X			X						1 3			Intruder spatial correlation / decorrelation encounters (#4 of 34): Spatial correlation. T1: ORNCT with ORNCT-ID. T2: ADS-B with anonymous Mode S. At t=54, the tracks spatially correlate and both IDs (ORNCT-ID of 111 and anonymous Mode S of 222) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3212002	321	X					X							Intruder spatial correlation / decorrelation encounters (#5 of 34): Spatial correlation. T1: ORNCT with ORNCT-ID. T2: V2V with V2V-UID. At t=58, the tracks spatially correlate and both IDs (ORNCT-ID of 111 and V2V-UID of 222) populate the intruder ID dictionary.
3212003	321	X					X				1 3			Intruder spatial correlation / decorrelation encounters (#6 of 34): Spatial correlation. T1: ORNCT with ORNCT-ID. T2: V2V with V2V-UID and ICAO Mode S. At t=58, the tracks spatially correlate and both IDs (ORNCT-ID of 111, V2V-UID of 222, ICAO Mode S of 333) populate the intruder ID dictionary.
3212004	321	X					X				1 3			Intruder spatial correlation / decorrelation encounters (#7 of 34): Spatial correlation. T1: ORNCT with ORNCT-ID. T2: V2V with V2V-UID and anonymous Mode S. At t=58, the tracks spatially correlate and both IDs (ORNCT-ID of 111, V2V-UID of 222, anonymous Mode S of 444) populate the intruder ID dictionary.
3213000	321		X		X						1 3 4			Intruder spatial correlation / decorrelation encounters (#8 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: ADS-B with ICAO Mode S. At t=36, the tracks spatially correlate and both IDs (AGT-ID of 111 and ICAO Mode S of 222) populate the intruder ID dictionary.
3213001	321		X		X						1 3 4			Intruder spatial correlation / decorrelation encounters (#9 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: ADS-B with anonymous Mode S. At t=36, the tracks spatially correlate and both IDs (AGT-ID of 111 and anonymous Mode S of 222) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3213002	321		X		X						1 3 4			Intruder spatial correlation / decorrelation encounters (#10 of 34): Spatial correlation. T1: AGT with AGT-ID and ICAO Mode S. T2: ADS-B with anonymous Mode S. At t=36, the tracks spatially correlate and all IDs (AGT-ID of 111, ICAO Mode S of 333, anonymous Mode S of 222) populate the intruder ID dictionary.
3213003	321		X		X						1 3 4			Intruder spatial correlation / decorrelation encounters (#11 of 34): Spatial correlation. T1: AGT with AGT-ID and anonymous Mode S. T2: ADS-B with ICAO Mode S. At t=36, the tracks spatially correlate and all IDs (AGT-ID of 111, anonymous Mode S of 333, ICAO Mode S of 222) populate the intruder ID dictionary.
3213004	321		X		X						1 3 4			Intruder spatial correlation / decorrelation encounters (#12 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: ADS-B with anonymous Mode S. At t=36, the tracks spatially correlate and all IDs (AGT-ID of 111, V2V-UID of 333, ICAO Mode S of 444, anonymous Mode S of 222) populate the intruder ID dictionary.
3213005	321		X		X						1 3 4			Intruder spatial correlation / decorrelation encounters (#13 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: ADS-B with ICAO Mode S. At t=36, the tracks spatially correlate and all IDs (AGT-ID of 111, V2V-UID of 333, anonymous Mode S of 444, ICAO Mode S of 222) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3213006	321		X				X							Intruder spatial correlation / decorrelation encounters (#14 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: V2V with V2V-UID. At t=42, the tracks spatially correlate and both IDs (AGT-ID of 111 and V2V-UID of 222) populate the intruder ID dictionary.
3213007	321		X				X			1000 5001	1 3	X		Intruder spatial correlation / decorrelation encounters (#15 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: V2V with V2V-UID and ICAO Mode S. At t=42, the tracks spatially correlate and all IDs (AGT-ID of 111, V2V-UID of 222, ICAO Mode S of 333) populate the intruder ID dictionary.
3213008	321		X				X			1000 5001	1 3	X		Intruder spatial correlation / decorrelation encounters (#16 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: V2V with V2V-UID and anonymous Mode S. At t=42, the tracks spatially correlate and all IDs (AGT-ID of 111, V2V-UID of 222, anonymous Mode S of 333) populate the intruder ID dictionary.
3213009	321		X				X			1000 5001	1 3	X		Intruder spatial correlation / decorrelation encounters (#17 of 34): Spatial correlation. T1: AGT with AGT-ID and ICAO Mode S. T2: V2V with V2V-UID and anonymous Mode S. At t=42, the tracks spatially correlate and all IDs (AGT-ID of 111, V2V-UID of 222, ICAO Mode S of 333, anonymous Mode S of 444) populate the intruder ID dictionary.
3213010	321		X				X			1000 5001	1 3	X		Intruder spatial correlation / decorrelation encounters (#18 of 34): Spatial correlation. T1: AGT with AGT-ID and anonymous Mode S. T2: V2V with V2V-UID and ICAO Mode S. At t=42, the tracks spatially correlate and all IDs (AGT-ID of 111, V2V-UID of 222, anonymous Mode S of 333, ICAO Mode S of 444) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3213011	321	X	X											Intruder spatial correlation / decorrelation encounters (#19 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: ORNCT with ORNCT-ID. At t=37, the tracks spatially correlate and both IDs (AGT-ID of 111, ORNCT-ID of 222) populate the intruder ID dictionary.
3213012	321	X	X							5001	1 3 4	X		Intruder spatial correlation / decorrelation encounters (#20 of 34): Spatial correlation. T1: AGT with AGT-ID and ICAO Mode S. T2: ORNCT with ORNCT-ID. At t=37, the tracks spatially correlate and both IDs (AGT-ID of 111, ORNCT-ID of 222, ICAO Mode S of 333) populate the intruder ID dictionary.
3213013	321	X	X							5001	1 3 4	X		Intruder spatial correlation / decorrelation encounters (#21 of 34): Spatial correlation. T1: AGT with AGT-ID and anonymous Mode S. T2: ORNCT with ORNCT-ID. At t=37, the tracks spatially correlate and both IDs (AGT-ID of 111, ORNCT-ID of 222, anonymous Mode S of 333) populate the intruder ID dictionary.
3213014	321	X	X											Intruder spatial correlation / decorrelation encounters (#22 of 34): Spatial correlation. T1: AGT with AGT-ID and V2V-UID. T2: ORNCT with ORNCT-ID. At t=37, the tracks spatially correlate and both IDs (AGT-ID of 111, ORNCT-ID of 222, V2V-UID of 333) populate the intruder ID dictionary.
3213015	321	X	X							5001	1 3 4	X		Intruder spatial correlation / decorrelation encounters (#23 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: ORNCT with ORNCT-ID. At t=37, the tracks spatially correlate and both IDs (AGT-ID of 111, ORNCT-ID of 222, V2V-UID of 333, ICAO Mode S of 444) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3213016	321	X	X							5001	1 3 4	X		Intruder spatial correlation / decorrelation encounters (#24 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: ORNCT with ORNCT-ID. At t=37, the tracks spatially correlate and both IDs (AGT-ID of 111, ORNCT-ID of 222, V2V-UID of 333, anonymous Mode S of 444) populate the intruder ID dictionary.
3213017	321		X											Intruder spatial correlation / decorrelation encounters (#25 of 34): Spatial correlation. T1: AGT with AGT-ID. T2: AGT with AGT-ID. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222) populate the intruder ID dictionary.
3213018	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#26 of 34): Spatial correlation. T1: AGT with AGT-ID and ICAO Mode S. T2: AGT with AGT-ID. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, ICAO Mode S of 333) populate the intruder ID dictionary.
3213019	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#27 of 34): Spatial correlation. T1: AGT with AGT-ID and anonymous Mode S. T2: AGT with AGT-ID. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, anonymous Mode S of 333) populate the intruder ID dictionary.
3213020	321		X											Intruder spatial correlation / decorrelation encounters (#28 of 34): Spatial correlation. T1: AGT with AGT-ID and V2V-UID. T2: AGT with AGT-ID. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3213021	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#29 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333, ICAO Mode S of 444) populate the intruder ID dictionary.
3213022	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#30 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333, anonymous Mode S of 444) populate the intruder ID dictionary.
3213023	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#31 of 34): Spatial correlation. T1: AGT with AGT-ID, anonymous Mode S. T2: AGT with AGT-ID, ICAO Mode S. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, anonymous Mode S of 333, ICAO Mode S of 444) populate the intruder ID dictionary.
3213024	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#32 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID, ICAO Mode S. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, anonymous Mode S of 333, ICAO Mode S of 444, V2V-UID of 555) populate the intruder ID dictionary.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3213025	321		X								1 3 4			Intruder spatial correlation / decorrelation encounters (#33 of 34): Spatial correlation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID, anonymous Mode S. At t=38, the tracks spatially correlate and both IDs (AGT-ID of 111, AGT-ID of 222, ICAO Mode S of 333, anonymous Mode S of 444, V2V-UID of 555) populate the intruder ID dictionary.
3214000	321				X									Intruder spatial correlation / decorrelation encounters (#34 of 34): Spatial correlation. T1: ADS-B with anonymous Mode S. T2: ADS-B with ICAO Mode S. At t=25, the tracks spatially correlate and both IDs (anonymous Mode S of 111, ICAO Mode S of 222) populate the intruder ID dictionary.
3221000	322				X		X				3			Intruder ICAO Mode S ID correlation encounters (#1 of 6): T1: V2V with V2V-UID and ICAO Mode S. T2: ADS-B with ICAO Mode S. Because the 2 tracks share the same ICAO Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (V2V-UID of 111, ICAO Mode S of 222) populate the intruder ID dictionary
3221001	322		X		X									Intruder ICAO Mode S ID correlation encounters (#2 of 6): T1: AGT with AGT-ID and ICAO Mode S. T2: ADS-B with ICAO Mode S. Because the 2 tracks share the same ICAO Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, ICAO Mode S of 222) populate the intruder ID dictionary

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3221002	322		X		X									Intruder ICAO Mode S ID correlation encounters (#3 of 6): T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: ADS-B with ICAO Mode S. Because the 2 tracks share the same ICAO Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, ICAO Mode S of 222, V2V-UID of 333) populate the intruder ID dictionary
3221003	322		X				X				1 3			Intruder ICAO Mode S ID correlation encounters (#4 of 6): T1: AGT with AGT-ID and ICAO Mode S. T2: V2V with V2V-UID and ICAO Mode S. Because the 2 tracks share the same ICAO Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, V2V-UID of 222, ICAO Mode S of 333) populate the intruder ID dictionary
3221004	322		X								1 3 4			Intruder ICAO Mode S ID correlation encounters (#5 of 6): T1: AGT with AGT-ID and ICAO Mode S. T2: AGT with AGT-ID and ICAO Mode S. Because the 2 tracks share the same ICAO Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, ICAO Mode S of 333) populate the intruder ID dictionary
3221005	322		X								1 3 4			Intruder ICAO Mode S ID correlation encounters (#6 of 6): T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID and ICAO Mode S. Because the 2 tracks share the same ICAO Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, ICAO Mode S of 333, V2V-UID of 444) populate the intruder ID dictionary

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3222000	322				X		X				3			Intruder anonymous Mode S ID correlation encounters (#1 of 6): T1: V2V with V2V-UID and anonymous Mode S. T2: ADS-B with anonymous Mode S. Because the 2 tracks share the same anonymous Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (V2V-UID of 111, anonymous Mode S of 222) populate the intruder ID dictionary
3222001	322		X		X									Intruder anonymous Mode S ID correlation encounters (#2 of 6): T1: AGT with AGT-ID and anonymous Mode S. T2: ADS-B with anonymous Mode S. Because the 2 tracks share the same anonymous Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, anonymous Mode S of 222) populate the intruder ID dictionary
3222002	322		X		X									Intruder anonymous Mode S ID correlation encounters (#3 of 6): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: ADS-B with anonymous Mode S. Because the 2 tracks share the same anonymous Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, anonymous Mode S of 222, V2V-UID of 333) populate the intruder ID dictionary
3222003	322		X				X				1 3			Intruder anonymous Mode S ID correlation encounters (#4 of 6): T1: AGT with AGT-ID and anonymous Mode S. T2: V2V with V2V-UID and anonymous Mode S. Because the 2 tracks share the same anonymous Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, V2V-UID of 222, anonymous Mode S of 333) populate the intruder ID dictionary

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3222004	322		X								1 3 4			Intruder anonymous Mode S ID correlation encounters (#5 of 6): T1: AGT with AGT-ID and anonymous Mode S. T2: AGT with AGT-ID and anonymous Mode S. Because the 2 tracks share the same anonymous Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, anonymous Mode S of 333) populate the intruder ID dictionary
3222005	322		X								1 3 4			Intruder anonymous Mode S ID correlation encounters (#6 of 6): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID and anonymous Mode S. Because the 2 tracks share the same anonymous Mode S address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, anonymous Mode S of 333, V2V-UID of 444) populate the intruder ID dictionary
3223000	322		X				X							Intruder V2V-UID ID correlation encounters (#1 of 5): T1: AGT with AGT-ID and V2V-UID. T2: V2V with V2V-UID. Because the 2 tracks share the same V2V-UID address, the tracks correlate via ID throughout the duration of the encounter. Both IDs (AGT-ID of 111 and V2V-UID of 222) populate the intruder ID dictionary
3223001	322		X				X				1 3			Intruder V2V-UID ID correlation encounters (#2 of 5): T1: AGT with AGT-ID, V2V-UID, and ICAO Mode S. T2: V2V with V2V-UID and anonymous Mode S. Because the 2 tracks share the same V2V-UID address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111 and V2V-UID of 222, ICAO Mode S of 333, anonymous Mode S of 444) populate the intruder ID dictionary

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3223002	322		X				X				1 3			Intruder V2V-UID ID correlation encounters (#3 of 5): T1: AGT with AGT-ID, V2V-UID, and anonymous Mode S. T2: V2V with V2V-UID and ICAO Mode S. Because the 2 tracks share the same V2V-UID address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111 and V2V-UID of 222, anonymous Mode S of 333, ICAO Mode S of 444) populate the intruder ID dictionary
3223003	322		X											Intruder V2V-UID ID correlation encounters (#4 of 5): T1: AGT with AGT-ID and V2V-UID. T2: AGT with AGT-ID and V2V-UID. Because the 2 tracks share the same V2V-UID address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333) populate the intruder ID dictionary
3223004	322		X								1 3 4			Intruder V2V-UID ID correlation encounters (#5 of 5): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID, V2V-UID, ICAO Mode S. Because the 2 tracks share the same V2V-UID address, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333, ICAO Mode S of 444, anonymous Mode S of 555) populate the intruder ID dictionary
3224000	322		X				X				1 3			Intruder V2V-UID and ICAO Mode S ID correlation encounters (#1 of 2): T1: AGT with AGT-ID, V2V-UID, and ICAO Mode S. T2: V2V with V2V-UID and ICAO Mode S. Because the 2 tracks share the same V2V-UID and ICAO Mode S addresses, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111 and V2V-UID of 222, ICAO Mode S of 333) populate the intruder ID dictionary

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3224001	322		X								1 3 4			Intruder V2V-UID and ICAO Mode S ID correlation encounters (#2 of 2): T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID, V2V-UID, ICAO Mode S. Because the 2 tracks share the same V2V-UID and ICAO Mode S addresses, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333, ICAO Mode S of 444) populate the intruder ID dictionary
3225000	322		X				X				1 3			Intruder V2V-UID and anonymous Mode S ID correlation encounters (#1 of 2): T1: AGT with AGT-ID, V2V-UID, and anonymous Mode S. T2: V2V with V2V-UID and anonymous Mode S. Because the 2 tracks share the same V2V-UID and anonymous Mode S addresses, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111 and V2V-UID of 222, anonymous Mode S of 333) populate the intruder ID dictionary
3225001	322		X								1 3 4			Intruder V2V-UID and ICAO Mode S ID correlation encounters (#2 of 2): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID, V2V-UID, anonymous Mode S. Because the 2 tracks share the same V2V-UID and anonymous Mode S addresses, the tracks correlate via ID throughout the duration of the encounter. All IDs (AGT-ID of 111, AGT-ID of 222, V2V-UID of 333, anonymous Mode S of 444) populate the intruder ID dictionary

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3231000	323				X		X							Intruder spatial correlation / decorrelation encounters (#1 of 34): Spatial decorrelation. T1: V2V with V2V-UID and ICAO Mode S. T2: ADS-B with anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3231001	323				X		X							Intruder spatial correlation / decorrelation encounters (#2 of 34): Spatial decorrelation. T1: V2V with V2V-UID and anonymous Mode S. T2: ADS-B with ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3232000	323	X			X									Intruder spatial correlation / decorrelation encounters (#3 of 34): Spatial decorrelation. T1: ORNCT with ORNCT-ID. T2: ADS-B with ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=57, the tracks spatially decorrelate.
3232001	323	X			X									Intruder spatial correlation / decorrelation encounters (#4 of 34): Spatial decorrelation. T1: ORNCT with ORNCT-ID. T2: ADS-B with anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=57, the tracks spatially decorrelate.
3232002	323	X					X							Intruder spatial correlation / decorrelation encounters (#5 of 34): Spatial decorrelation. T1: ORNCT with ORNCT-ID. T2: V2V with V2V-UID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=47, the tracks spatially decorrelate.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3232003	323	X					X							Intruder spatial correlation / decorrelation encounters (#6 of 34): Spatial decorrelation. T1: ORNCT with ORNCT-ID. T2: V2V with V2V-UID and ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=47, the tracks spatially decorrelate.
3232004	323	X					X							Intruder spatial correlation / decorrelation encounters (#7 of 34): Spatial decorrelation. T1: ORNCT with ORNCT-ID. T2: V2V with V2V-UID and anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=47, the tracks spatially decorrelate.
3233000	323		X		X									Intruder spatial correlation / decorrelation encounters (#8 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: ADS-B with ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3233001	323		X		X									Intruder spatial correlation / decorrelation encounters (#9 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: ADS-B with anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3233002	323		X		X									Intruder spatial correlation / decorrelation encounters (#10 of 34): Spatial decorrelation. T1: AGT with AGT-ID and ICAO Mode S. T2: ADS-B with anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3233003	323		X		X									Intruder spatial correlation / decorrelation encounters (#11 of 34): Spatial decorrelation. T1: AGT with AGT-ID and anonymous Mode S. T2: ADS-B with ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3233004	323		X		X									Intruder spatial correlation / decorrelation encounters (#12 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: ADS-B with anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3233005	323		X		X									Intruder spatial correlation / decorrelation encounters (#13 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: ADS-B with ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=55, the tracks spatially decorrelate.
3233006	323		X				X							Intruder spatial correlation / decorrelation encounters (#14 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: V2V with V2V-UID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=39, the tracks spatially decorrelate.
3233007	323		X				X							Intruder spatial correlation / decorrelation encounters (#15 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: V2V with V2V-UID and ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=39, the tracks spatially decorrelate.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3233008	323		X				X							Intruder spatial correlation / decorrelation encounters (#16 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: V2V with V2V-UID and anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=39, the tracks spatially decorrelate.
3233009	323		X				X							Intruder spatial correlation / decorrelation encounters (#17 of 34): Spatial decorrelation. T1: AGT with AGT-ID and ICAO Mode S. T2: V2V with V2V-UID and anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=39, the tracks spatially decorrelate.
3233010	323		X				X							Intruder spatial correlation / decorrelation encounters (#18 of 34): Spatial decorrelation. T1: AGT with AGT-ID and anonymous Mode S. T2: V2V with V2V-UID and ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=39, the tracks spatially decorrelate.
3233011	323	X	X											Intruder spatial correlation / decorrelation encounters (#19 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: ORNCT with ORNCT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=44, the tracks spatially decorrelate.
3233012	323	X	X											Intruder spatial correlation / decorrelation encounters (#20 of 34): Spatial decorrelation. T1: AGT with AGT-ID and ICAO Mode S. T2: ORNCT with ORNCT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=44, the tracks spatially decorrelate.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3233013	323	X	X											Intruder spatial correlation / decorrelation encounters (#21 of 34): Spatial decorrelation. T1: AGT with AGT-ID and anonymous Mode S. T2: ORNCT with ORNCT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=44, the tracks spatially decorrelate.
3233014	323	X	X											Intruder spatial correlation / decorrelation encounters (#22 of 34): Spatial decorrelation. T1: AGT with AGT-ID and V2V-UID. T2: ORNCT with ORNCT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=44, the tracks spatially decorrelate.
3233015	323	X	X											Intruder spatial correlation / decorrelation encounters (#23 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: ORNCT with ORNCT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=44, the tracks spatially decorrelate.
3233016	323	X	X											Intruder spatial correlation / decorrelation encounters (#24 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: ORNCT with ORNCT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=44, the tracks spatially decorrelate.
3233017	323		X											Intruder spatial correlation / decorrelation encounters (#25 of 34): Spatial decorrelation. T1: AGT with AGT-ID. T2: AGT with AGT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3233018	323		X											Intruder spatial correlation / decorrelation encounters (#26 of 34): Spatial decorrelation. T1: AGT with AGT-ID and ICAO Mode S. T2: AGT with AGT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3233019	323		X											Intruder spatial correlation / decorrelation encounters (#27 of 34): Spatial decorrelation. T1: AGT with AGT-ID and anonymous Mode S. T2: AGT with AGT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3233020	323		X											Intruder spatial correlation / decorrelation encounters (#28 of 34): Spatial decorrelation. T1: AGT with AGT-ID and V2V-UID. T2: AGT with AGT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3233021	323		X											Intruder spatial correlation / decorrelation encounters (#29 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3233022	323		X											Intruder spatial correlation / decorrelation encounters (#30 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3233023	323		X											Intruder spatial correlation / decorrelation encounters (#31 of 34): Spatial decorrelation. T1: AGT with AGT-ID and anonymous Mode S. T2: AGT with AGT-ID and ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3233024	323		X											Intruder spatial correlation / decorrelation encounters (#32 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID and ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3233025	323		X											Intruder spatial correlation / decorrelation encounters (#33 of 34): Spatial decorrelation. T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID and anonymous Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=38, the tracks spatially decorrelate.
3234000	323				X									Intruder spatial correlation / decorrelation encounters (#34 of 34): Spatial decorrelation. T1: ADS-B with anonymous Mode S. T2: ADS-B with ICAO Mode S. The two tracks are so close together that they correlate spatially. As the tracks drift apart, at t=62, the tracks spatially decorrelate.
3241000	324				X		X				3		X	Intruder ICAO Mode S ID correlation encounters (#1 of 6): T1: V2V with V2V-UID and ICAO Mode S. T2: ADS-B with ICAO Mode S. Because the 2 tracks have different ICAO Mode S addresses (222 and 333), the tracks do not correlate throughout the duration of the encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3241001	324		X		X						1 3		X	Intruder ICAO Mode S ID correlation encounters (#2 of 6): T1: AGT with AGT-ID and ICAO Mode S. T2: ADS-B with ICAO Mode S. Because the 2 tracks have different ICAO Mode S addresses (222 and 333), the tracks do not correlate throughout the duration of the encounter.
3241002	324		X		X						1 3		X	Intruder ICAO Mode S ID correlation encounters (#3 of 6): T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: ADS-B with ICAO Mode S. Because the 2 tracks have different ICAO Mode S addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3241003	324		X				X			1000 5001	1 3	X	X	Intruder ICAO Mode S ID correlation encounters (#4 of 6): T1: AGT with AGT-ID and ICAO Mode S. T2: V2V with V2V-UID and ICAO Mode S. Because the 2 tracks have different ICAO Mode S addresses (333 and 777), the tracks do not correlate throughout the duration of the encounter.
3241004	324		X								3 4		X	Intruder ICAO Mode S ID correlation encounters (#5 of 6): T1: AGT with AGT-ID and ICAO Mode S. T2: AGT with AGT-ID and ICAO Mode S. Because the 2 tracks have different ICAO Mode S addresses (333 and 777), the tracks do not correlate throughout the duration of the encounter.
3241005	324		X								3 4		X	Intruder ICAO Mode S ID correlation encounters (#6 of 6): T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID and ICAO Mode S. Because the 2 tracks have different ICAO Mode S addresses (333 and 777), the tracks do not correlate throughout the duration of the encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3242000	324				X		X						X	Intruder anonymous Mode S ID correlation encounters (#1 of 6): T1: V2V with V2V-UID and anonymous Mode S. T2: ADS-B with anonymous Mode S. Because the 2 tracks have different anonymous Mode S addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3242001	324		X		X						1 3		X	Intruder anonymous Mode S ID correlation encounters (#2 of 6): T1: AGT with AGT-ID and anonymous Mode S. T2: ADS-B with anonymous Mode S. Because the 2 tracks have different anonymous Mode S addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3242002	324		X		X						1 3		X	Intruder anonymous Mode S ID correlation encounters (#3 of 6): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: ADS-B with anonymous Mode S. Because the 2 tracks have different anonymous Mode S addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3242003	324		X				X			1000 5001	1 3	X	X	Intruder anonymous Mode S ID correlation encounters (#4 of 6): T1: AGT with AGT-ID and anonymous Mode S. T2: V2V with V2V-UID and anonymous Mode S. Because the 2 tracks have different anonymous Mode S addresses (333 and 777), the tracks do not correlate throughout the duration of the encounter.
3242004	324		X								3 4		X	Intruder anonymous Mode S ID correlation encounters (#5 of 6): T1: AGT with AGT-ID and anonymous Mode S. T2: AGT with AGT-ID and anonymous Mode S. Because the 2 tracks have different anonymous Mode S address (333 and 777), the tracks do not correlate throughout the duration of the encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3242005	324		X								3 4		X	Intruder anonymous Mode S ID correlation encounters (#6 of 6): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID and anonymous Mode S. Because the 2 tracks have different anonymous Mode S address (333 and 777), the tracks do not correlate throughout the duration of the encounter.
3243000	324		X				X						X	Intruder V2V-UID ID correlation encounters (#1 of 5): T1: AGT with AGT-ID and V2V-UID. T2: V2V with V2V-UID. Because the 2 tracks have different V2V-UID addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3243001	324		X				X			1000 5001	1 3	X	X	Intruder V2V-UID ID correlation encounters (#2 of 5): T1: AGT with AGT-ID, V2V-UID, and ICAO Mode S. T2: V2V with V2V-UID and anonymous Mode S. Because the 2 tracks have different V2V-UID addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3243002	324		X				X			1000 5001	1 3	X	X	Intruder V2V-UID ID correlation encounters (#3 of 5): T1: AGT with AGT-ID, V2V-UID, and anonymous Mode S. T2: V2V with V2V-UID and ICAO Mode S. Because the 2 tracks have differenet V2V-UID addresses (222 and 777), the tracks do not correlate throughout the duration of the encounter.
3243003	324		X										X	Intruder V2V-UID ID correlation encounters (#4 of 5): T1: AGT with AGT-ID and V2V-UID. T2: AGT with AGT-ID and V2V-UID. Because the 2 tracks have different V2V-UID addresses (333 and 77), the tracks do not correlate throughout the duration of the encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3243004	324		X								3 4		X	Intruder V2V-UID ID correlation encounters (#5 of 5): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID, V2V-UID, ICAO Mode S. Because the 2 tracks have different V2V-UID addresses (333 and 777), the tracks do not correlate throughout the duration of the encounter.
3244000	324		X				X			1000 5001	1 3	X	X	Intruder V2V-UID and ICAO Mode S ID correlation encounters (#1 of 2): T1: AGT with AGT-ID, V2V-UID, and ICAO Mode S. T2: V2V with V2V-UID and ICAO Mode S. Because the 2 tracks have different V2V-UID and ICAO Mode S addresses, the tracks do not correlate via ID throughout the duration of the encounter.
3244001	324		X								3 4		X	Intruder V2V-UID and ICAO Mode S ID correlation encounters (#2 of 2): T1: AGT with AGT-ID, V2V-UID, ICAO Mode S. T2: AGT with AGT-ID, V2V-UID, ICAO Mode S. Because the 2 tracks have different V2V-UID and ICAO Mode S addresses, the tracks do not correlate throughout the duration of the encounter.
3245000	324		X				X			1000 5001	1 3	X	X	Intruder V2V-UID and anonymous Mode S ID correlation encounters (#1 of 2): T1: AGT with AGT-ID, V2V-UID, and anonymous Mode S. T2: V2V with V2V-UID and anonymous Mode S. Because the 2 tracks have different V2V-UID and anonymous Mode S addresses, the tracks do not correlate throughout the duration of the encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
3245001	324		X								3 4		X	Intruder V2V-UID and ICAO Mode S ID correlation encounters (#2 of 2): T1: AGT with AGT-ID, V2V-UID, anonymous Mode S. T2: AGT with AGT-ID, V2V-UID, anonymous Mode S. Because the 2 tracks have different V2V-UID and anonymous Mode S addresses, the tracks do not correlate throughout the duration of the encounter.

Table 8. Test Group 40

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4110010	41	X	X		X		X			4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4110011	41		X		X		X			4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4110012	41				X		X			4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4110020	41	X	X		X		X			4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4110021	41	X	X		X					4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4110030	41		X		X		X			4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4110031	41				X		X			4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.
4110032	41		X		X					4010 6020	3 4			Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4110033	41				X					4010 6020	3 4		X	Nominal inputs table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: unavailable.
4110040	41	X	X		X		X			1000 4010 6020	3 4			Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4110041	41	X	X				X			1000 4010 6020	3 4			Nominal inputs table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4110050	41		X		X		X			1000 4010 6020	3 4			Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4110051	41				X		X			1000 4010 6020	3 4			Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4110052	41		X				X			1000 4010 6020	3 4			Nominal inputs table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4110053	41						X			1000 4010 6020	3 4		X	Nominal inputs table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4110060	41	X	X		X		X			1000 4010 6020	2 3 4			Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4110061	41	X	X		X					1000 4010 6020	2 3 4			Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4110062	41	X	X				X			1000 4010 6020	2 3 4			Nominal inputs table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4110063	41	X	X							1000 4010 6020	2 3 4			Nominal inputs table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4110070	41		X		X		X							Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4110071	41				X		X							Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.
4110080	41		X		X									Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4110090	41				X								X	Nominal inputs table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: unavailable.
4110101	41		X				X							Nominal inputs table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4110110	41						X						X	Nominal inputs table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.
4110120	41		X										X	Nominal inputs table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4110130	41												X	Nominal inputs table - ADSB: unavailable; V2V: unavailable; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4120010	41	X	X		X		X			6020 1000 5001 4010 4210 6002	2 3 4	X		NAR ADS-B table - ADSB: available, validated, NAR; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4120011	41		X		X		X			1000 4010 6020	3 4			NAR ADS-B table - ADSB: available, validated, NAR; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4120012	41				X		X			1000 4010 6020	3 4			NAR ADS-B table - ADSB: available, validated, NAR; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4120020	41	X	X		X		X			1000 4010	2 3 4	X		NAR ADS-B table - ADSB: available, validated, NAR; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4120021	41	X	X		X					1000 4010	2 3 4	X		NAR ADS-B table - ADSB: available, validated, NAR; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4120030	41		X		X		X				3 4			NAR ADS-B table - ADSB: available, validated, NAR; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4120031	41				X		X				3 4			NAR ADS-B table - ADSB: available, validated, NAR; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4120032	41		X		X						3 4			NAR ADS-B table - ADSB: available, validated, NAR; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4120033	41				X						3 4		X	NAR ADS-B table - ADSB: available, validated, NAR; V2V: unavailable; Pre-tracked: unavailable.
4120040	41	X	X		X		X			6020 1000 5001 4010 4210 6002	2 3 4	X		NAR ADS-B table - ADSB: available, not validated, NAR; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4120041	41	X	X				X			1000 4010 6020	3 4			NAR ADS-B table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4120050	41		X		X		X			1000 4010 6020	3 4			NAR ADS-B table - ADSB: available, not validated, NAR; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4120051	41				X		X			1000 4010 6020	3 4			NAR ADS-B table - ADSB: available, not validated, NAR; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4120052	41		X				X			1000 4010 6020	3 4			NAR ADS-B table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4120053	41						X			1000 4010 6020	3 4		X	NAR ADS-B table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4120060	41	X	X		X		X			1000 4010	2 3 4	X		NAR ADS-B table - ADSB: available, not validated, NAR; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4120061	41	X	X		X					1000 4010	2 3 4	X		NAR ADS-B table - ADSB: available, not validated, NAR; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4120062	41	X	X				X			1000 4010 6020	2 3 4			NAR ADS-B table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, altitude reporting.
4120063	41	X	X							1000 4010 6020	2 3 4			NAR ADS-B table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4120070	41		X		X		X							NAR ADS-B table - ADSB: available, not validated, NAR; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.
4120071	41				X		X							NAR ADS-B table - ADSB: available, not validated, NAR; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.
4120080	41		X		X									NAR ADS-B table - ADSB: available, not validated, NAR; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4120090	41				X								X	NAR ADS-B table - ADSB: available, not validated, NAR; V2V: unavailable; Pre-tracked: unavailable.
4120100	41		X				X							NAR ADS-B table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, altitude reporting.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4120110	41						X						X	NAR ADS-B table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.
4120120	41		X										X	NAR ADS-B table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4120130	41												X	NAR ADS-B table - ADSB: unavailable; V2V: unavailable; Pre-tracked: unavailable.
4130010	41	X	X		X		X			4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: available, validated, NAR; Pre-tracked: available, validated, altitude reporting.
4130011	41		X		X		X			4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: available, validated, NAR; Pre-tracked: available, not validated, altitude reporting.
4130012	41				X		X			4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: available, validated, NAR; Pre-tracked: unavailable.
4130020	41	X	X		X		X			4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: available, not validated, NAR; Pre-tracked: available, validated, altitude reporting.
4130021	41	X	X		X					4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4130030	41		X		X		X			4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: available, not validated, NAR; Pre-tracked: available, not validated, altitude reporting.
4130031	41				X		X			4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: available, not validated, NAR; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4130032	41		X		X					4010 6020	3 4			NAR V2V table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4130033	41				X					4010 6020	3 4		X	NAR V2V table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: unavailable.
4130040	41	X	X		X		X			1000 4010 6020	2 3 4			NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: available, validated, NAR; Pre-tracked: available, validated, altitude reporting.
4130041	41	X	X				X			1000 4010	2 3 4	X		NAR V2V table - ADSB: unavailable; V2V: available, validated, NAR; Pre-tracked: available, validated, altitude reporting.
4130050	41		X		X		X				3 4			NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: available, validated, NAR; Pre-tracked: available, not validated, altitude reporting.
4130051	41				X		X				3 4			NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: available, validated, NAR; Pre-tracked: unavailable.
4130052	41		X				X				3 4			NAR V2V table - ADSB: unavailable; V2V: available, validated, NAR; Pre-tracked: available, not validated, altitude reporting.
4130053	41						X				3 4		X	NAR V2V table - ADSB: unavailable; V2V: available, validated, NAR; Pre-tracked: unavailable.
4130060	41	X	X		X		X			1000 4010 6020	2 3 4			NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, NAR; Pre-tracked: available, validated, altitude reporting.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4130061	41	X	X		X					1000 4010 6020	2 3 4			NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4130062	41	X	X				X			1000 4010	2 3 4	X		NAR V2V table - ADSB: unavailable; V2V: available, not validated, NAR; Pre-tracked: available, validated, altitude reporting.
4130063	41	X	X							1000 4010 6020	2 3 4			NAR V2V table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, validated, altitude reporting.
4130070	41		X		X		X							NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, NAR; Pre-tracked: available, not validated, altitude reporting.
4130071	41				X		X							NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, NAR; Pre-tracked: unavailable.
4130080	41		X		X									NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4130090	41				X								X	NAR V2V table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: unavailable.
4130100	41		X				X							NAR V2V table - ADSB: unavailable; V2V: available, not validated, NAR; Pre-tracked: available, not validated, altitude reporting.
4130110	41						X						X	NAR V2V table - ADSB: unavailable; V2V: available, not validated, NAR; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4130120	41		X										X	NAR V2V table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, not validated, altitude reporting.
4130130	41												X	NAR V2V table - ADSB: unavailable; V2V: unavailable; Pre-tracked: unavailable.
4140010	41				X		X			4010 6020	2 3 4	X		Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, NAR.
4140011	41				X		X			4010 6020	3 4			Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, NAR.
4140012	41				X		X			4010 6020	3 4			Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4140020	41				X		X			4010 6020	2 3 4	X		Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, NAR.
4140021	41				X					4010 6020	2 3 4	X		Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: available, validated, NAR.
4140030	41				X		X			4010 6020	3 4			Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, NAR.
4140031	41				X		X			4010 6020	3 4			Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4140032	41				X					4010 6020	3 4			Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: available, not validated, NAR.
4140033	41				X					4010 6020	3 4		X	Nar pre-tracked table - ADSB: available, validated, altitude reporting; V2V: unavailable; Pre-tracked: unavailable.
4140040	41				X		X			1000 4010 6020	2 3 4	X		Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, NAR.
4140041	41						X			1000 4010 6020	2 3 4	X		Nar pre-tracked table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: available, validated, NAR.
4140050	41				X		X			1000 4010 6020	3 4			Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, NAR.
4140051	41				X		X			1000 4010 6020	3 4			Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4140052	41						X			1000 4010 6020	3 4			Nar pre-tracked table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: available, not validated, NAR.
4140053	41						X			1000 4010 6020	3 4		X	Nar pre-tracked table - ADSB: unavailable; V2V: available, validated, altitude reporting; Pre-tracked: unavailable.
4140060	41				X		X				2 3 4			Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, NAR.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4140061	41				X						2 3 4			Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: available, validated, NAR.
4140062	41						X				2 3 4			Nar pre-tracked table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: available, validated, NAR.
4140063	41										2 3 4		X	Nar pre-tracked table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, validated, NAR.
4140070	41				X		X							Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, NAR.
4140071	41				X		X							Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.
4140080	41				X									Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: available, not validated, NAR.
4140090	41				X								X	Nar pre-tracked table - ADSB: available, not validated, altitude reporting; V2V: unavailable; Pre-tracked: unavailable.
4140100	41						X							Nar pre-tracked table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: available, not validated, NAR.
4140110	41						X						X	Nar pre-tracked table - ADSB: unavailable; V2V: available, not validated, altitude reporting; Pre-tracked: unavailable.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4140120	41												X	Nar pre-tracked table - ADSB: unavailable; V2V: unavailable; Pre-tracked: available, not validated, NAR.
4140130	41												X	Nar pre-tracked table - ADSB: unavailable; V2V: unavailable; Pre-tracked: unavailable.
4210000	42				X					5001			X	Altitude Selection Encounters with ADS-B 1090ES (#1 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4210001	42				X					5001			X	Altitude Selection Encounters with ADS-B 1090ES (#2 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4210002	42				X					5001			X	Altitude Selection Encounters with ADS-B 1090ES (#3 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has no Pressure Altitude and Valid HAE, HAE saved.
4210003	42				X					5001			X	Altitude Selection Encounters with ADS-B 1090ES (#4 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has no Pressure Altitude and Valid HAE, HAE saved.
4210004	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#5 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Invalid Pressure Altitude and No HAE.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4210005	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#6 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Invalid Pressure Altitude and No HAE.
4210006	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#7 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has no Pressure Altitude and Invalid HAE.
4210007	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#8 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has no Pressure Altitude and Invalid HAE.
4210008	42				X					5001			X	Altitude Selection Encounters with ADS-B 1090ES (#9 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4210009	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#10 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Pressure Altitude but Valid HAE, HAE saved.
4210010	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#11 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Invalid Pressure Altitude and No HAE.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4210011	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#12 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has no Pressure Altitude and Invalid HAE.
4210012	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#13 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4210013	42				X					5001			X	Altitude Selection Encounters with ADS-B 1090ES (#14 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has No Pressure Altitude but Valid HAE, HAE saved.
4210014	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#15 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has Invalid Pressure Altitude and No HAE.
4210015	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#16 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has no Pressure Altitude and Invalid HAE.
4210016	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#17 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has Valid Pressure Altitude and No HAE, pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4210017	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#18 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has No Pressure Altitude but Valid HAE, HAE saved.
4210018	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#19 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has Invalid Pressure Altitude and No HAE.
4210019	42				X								X	Altitude Selection Encounters with ADS-B 1090ES (#20 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has no Pressure Altitude and Invalid HAE.
4220000	42					X				5001			X	Altitude Selection Encounters with ADS-B UAT (#1 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4220001	42					X				5001			X	Altitude Selection Encounters with ADS-B UAT (#2 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4220002	42					X				5001			X	Altitude Selection Encounters with ADS-B UAT (#3 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has no Pressure Altitude and Valid HAE, HAE saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4220003	42					X				5001			X	Altitude Selection Encounters with ADS-B UAT (#4 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has no Pressure Altitude and Valid HAE, HAE saved.
4220004	42					X							X	Altitude Selection Encounters with ADS-B UAT (#5 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Invalid Pressure Altitude and No HAE.
4220005	42					X							X	Altitude Selection Encounters with ADS-B UAT (#6 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Invalid Pressure Altitude and No HAE.
4220006	42					X							X	Altitude Selection Encounters with ADS-B UAT (#7 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has no Pressure Altitude and Invalid HAE.
4220007	42					X							X	Altitude Selection Encounters with ADS-B UAT (#8 of 20): Ownship has Valid Pressure Altitude and Valid HAE. vfom_m set to 4.5 and Ownship HAE Saved. Intruder has no Pressure Altitude and Invalid HAE.
4220008	42					X				5001			X	Altitude Selection Encounters with ADS-B UAT (#9 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4220009	42					X							X	Altitude Selection Encounters with ADS-B UAT (#10 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Pressure Altitude but Valid HAE, HAE saved.
4220010	42					X							X	Altitude Selection Encounters with ADS-B UAT (#11 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Invalid Pressure Altitude and No HAE.
4220011	42					X							X	Altitude Selection Encounters with ADS-B UAT (#12 of 20): Ownship has Valid Pressure Altitude but No HAE. vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has no Pressure Altitude and Invalid HAE.
4220012	42					X							X	Altitude Selection Encounters with ADS-B UAT (#13 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4220013	42					X				5001			X	Altitude Selection Encounters with ADS-B UAT (#14 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has No Pressure Altitude but Valid HAE, HAE saved.
4220014	42					X							X	Altitude Selection Encounters with ADS-B UAT (#15 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has Invalid Pressure Altitude and No HAE.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4220015	42					X							X	Altitude Selection Encounters with ADS-B UAT (#16 of 20): Ownship has No Valid Pressure Altitude and Valid HAE. vfom_m set to 8 and Ownship HAE Saved. Intruder has no Pressure Altitude and Invalid HAE.
4220016	42					X							X	Altitude Selection Encounters with ADS-B UAT (#17 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has Valid Pressure Altitude and No HAE, pressure saved.
4220017	42					X							X	Altitude Selection Encounters with ADS-B UAT (#18 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has No Pressure Altitude but Valid HAE, HAE saved.
4220018	42					X							X	Altitude Selection Encounters with ADS-B UAT (#19 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has Invalid Pressure Altitude and No HAE.
4220019	42					X							X	Altitude Selection Encounters with ADS-B UAT (#20 of 20): Ownship has No Valid Pressure Altitude but No Valid HAE. vfom_m set to 8 but no altitude type able to be saved in ownship. Intruder has no Pressure Altitude and Invalid HAE.
4230000	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#1 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 8 and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4230001	42						X			5001			X	Altitude Selection Encounters with V2V (#2 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 8 and pressure saved.
4230002	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#3 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 4.9 and HAE saved.
4230003	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#4 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 4.9 and HAE saved.
4230004	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#5 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and pressure saved.
4230005	42						X			5001			X	Altitude Selection Encounters with V2V (#6 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4230006	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#7 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude but Valid HAE, vfom_m set to 4.9 and HAE saved.
4230007	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#8 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude but Valid HAE, vfom_m set to 4.9 and HAE saved.
4230008	42						X						X	Altitude Selection Encounters with V2V (#9 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Altitude and No Valid HAE, vfom_m set to 8 and no altitude saved.
4230009	42						X						X	Altitude Selection Encounters with V2V (#10 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and no altitude saved.
4230010	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#11 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 8 and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4230011	42						X						X	Altitude Selection Encounters with V2V (#12 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 4.5 and HAE saved.
4230012	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#13 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No Valid HAE,vfom_m set to 8 and pressure saved.
4230013	42						X						X	Altitude Selection Encounters with V2V (#14 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude but Valid HAE,vfom_m set to 4.5 and HAE saved.
4230014	42						X						X	Altitude Selection Encounters with V2V (#15 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and no altitude saved.
4230015	42						X						X	Altitude Selection Encounters with V2V (#16 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 8 and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4230016	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#17 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 4.5 and HAE saved.
4230017	42						X						X	Altitude Selection Encounters with V2V (#18 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and pressure saved.
4230018	42						X			5001 6002			X	Altitude Selection Encounters with V2V (#19 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude but Valid HAE, vfom_m set to 4.5 and HAE saved.
4230019	42						X						X	Altitude Selection Encounters with V2V (#20 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and no altitude saved.
4230020	42						X						X	Altitude Selection Encounters with V2V (#21 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 8 and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4230021	42						X						X	Altitude Selection Encounters with V2V (#22 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, vfom_m set to 4.5 and HAE saved.
4230022	42						X						X	Altitude Selection Encounters with V2V (#23 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and pressure saved.
4230023	42						X						X	Altitude Selection Encounters with V2V (#24 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude but Valid HAE, vfom_m set to 4.5 and HAE saved.
4230024	42						X						X	Altitude Selection Encounters with V2V (#25 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, vfom_m set to 8 and no altitude saved.
4240000	42		X	X									X	Altitude Selection Encounters with AGT (#1 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance lower than HAE covariance and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4240001	42		X	X									X	Altitude Selection Encounters with AGT (#2 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance lower than HAE covariance and pressure saved.
4240002	42		X	X									X	Altitude Selection Encounters with AGT (#3 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240003	42		X	X									X	Altitude Selection Encounters with AGT (#4 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240004	42			X									X	Altitude Selection Encounters with AGT (#5 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance and pressure saved.
4240005	42			X									X	Altitude Selection Encounters with AGT (#6 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4240006	42		X										X	Altitude Selection Encounters with AGT (#7 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude but Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240007	42		X										X	Altitude Selection Encounters with AGT (#8 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude but Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240008	42												X	Altitude Selection Encounters with AGT (#9 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Altitude and No Valid HAE, pressure covariance lower than HAE covariance but no altitude saved.
4240009	42												X	Altitude Selection Encounters with AGT (#10 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance but no altitude saved.
4240010	42		X	X									X	Altitude Selection Encounters with AGT (#11 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance lower than HAE covariance and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4240011	42		X	X									X	Altitude Selection Encounters with AGT (#12 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240012	42			X									X	Altitude Selection Encounters with AGT (#13 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No Valid HAE,pressure covariance lower than HAE covariance and pressure saved.
4240013	42		X										X	Altitude Selection Encounters with AGT (#14 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude but Valid HAE,pressure covariance higher than HAE covariance and HAE saved
4240014	42												X	Altitude Selection Encounters with AGT (#15 of 25): Ownship has Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance but no altitude saved.
4240015	42		X	X									X	Altitude Selection Encounters with AGT (#16 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance lower than HAE covariance and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4240016	42		X	X									X	Altitude Selection Encounters with AGT (#17 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240017	42			X									X	Altitude Selection Encounters with AGT (#18 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance and pressure saved.
4240018	42		X										X	Altitude Selection Encounters with AGT (#19 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude but Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240019	42												X	Altitude Selection Encounters with AGT (#20 of 25): Ownship has Valid Pressure Altitude and Valid HAE. ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance but no altitude saved.
4240020	42		X	X									X	Altitude Selection Encounters with AGT (#21 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance lower than HAE covariance and pressure saved.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4240021	42		X	X									X	Altitude Selection Encounters with AGT (#22 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240022	42			X									X	Altitude Selection Encounters with AGT (#23 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance and pressure saved.
4240023	42		X										X	Altitude Selection Encounters with AGT (#24 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude but Valid HAE, pressure covariance higher than HAE covariance and HAE saved
4240024	42												X	Altitude Selection Encounters with AGT (#25 of 25): Ownship has No Valid Pressure Altitude and No Valid HAE. ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Pressure Altitude and No Valid HAE, pressure covariance lower than HAE covariance but no altitude saved.
4250000	42	X								5001 6002	2 4		X	Altitude Selection Encounters with ORNCT (#1 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Relative Altitude.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4250001	42	X								5001 6002	2 4		X	Altitude Selection Encounters with ORNCT (#2 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Relative Altitude.
4250002	42	X											X	Altitude Selection Encounters with ORNCT (#3 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No valid Relative Altitude.
4250003	42	X											X	Altitude Selection Encounters with ORNCT (#4 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No valid Relative Altitude.
4250004	42	X								5001 6002	2 4		X	Altitude Selection Encounters with ORNCT (#5 of 10): Ownship has Valid Pressure Altitude but No Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has Valid Relative Altitude.
4250005	42	X											X	Altitude Selection Encounters with ORNCT (#6 of 10): Ownship has Valid Pressure Altitude but No Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Intruder has No Valid Relative Altitude.
4250006	42	X								5001 6002	2 4		X	Altitude Selection Encounters with ORNCT (#7 of 10): Ownship has No Valid Pressure Altitude but Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has Valid Relative Altitude.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4250007	42	X											X	Altitude Selection Encounters with ORNCT (#8 of 10): Ownship has No Valid Pressure Altitude but Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Intruder has No Valid Relative Altitude.
4250008	42	X											X	Altitude Selection Encounters with ORNCT (#9 of 10): Ownship has No Valid Pressure Altitude and No HAE. Ownship vfom_m set to 8 and no altitude Saved. Intruder has Valid Relative Altitude.
4250009	42	X											X	Altitude Selection Encounters with ORNCT (#10 of 10): Ownship has No Valid Pressure Altitude and No HAE. Ownship vfom_m set to 8 and no altitude Saved. Intruder has No Valid Relative Altitude.
4260000	42							X			2		X	Altitude Selection Encounters with POA (#1 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Obstacle has Valid HAE.
4260001	42							X			2		X	Altitude Selection Encounters with POA (#2 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Obstacle has Valid HAE.
4260002	42							X					X	Altitude Selection Encounters with POA (#3 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Obstacle has Invalid HAE.
4260003	42							X					X	Altitude Selection Encounters with POA (#4 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Obstacle has Invalid HAE.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4260004	42							X					X	Altitude Selection Encounters with POA (#5 of 10): Ownship has Valid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Obstacle has Valid HAE.
4260005	42							X					X	Altitude Selection Encounters with POA (#6 of 10): Ownship has Valid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. Obstacle has Invalid HAE.
4260006	42							X			2		X	Altitude Selection Encounters with POA (#7 of 10): Ownship has Invalid Pressure and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Obstacle has Valid HAE.
4260007	42							X					X	Altitude Selection Encounters with POA (#8 of 10): Ownship has Invalid Pressure and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. Obstacle has Invalid HAE.
4260008	42							X					X	Altitude Selection Encounters with POA (#9 of 10): Ownship has Invalid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and no altitude saved. Obstacle has Valid HAE.
4260009	42							X					X	Altitude Selection Encounters with POA (#10 of 10): Ownship has Invalid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and no altitude saved. Obstacle has Invalid HAE.
4270000	42												X	Altitude Selection Encounters with GPOA (#1 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. AGL Valid.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4270001	42												X	Altitude Selection Encounters with GPOA (#2 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. AGL Valid.
4270002	42												X	Altitude Selection Encounters with GPOA (#3 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. AGL Not Valid.
4270003	42												X	Altitude Selection Encounters with GPOA (#4 of 10): Ownship has Valid Pressure Altitude and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. AGL Not Valid.
4270004	42												X	Altitude Selection Encounters with GPOA (#5 of 10): Ownship has Valid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. AGL Valid.
4270005	42												X	Altitude Selection Encounters with GPOA (#6 of 10): Ownship has Valid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and Ownship Pressure Altitude Saved. AGL Not Valid.
4270006	42												X	Altitude Selection Encounters with GPOA (#7 of 10): Ownship has Invalid Pressure and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. AGL Valid.
4270007	42												X	Altitude Selection Encounters with GPOA (#8 of 10): Ownship has Invalid Pressure and Valid HAE. Ownship vfom_m set to 4.5 and Ownship HAE Saved. AGL Not Valid.
4270008	42												X	Altitude Selection Encounters with GPOA (#9 of 10): Ownship has Invalid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and no altitude saved. AGL Valid.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4270009	42												X	Altitude Selection Encounters with GPOA (#10 of 10): Ownship has Invalid Pressure Altitude and Invalid HAE. Ownship vfom_m set to 8 and no altitude saved. AGL Not Valid.
4280101	42				X					1000 4010 6020	1 2 4		X	Altitude prioritization: ADS-B Encounters (#1 of 6). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280102	42				X					1000 5001 6002	1 2 4		X	Altitude prioritization: ADS-B Encounters (#2 of 6). Intruder: Small, Is_Alt_Geo_Hae: True, Alt_Value_Valid: Valid, GVA: 2
4280103	42				X						1 2 4		X	Altitude prioritization: ADS-B Encounters (#3 of 6). Intruder: Small, Is_Alt_Geo_Hae: True, Alt_Value_Valid: Valid, GVA: NaN
4280104	42				X						1 2 4		X	Altitude prioritization: ADS-B Encounters (#4 of 6). Intruder: Small, Is_Alt_Geo_Hae: True, Alt_Value_Valid: Valid, GVA: 1
4280105	42				X						1 2 4		X	Altitude prioritization: ADS-B Encounters (#5 of 6). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: NaN, GVA: 2
4280106	42				X						1 2 4		X	Altitude prioritization: ADS-B Encounters (#6 of 6). Intruder: Small, Is_Alt_Geo_Hae: True, Alt_Value_Valid: NaN, GVA: 2
4280201	42						X			1000 4010 6020	1 2		X	Altitude prioritization: V2V Encounters (#1 of 28). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: False

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280202	42						X						X	Altitude prioritization: V2V Encounters (#2 of 28). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280203	42						X			1000 4010 6020	1 2		X	Altitude prioritization: V2V Encounters (#3 of 28). Intruder: Small, PA: Valid, HAE: NaN, Sigm_VEPU < Sigma_AEM: False
4280204	42						X			1000 4010 6020	1 2		X	Altitude prioritization: V2V Encounters (#4 of 28). Intruder: Small, PA: NaN, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280205	42						X			1000 4010 6020	1 2		X	Altitude prioritization: V2V Encounters (#5 of 28). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280206	42						X			1000 4010 6020	1 2		X	Altitude prioritization: V2V Encounters (#6 of 28). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280207	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#7 of 28). Intruder: Small, PA: NaN, HAE: NaN, Sigm_VEPU < Sigma_AEM: True
4280208	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#8 of 28). Intruder: Small, PA: NaN, HAE: Set outside valid range, Sigm_VEPU < Sigma_AEM: True
4280209	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#9 of 28). Intruder: Small, PA: NaN, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280210	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#10 of 28). Intruder: Small, PA: NaN, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280211	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#11 of 28). Intruder: Small, PA: Outside of Valid Range, HAE: NaN, Sigm_VEPU < Sigma_AEM: True
4280212	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#12 of 28). Intruder: Small, PA: Outside of Valid Range, HAE: Outside of Valid Range, Sigm_VEPU < Sigma_AEM: True
4280213	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#13 of 28). Intruder: Small, PA: Outside of Valid Range, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280214	42						X				1 2 3		X	Altitude prioritization: V2V Encounters (#14 of 28). Intruder: Small, PA: Outside of Valid Range, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280215	42						X			1000 4010	1 2 4		X	Altitude prioritization: V2V Encounters (#15 of 28). Intruder: Large, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: False
4280216	42						X			1000 5001	1 2 4		X	Altitude prioritization: V2V Encounters (#16 of 28). Intruder: Large, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280217	42						X			1000 4010	1 2 4		X	Altitude prioritization: V2V Encounters (#17 of 28). Intruder: Large, PA: Valid, HAE: NaN, Sigm_VEPU < Sigma_AEM: False

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280218	42						X			1000 4010	1 2 4		X	Altitude prioritization: V2V Encounters (#18 of 28). Intruder: Large, PA: NaN, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280219	42						X			1000 4010	1 2 4		X	Altitude prioritization: V2V Encounters (#19 of 28). Intruder: Large, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280220	42						X			1000 4010	1 2 4		X	Altitude prioritization: V2V Encounters (#20 of 28). Intruder: Large, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280221	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#21 of 28). Intruder: Large, PA: NaN, HAE: NaN, Sigm_VEPU < Sigma_AEM: True
4280222	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#22 of 28). Intruder: Large, PA: NaN, HAE: Set outside valid range, Sigm_VEPU < Sigma_AEM: True
4280223	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#23 of 28). Intruder: Large, PA: NaN, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280224	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#24 of 28). Intruder: Large, PA: NaN, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280225	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#25 of 28). Intruder: Large, PA: Outside of Valid Range, HAE: NaN, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280226	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#26 of 28). Intruder: Large, PA: Outside of Valid Range, HAE: Outside of Valid Range, Sigm_VEPU < Sigma_AEM: True
4280227	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#27 of 28). Intruder: Large, PA: Outside of Valid Range, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280228	42						X				1 2 4		X	Altitude prioritization: V2V Encounters (#28 of 28). Intruder: Large, PA: Outside of Valid Range, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280301	42		X	X						1000 4010	1 2 3		X	Altitude prioritization: AGT Encounters (#1 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: False
4280302	42		X	X						1000 5001 6002	1 2		X	Altitude prioritization: AGT Encounters (#2 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280303	42			X						1000 4010	1 2 3		X	Altitude prioritization: AGT Encounters (#3 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280304	42		X	X						1000 4010	1 2 3		X	Altitude prioritization: AGT Encounters (#4 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280305	42		X	X						1000 4010	1 2 3		X	Altitude prioritization: AGT Encounters (#5 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280306	42		X	X						1000 4010	1 2 3		X	Altitude prioritization: AGT Encounters (#6 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280307	42		X							1000 5001 6002	1 2		X	Altitude prioritization: AGT Encounters (#7 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280308	42		X	X						1000 5001 6002	1 2		X	Altitude prioritization: AGT Encounters (#8 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280309	42		X	X						1000 5001 6002	1 2		X	Altitude prioritization: AGT Encounters (#9 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280310	42		X	X						1000 5001 6002	1 2		X	Altitude prioritization: AGT Encounters (#10 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280311	42										1 2 3		X	Altitude prioritization: AGT Encounters (#11 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280312	42		X								1 2 3		X	Altitude prioritization: AGT Encounters (#12 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280313	42		X								1 2 3		X	Altitude prioritization: AGT Encounters (#13 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280314	42		X								1 2 3		X	Altitude prioritization: AGT Encounters (#14 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280315	42			X							1 2 3		X	Altitude prioritization: AGT Encounters (#15 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280316	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#16 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280317	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#17 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280318	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#18 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280319	42			X							1 2 3		X	Altitude prioritization: AGT Encounters (#19 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280320	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#20 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280321	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#21 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280322	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#22 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280323	42			X							1 2 3		X	Altitude prioritization: AGT Encounters (#23 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280324	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#24 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280325	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#25 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280326	42		X	X							1 2 3		X	Altitude prioritization: AGT Encounters (#26 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280327	42		X	X						1000 5001 4210 4010 5201	1 2 4		X	Altitude prioritization: AGT Encounters (#27 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: False
4280328	42		X	X						1000 5001	1 2 4		X	Altitude prioritization: AGT Encounters (#28 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280329	42			X						1000 5001 4210 4010 5201	1 2 4		X	Altitude prioritization: AGT Encounters (#29 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280330	42		X	X						1000 5001 4210 4010 5201	1 2 4		X	Altitude prioritization: AGT Encounters (#30 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280331	42		X	X						1000 5001 4210 4010 5201	1 2 4		X	Altitude prioritization: AGT Encounters (#31 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280332	42		X	X						1000 5001 4210 4010 5201	1 2 4		X	Altitude prioritization: AGT Encounters (#32 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280333	42		X							1000 5001	1 2 4		X	Altitude prioritization: AGT Encounters (#33 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280334	42		X	X						1000 5001	1 2 4		X	Altitude prioritization: AGT Encounters (#34 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280335	42		X	X						1000 5001	1 2 4		X	Altitude prioritization: AGT Encounters (#35 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280336	42		X	X						1000 5001	1 2 4		X	Altitude prioritization: AGT Encounters (#36 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280337	42										1 2 4		X	Altitude prioritization: AGT Encounters (#37 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280338	42		X								1 2 4		X	Altitude prioritization: AGT Encounters (#38 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280339	42		X								1 2 4		X	Altitude prioritization: AGT Encounters (#39 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280340	42		X								1 2 4		X	Altitude prioritization: AGT Encounters (#40 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280341	42			X							1 2 4		X	Altitude prioritization: AGT Encounters (#41 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280342	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#42 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280343	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#43 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280344	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#44 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280345	42			X							1 2 4		X	Altitude prioritization: AGT Encounters (#45 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280346	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#46 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280347	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#47 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280348	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#48 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280349	42			X							1 2 4		X	Altitude prioritization: AGT Encounters (#49 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280350	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#50 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280351	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#51 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280352	42		X	X							1 2 4		X	Altitude prioritization: AGT Encounters (#52 of 52). Intruder: Small, PA: Valid, HAE: Valid, Sigm_VEPU < Sigma_AEM: True
4280401	42				X					1000 4010 6020	1 2 4		X	Altitude prioritization: Ownship Encounters (#1 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280402	42				X					1000 4010 6020	1 2 4		X	Altitude prioritization: Ownship Encounters (#2 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280403	42				X					1000 4010 6020	1 2 4		X	Altitude prioritization: Ownship Encounters (#3 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280404	42				X					1000 4010 6020	1 2 4		X	Altitude prioritization: Ownship Encounters (#4 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280405	42				X					1000 4010 6020	1 2 4		X	Altitude prioritization: Ownship Encounters (#5 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280406	42				X						1 2 4		X	Altitude prioritization: Ownship Encounters (#6 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4280407	42				X								X	Altitude prioritization: Ownship Encounters (#7 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280408	42				X								X	Altitude prioritization: Ownship Encounters (#8 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280409	42				X						1 2 4		X	Altitude prioritization: Ownship Encounters (#9 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280410	42				X								X	Altitude prioritization: Ownship Encounters (#10 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280411	42				X								X	Altitude prioritization: Ownship Encounters (#11 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280412	42				X								X	Altitude prioritization: Ownship Encounters (#12 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4280413	42				X								X	Altitude prioritization: Ownship Encounters (#13 of 13). Intruder: Small, Is_Alt_Geo_Hae: False, Alt_Value_Valid: Valid, GVA: 2
4310000	431				X					5001	2		X	ADS-B validation encounters (#1 of 4): Base encounter with validatd ADS-B throughout.
4310001	431				X					5001	1 2		X	ADS-B validation encounters (#2 of 4): ADS-B starts out validated and midway through the encounter becomes invalidated.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4310002	431				X					5001	2		X	ADS-B validation encounters (#3 of 4): ADS-B starts out invalidated and midway through the encounter becomes validated.
4310003	431				X					5001	1 2		X	ADS-B validation encounters (#4 of 4): ADS-B starts out validated, midway through the encounter becomes invalidated, and 10 seconds later is validated again.
4320000	432				X					5001	2		X	ADS-B quality encounters (#1 of 4): Base encounter that passes init_M of init_N test and M of N test throughout. A descend RA and a turn right RA are issued from t=4 to the end of the encounter.
4320001	432				X					5001	2		X	ADS-B quality encounters (#2 of 4): Encounter fails init_M of init_N test. Intruder's NACp is reported as 6 (less than minimum required value of 7) for the first 5 cycles. Then intruder's NACp is restored to 8 and the track passes the M of N test. Descend RA and turn right RA are not issued until t=9.
4320002	432				X					1000 5001	2		X	ADS-B quality encounters (#3 of 4): Encounter passes init_M of init_N test and subsequent M of N tests. At second 19.999, the intruder's adsb_version is reported as 1 (less than minimum required value of 2). This is maintained throughout the rest of the encounter. The descend RA and turn right RA disappear at t=21 and do not return.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4320003	432				X					1000 5001	2		X	ADS-B quality encounters (#4 of 4): Encounter passes init_M of init_N test and subsequent M of N tests. At second 19.999, the intruder's NIC is reported as 5 (less than minimum required value of 6). This is maintained for 10 seconds before being restored to 8. The descend RA and turn right RA disappear at t=21 and return at t=34 when the ADS-B is revalidated and passes the M_revalidate test.
4330000	433						X			4010	3		X	V2V validation encounters (#1 of 4): Base encounter with validatd V2V throughout.
4330001	433						X			4010	1 3		X	V2V validation encounters (#2 of 4): V2V starts out validated and midway through the encounter becomes invalidated.
4330002	433						X			4010	3		X	V2V validation encounters (#3 of 4): V2V starts out invalidated and midway through the encounter becomes validated.
4330003	433						X			4010	1 3		X	V2V validation encounters (#4 of 4): V2V starts out validated, midway through the encounter becomes invalidated, and 10 seconds later is validated again.
4340000	434						X			5001	2		X	V2V quality encounters (#1 of 4): Base encounter that passes init_M of init_N test and M of N test throughout. A descend RA and a turn right RA are issued from t=4 to the end of the encounter.
4340001	434						X			5001	2		X	V2V quality encounters (#2 of 4): Encounter fails init_M of init_N test. Intruder's NACp is reported as 6 (less than minimum required value of 7) for the first 5 cycles. Then intruder's NACp is restored to 10 and the track passes the M of N test. Descend RA and turn right RA are not issued until t=9.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4340002	434						X			1000 5001	2		X	V2V quality encounters (#3 of 4): Encounter passes init_M of init_N test and subsequent M of N tests. At second 19.997, the intruder's NACv is reported as 0 (less than minimum required value of 1). This is maintained throughout the rest of the encounter. The descend RA and turn right RA disappear at t=21 and do not return.
4340003	434						X			1000 5001 6002	2		X	V2V quality encounters (#4 of 4): Encounter passes init_M of init_N test and subsequent M of N tests. At second 19.997, the intruder's NACp is reported as 6 (less than minimum required value of 7). This is maintained for 10 seconds before being restored to 10. The descend / do not climb RA and turn right RA disappear at t=21 and return at t=33 when the V2V track again passes the M of N test.
4350000	435						X			4010 6020	2		X	OwnWGS84 quality encounters (#1 of 4): Base encounter where ownship maintains NACp and NACv above minimum quality requirements throughout. Climb and do not descend RAs and a turn right RA are issued from t=4 to the end of the encounter.
4350001	435						X			4010 6020	2		X	OwnWGS84 quality encounters (#2 of 4): Ownship's NACp is reported as 6 (less than minimum required value of 7) for the first 5 cycles. Then ownship's NACp is restored to 10. Climb RA and turn right RA are not issued until t=8.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4350002	435						X			1000 4010 6020	2		X	OwnWGS84 quality encounters (#3 of 4): Ownship maintains NACp and NACv above minimum quality requirements until t=19.997 when ownship's NACp value is reported as 0 (below the minimum required value of 1) and NACv is reported as 6 (below the minimum required value of 7). This is maintained throughout the rest of the encounter. The do not descend RA and turn right RA disappear at t=22 and do not return.
4350003	435						X			1000 4010 6020	2		X	OwnWGS84 quality encounters (#4 of 4): Ownship maintains NACp and NACv above minimum quality requirements until t=14.997 when ownship's NACv value is reported as 0 (below the minimum required value of 1). This is maintained for 10 seconds before being restored to 1. The do not descend RA and turn right RA disappear at t=17. The turn right RA returns at t=27 when ownship's data again meets quality requirements.
4410001	441						X	X					X	Downselect with 35 intruders, no coordination, and no RAs.
4410002	441						X	X					X	Downselect with 5 intruders, no coordination, and no RAs.
4410003	441						X						X	Downselect with 20 intruders, no coordination, and no RAs.
4410004	441						X	X	X				X	Downselect with 34 intruders, 1 coordination intruder, and no RAs.
4410005	441				X		X	X	X				X	Downselect with 31 intruders, 5 coordination intruder, and no RAs.
4410006	441						X	X	X				X	Downselect with 31 intruders, 19 coordination intruder, and no RAs.
4410007	441						X	X	X				X	Downselect with 31 intruders, 31 coordination intruder, and no RAs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
4410008	441						X	X	X	6002			X	Downselect with 31 intruders, 31 coordination intruder, and 1 RA Intruder.
4420001	442						X			4010	2	X	X	Downselect for HTRM with 6 intruders.
4420002	442						X				2		X	Downselect for HTRM with 6 intruders. Intruder with address of 7, caused the original RA, will be NAR.

Table 9. Test Group 50

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5100000	51						X			5001 6002	3		X	Manual mode switching (#1 of 3): Base encounter. Single V2V intruder, ownship is in RA mode throughout. Vertical RAs begin at t=7 and continue through the end of the encounter (5001 to 6002).
5100001	51						X			5001	3		X	Manual mode switching (#2 of 3): Mode-switching encounter. Single V2V intruder, ownship is in RA mode from t=0 to t=19, SURVEILLANCE_ONLY mode from t=20 to t=34, and STANDBY mode from t=35 to t=50. Vertical RA (5001) begins at t=7 and drops at t=20 when ownship mode switches to SURVEILLANCE_ONLY.
5100002	51						X						X	Manual mode switching (#3 of 3): Mode-switching encounter. Single V2V intruder, ownship is in STANDBY mode from t=0 to t=19, SURVEILLANCE_ONLY mode from t=20 to t=34, and RA mode from t=35 to t=50. No RAs are issued.
5100100	51						X						X	Removal of automatic mode switching (#1 of 11): Old automatic mode transition encounter. Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL (10 ft) and its 'belief_vert' is set to its pressure altitude (0.5 ft).

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5100101	51						X						X	Removal of automatic mode switching (#2 of 11): Old automatic mode transition encounter (encounter no longer enters STANDBY mode). Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL (0.5 ft) and its 'belief_vert' is set to its pressure altitude (0.5 ft).
5100102	51						X						X	Removal of automatic mode switching (#3 of 11): Old automatic mode transition encounter. Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL (65 ft) and its 'belief_vert' is set to its pressure altitude (15 ft).
5100103	51						X						X	Removal of automatic mode switching (#4 of 11): Old automatic mode transition encounter (encounter no longer enters SURVILLENACE_ONLY mode). Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL (15 ft) and its 'belief_vert' is set to its pressure altitude (15 ft).
5100104	51						X			5001	3		X	Removal of automatic mode switching (#5 of 11): Old automatic mode transition encounter. Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL (65 ft) and its 'belief_vert' is set to its pressure altitude (200 ft). Ownship has 5001 descend RA.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5100105	51						X			5001	3		X	Removal of automatic mode switching (#6 of 11): Old automatic mode transition encounter (encounter no longer has automatic mode transitions). Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL and descends from 65 ft to 0.5 ft. Ownship's pressure altitude stays steady at 200 ft and is set in the 'belief_vert' field in the STM report. Because mode transitions do not occur, the 5001 descend RA is maintained throughout the encounter.
5100106	51						X			5001	3		X	Removal of automatic mode switching (#7 of 11): Old automatic mode transition encounter (encounter no longer has automatic mode transitions). Because the requested_opmode field in OWNSHIP_DISCRETES is set to 3 (OPMODE_RA), ownship remains in RA mode the entire time. In the STM report, ownship's 'h' is set to its AGL and ascends from 0.5 ft to 65 ft. Ownship's pressure altitude stays steady at 200 ft and is set in the 'belief_vert' field in the STM report. Because mode transitions do not occur, the 5001 descend RA is maintained throughout the encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5100107	51				X					5001 4210 4410	3		X	Removal of automatic mode switching (#8 of 11): Old automatic mode transition encounter (encounter no longer has automatic mode transitions). Ownship is ascending with no point obstacles. Ownship starts at 0.5 ft, ascends to 6 ft, 30 ft, and finally to 65 ft. Ownship remains in OPMODE_RA the whole time due to requested_opmode field in OWNSHIP_DISCRETES. 5001 descend RA changes into a climb RA at t=19.
5100108	51				X			X		5001 6002 4410 4210	3	X	X	Removal of automatic mode switching (#9 of 11): Old automatic mode transition encounter (encounter no longer has automatic mode transitions). Ownship is ascending with point obstacles. Ownship starts at 0.5 ft, ascends to 6 ft, 30 ft, and finally to 65 ft. Ownship remains in OPMODE_RA the whole time due to requested_opmode field in OWNSHIP_DISCRETES. 5001 descend RA changes into a maintain RA at t=13. At t=15, the vertical RA changes back to 5001 descend, and at t=19, the vertical RA changes to climb.
5100109	51				X					5001 4210 6020 4010 5201	3	X	X	Removal of automatic mode switching (#10 of 11): Old automatic mode transition encounter (encounter no longer has automatic mode transitions). Ownship is descending with no point obstacles. Ownship starts at 61 ft, descends to 35 ft, 10 ft, and finally to 0.5 ft. Ownship remains in OPMODE_RA the whole time due to requested_opmode field in OWNSHIP_DISCRETES. The vertical RA oscillates between climb, descend, and maintain.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5100110	51				X			X		5201 4010 4020 4210 5001	3	X	X	Removal of automatic mode switching (#11 of 11): Old automatic mode transition encounter (encounter no longer has automatic mode transitions). Ownship is descending with point obstacles. Ownship starts at 61 ft, descends to 35 ft, 10 ft, and finally to 0.5 ft. Ownship remains in OPMODE_RA the whole time due to requested_opmode field in OWNSHIP_DISCRETES. The vertical RA oscillates between climb and descend.
5200000	52												X	Obstacle and terrain scenarios. No AGL feed, no POA feed, no intruder feed, POA and GPOA disabled. No protection provided. Ownship descends.
5200001	52												X	Obstacle and terrain scenarios. No AGL feed, no POA feed, no intruder feed, POA and GPOA enabled. No protection provided. Ownship descends.
5200002	52				X					5001	2		X	Obstacle and terrain scenarios. No AGL feed, no POA feed, intruder feed available, POA and GPOA disabled. Intruder protection provided (vertical alert = DESCEND). Intruder above and head-on to ownship.
5200003	52				X					5001	2		X	Obstacle and terrain scenarios. No AGL feed, no POA feed, intruder feed available, POA and GPOA enabled. Intruder protection provided (vertical alert = DESCEND). Intruder above and head-on to ownship.
5200004	52							X					X	Obstacle and terrain scenarios. No AGL feed, POA feed available, no intruder feed, POA and GPOA disabled. No protection provided.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5200005	52							X		1000 4010 6020	2 3 4	X	X	Obstacle and terrain scenarios. No AGL feed, POA feed available, no intruder feed, POA and GPOA enabled. Terrain/obstacle protection provided (vertical alert = CLIMB, DO NOT DESCEND).
5200006	52				X			X		5001	2		X	Obstacle and terrain scenarios. No AGL feed, POA feed available, intruder feed available, POA nad GPOA disabled. Intruder protection provided (vertical alert = DESCEND). Intruder above and head-on to ownship.
5200007	52				X			X		5201 6020 4110 5001	2	X	X	Obstacle and terrain scenarios. No AGL feed, POA feed available, intruder feed available, POA and GPOA enabled. Intruder and terrain/obstacle protection provided (vertical alert = CLIMB, DO NOT DESCEND, DESCEND).
5200008	52												X	Obstacle and terrain scenarios. AGL feed available, no POA feed, no intruder feed, POA and GPOA disabled. No protection provided. Ownship descends and travels over a cliff at end of encounter with mistmatched pressure and AGL altitudes.
5200009	52									4010 6020			X	Obstacle and terrain scenarios. AGL feed available, no POA feed, no intruder feed, POA and GPOA enabled. Ground protection provided (vertical alert = CLIMB, DO NOT DESCEND). Ownship descends and travels over a cliff at end of encounter with mistmatched pressure and AGL altitudes.
5200010	52				X					5001	2		X	Obstacle and terrain scenarios. AGL feed available, no POA feed, intruder feed available, POA and GPOA disabled. Intruder protection provided (vertical alert = DESCEND). Intruder above and head-on to ownship. Ownship flies over cliff with mismatched pressure and AGL altitudes starting at t=25.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5200011	52				X					5001	2	X	X	Obstacle and terrain scenarios. AGL feed available, no POA feed, intruder feed available, POA and GPOA enabled. Intruder protection provided (vertical alert = DESCEND). Intruder above and head-on to ownship. Ownship flies over cliff with mismatched pressure and AGL altitudes starting at t=25.
5200012	52							X					X	Obstacle and terrain scenarios. AGL feed available, POA feed available, no intruder feed, POA and GPOA disabled. No protection provided. Ownship first flies over point obstacles set at an altitude of 100 ft. Then ownship descends and flies over cliff.
5200013	52							X		1000 4010 6020	2 3 4	X	X	Obstacle and terrain scenarios. AGL feed available, POA feed available, no intruder feed, POA and GPOA enabled. Obstacle/terrain and ground protection provided (vertical alert = DO NOT DESCEND, CLIMB, DO NOT DESCEND). Ownship first flies over point obstacles set at an altitude of 100 ft. Then ownship descends and flies over cliff.
5200014	52				X			X		5001	2		X	Obstacle and terrain scenarios. AGL feed available, POA feed available, intruder feed available, POA and GPOA disabled. Intruder protection provided (vertical alert = DESCEND). Intruder is above and head-on towards ownship.
5200015	52				X			X		5001	2	X	X	Obstacle and terrain scenarios. AGL feed available, POA feed available, intruder feed available, POA and GPOA enabled. Intruder and obstacle/terrain protection provided (vertical alert = DESCEND). Intruder is above and head-on towards ownship.
5200100	52							X		1000 6020	2 3 4	X	X	Using USGS terrain data: Sparce terrain with intruder far away.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5200101	52							X		1000 6020	2 3 4	X	X	Using USGS terrain data: Dense terrain with intruder far away.
5200102	52		X		X			X		6020 1000 5001 4010 5201 6002	2	X		Using USGS terrain data: Dense terrain with large intruder.
5200103	52		X				X	X		5201 6002 6020 5001	3	X		Using USGS terrain data: Dense terrain with small intruder.
5200104	52		X		X		X	X		5201 6002 6020 5001	2	X		Using USGS terrain data: Dense terrain with large and small intruders.
5200201	52						X		X	1000 4010 6020	1 2 3 4	X	X	Ground proximity seems to cause 1s RAs in sXu 4.0 (#1 of 6). 1s RA present in this encounter.
5200202	52						X		X	1000 5001 6002	1 3 4		X	Ground proximity seems to cause 1s RAs in sXu 4.0 (#2 of 6). V2V coordination message added, 1s RA not present in this encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
5200203	52						X		X	1000 4010 6020	1 2 3 4		X	Ground proximity seems to cause 1s RAs in sXu 4.0 (#3 of 6). GPOA disabled for ownship, 1s RA not present in this encounter.
5200204	52						X		X	1000 4010 6020	1 2 3 4		X	Ground proximity seems to cause 1s RAs in sXu 4.0 (#4 of 6). AGL increased by 100 ft, 1s RA not present in this encounter.
5200205	52						X		X	1000 4010 6020	1 2 3 4		X	Ground proximity seems to cause 1s RAs in sXu 4.0 (#5 of 6). AGL increased by 100 ft with a couple of AGL messages reporting very low, 1s RA not present in this encounter.
5200206	52									1000 6020			X	Ground proximity seems to cause 1s RAs in sXu 4.0 (#6 of 6). Intruder removed, 1s RA present in this encounter.

Table 10. Test Group 60

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6100000	61	X	X		X		X			5201 4010 6002 4210 5001	2 3 4	X		Wind-relative: Base encounter, ownship with wgs84, intruder 1 with AGT, validated ADSB and ORNCT, and intruder 2 with V2V.
6100001	61	X	X		X		X						X	Wind-relative: Ownship without wgs84, intruder 1 with AGT, validated ADSB and ORNCT, and intruder 2 with V2V. TRM_VELMODE is set to TRM_VELMODE_NONE.
6100002	61	X	X		X		X			5201 4010 6002 4210 5001	2 3 4	X		Wind-relative: Ownship with wgs84 and airspeed, intruder 1 with AGT, validated ADSB and ORNCT, and intruder 2 with V2V. TRM_VELMODE is set to TRM_VELMODE_GROUNDSPEED_TRACKANGLE. Airspeed is utilized in tracking calculations.
6100003	61	X	X		X		X						X	Wind-relative: Ownship without wgs84, with airspeed, intruder 1 with AGT, validated ADSB and ORNCT, and intruder 2 with V2V. TRM_VELMODE is set to TRM_VELMODE_AIRSPEED_HEADING.
6100004	61	X	X		X		X			5201 4010 6002 4210 5001	3 4	X		Wind-relative: Ownship with wgs84 and airspeed, prefer_wind_relative flag set to true, intruder 1 with AGT, validated ADSB and ORNCT, and intruder 2 with V2V.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6100005	61	X	X		X		X			5001	2			Wind-relative: Ownship hovering at 5 fps without airspeed messages. Ownship is in wind-relative mode with wind_relative = true and TRM_VELMODE = GROUNDSPEED_HEADING until its speed increases above 30 fps. Then ownship exits wind-relative mode with wind_relative = false and TRM_VELMODE = GROUNDSPEED_TRACKANGLE.
6100006	61	X	X		X		X			5001	2			Wind-relative: Ownship hovering at 5 fps with airspeed messages. Ownship is in wind-relative mode with wind_relative = true and TRM_VELMODE = AIRSPEED_HEADING until its speed increases above 30 fps. Then ownship exits wind-relative mode with wind_relative = false and TRM_VELMODE = GROUNDSPEED_TRACKANGLE.
6200001	62						X			1000 4010 6020	1 2 4		X	Turning encounter: Ownship climbing, flying straight South. V2V intruder descending, starting South of ownship, turning clockwise (right), starting going East.
6200002	62						X			1000 4010 6020	1 3 4		X	Turning encounter: Ownship climbing, flying straight South. V2V intruder descending, starting South of ownship, turning counter-clockwise (left), starting going West.
6200003	62						X			1000 4010 6020	1 2 3 4		X	Turning encounter: Ownship climbing, flying straight South. V2V intruder descending, starting South-East of ownship, going parallel (South), then turning clockwise (right) towards ownship.
6200004	62						X			1000 4010 6020	1 2 3 4		X	Turning encounter: Ownship climbing, flying straight South. V2V intruder descending, starting South-West of ownship, going parallel (South), then turning counter-clockwise (left) towards ownship.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6200005	62						X			1000 4010 6020	1 2 3 4		X	Turning encounter: Ownship climbing, turning clockwise (right), starting going East. V2V intruder descending, starting North of ownship, flying straight South.
6200006	62						X			1000 4010 6020	1 2 3 4		X	Turning encounter: Ownship climbing, turning counter-clockwise (left), starting going West. V2V intruder descending, starting North of ownship, flying straight South.
6200007	62						X			1000 4010 6020	1 3 4		X	Turning encounter: V2V Intruder descending, flying straight South. Ownship climbing, starting South-East of intruder, going parallel (South), then turning clockwise (right) towards intruder.
6200008	62						X			1000 4010 6020	1 2 4		X	Turning encounter: V2V intruder descending, flying straight South. Ownship climbing, starting South-West of intruder, going parallel (South), then turning counter-clockwise (left) towards intruder.
6200009	62						X			4010 6020	3		X	Turning encounter: Ownship climbing, turning counter-clockwise (left) starting towards west, intruder descending, turning clockwise (right) starting towards east. Both same latitude, separated in longitude.
6200010	62						X			4010 6020	2 4		X	Turning encounter: Ownship climbing, turning clockwise (right) starting towards east, intruder descending, turning counter-clockwise (left) starting towards west. Both same latitude, separated in longitude.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6200011	62						X			1000 5001 6002 4210 4010	1 2 4		X	Turning encounter: Ownship climbing, turning counter-clockwise (left) starting towards west, intruder descending, turning clockwise (right) starting towards east. Both same latitude, overlapping in longitude.
6200012	62						X			1000 5001 6002 4210 4010	1 3 4		X	Turning encounter: Ownship climbing, turning clockwise (right) starting towards east, intruder descending, turning counter-clockwise (left) starting towards west. Both same latitude, overlapping in longitude.
6200013	62						X				2		X	Turning encounter: Ownship climbing, starting towards west, intruder descending, starting towards east. Both turning counter-clockwise (left) towards each other. Both separate in latitude and longitude.
6200014	62						X				3		X	Turning encounter: Ownship climbing, starting towards east, intruder descending, starting towards west. Both turning clockwise (right) towards each other. Both separate in latitude and longitude.
6200015	62						X				3		X	Turning encounter: Ownship climbing, starting towards west, intruder descending, starting towards east. Both turning clockwise (right) towards each other. Both separate in latitude and longitude.
6200016	62						X				2		X	Turning encounter: Ownship climbing, starting towards east, intruder descending, starting towards west. Both turning counter-clockwise (left) towards each other. Both separate in latitude and longitude.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6200017	62						X						X	Turning encounter: Ownship climbing, starting towards west, intruder descending, starting towards east. Both turning counter-clockwise (left) towards each other. Separate in longitude, overlapping in latitude.
6200018	62						X						X	Turning encounter: Ownship climbing, starting towards east, intruder descending, starting towards west. Both turning clockwise (right) towards each other. Separate in longitude, overlapping in latitude.
6200019	62						X						X	Turning encounter: Ownship climbing, starting towards west, intruder descending, starting towards east. Both turning clockwise (right) towards each other. Separate in longitude, overlapping in latitude.
6200020	62						X						X	Turning encounter: Ownship climbing, starting towards east, intruder descending, starting towards west. Both turning counter-clockwise (left) towards each other. Separate in longitude, overlapping in latitude.
6300001	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Base Encounter.
6300002	63						X	X		4010 6020	2		X	Test for Ownship Discretes. Opflg set to false during RA.
6300003	63						X	X					X	Test for Ownship Discretes. Requested_opmode is set to OPMODE_STANDBY(1).
6300004	63						X	X					X	Test for Ownship Discretes. Requested_opmode is set to OPMODE_STANDBY (2).

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6300005	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Use airspeed with wind relative calculations.
6300006	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Low turn rate limit.
6300007	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Very high turn rate limit.
6300008	63						X	X		1000 4010 6020	1 3 4		X	Test for Ownship Discretes. Low climb rate limit.
6300009	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Very high climb rate limit.
6300010	63						X	X		1000 4010 6020	1 3 4		X	Test for Ownship Discretes. Low turn rate limit and low climb rate limit.
6300011	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Very high turn rate limit and very high climb rate limit.
6300012	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Low turn rate limit and very high climb rate limit.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6300013	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Very high turn rate limit and low climb rate limit.
6300014	63						X	X		1000 4010 6020	1 2 4		X	Test for Ownship Discretes. Enable Obstacle POA.
6300015	63						X	X		4010 6020	1 2 4	X	X	Test for Ownship Discretes. Enable Ground proximity GPOA.
6300016	63						X	X		4010 6020	1 2 4	X	X	Test for Ownship Discretes. Enable Point Obstracle POA and Ground proximity GPOA.
6400000	64						X			1000 5001 6002	1 3 4		X	Improper or degraded inputs: Base encounter V2V.
6400001	64						X			1000 5001 6002	3		X	Improper or degraded inputs: V2V has invalid NACv = 5. V2V is still tracked and RAs are issued.
6400002	64						X						X	Improper or degraded inputs: V2V has NACv = 0. V2V is not tracked.
6400003	64						X			1000 5001 6002	1 3 4		X	Improper or degraded inputs: Multiple V2V messages from the same source in once cycle.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6400004	64						X			1000 5001			X	Improper or degraded inputs: At 15 seconds WGS84 has NaN inputs for position, velocity and altitude while receiving V2V messages. WGS84 is coasted for 2 seconds and still tracks V2V, also issuing RAs. After these 2 seconds WGS84 becomes invalid, no more tracking of V2V.
6400010	64				X					5001 4210 4010	3 4		X	Improper or degraded inputs: Base encounter ADS-B.
6400011	64				X					5001 4210 4010	3 4		X	Improper or degraded inputs: ADS-B has invalid NACv = 5. ADS-B is still tracked and RAs are issued.
6400012	64				X								X	Improper or degraded inputs: ADS-B has NACv = 0. ADS-B is not tracked.
6400013	64				X					5001 4210 4010	3 4		X	Improper or degraded inputs: Multiple ADS-B messages from the same source in once cycle.
6400014	64				X					1000 5001	3		X	Improper or degraded inputs: At 15 seconds WGS has NaN inputs for position, velocity and altitude while receiving ADS-B messages. WGS84 is coasted for 2 seconds and still tracks ADS-B, also issuing RAs. After these 2 seconds WGS84 becomes invalid, no more tracking of ADS-B.
6500000	65						X			4010 6020	2 4		X	Priority values in V2V capability report. Base encounter, intruder priority always set to 7, no coordination messages. Climb and do not descend RAs are issued.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
6500001	65						X		X	5001	2 4		X	Priority values in V2V capability report. Intruder sending a DO NOT CLIMB coordination message. Priority value cycles through 5, 6, 7, 8, and 9.

Table 11. Test Group 70

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000000	70	X	X		X	X	X			5001				Branch coverage: Reaches elseif branch in VerticalTRMUpdatePrep. Ownship is in RA_MODE or SURV_ONLY mode, ownship is about nars_threshold, and intruder is degraded surveillance NAR.
7000001	70				X					1000 4410			X	Branch coverage: Reaches IMP_ELSE branches in TauEstimation. There is 1 ADS-B intruder.
7000002	70						X		X				X	Branch coverage: Reaches branches in UpdateIntruderInputs. There are 2 head-on V2V intruders to ownship. At first, they send conflicting VRCs and then they send the same VRC.
7000003	70				X								X	Branch coverage: Reaches branches in IndividualCostEstimation and UpdateAltitudeInhibitCState. Ownship's height is greater than 2350 ft at the start of the encounter. It then dips below -10000 ft. Ownship loses AGL for 4 cycles mid-encounter.
7000004	70				X			X		5001	2		X	Branch coverage: Reaches branches in PointObsPreventEarlyCOCCost. Point obstacles altitude are set to very high and very low values, ranging from -830 ft to 830 ft, while ownship stays around 106 ft to 110 ft.
7000005	70				X			X		5001	2		X	Branch coverage: Reaches branch in PointObstacleTauEstimation. Point obstacle is set to be far away from ownship, causing the calculated mod_tau_estimate value to be greater that T_max_tau (60) due to large relative range (r_ground).
7000006	70						X		X	1000 6020 4110	1 3 4		X	Branch coverage: Reaches branch in CompatibilityCost. Vertical distance between ownship and intruder causes IsCrossing() to return true.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000007	70		X	X			X		X	1000 5001 4010 4210 5201 6002 4020 6422	1 3 4	X		Branch coverage: Multiple intruders with a V2V intruder sending coordination messages. Developed to reach branches in CompatibilityCost and DetermineResponse.
7000008	70		X	X			X		X	1000 6422 6002 4210 4110	1 3	X		Branch coverage: Reaches elseif branch in CompatibilityCost. Ownship receives several vertical coordination messages from one of the V2V intruders whose altitude is less than ownship's by more than 100 ft which satisfies the IsCrossing() conditions.
7000009	70				X								X	Branch coverage: Encounter to cover branches in DetermineVerticalRateReductionFactor, DetermineVerticalProximateFactor, and DetermineAltitudeDependentCOCFactor. Ownship is at extremely high altitudes ($>$ 20000 ft) to reach some branches.
7000010	70				X								X	Branch coverage: Encounter to cover branch in DetermineVerticalProximateFactor. Ownship is at extremely high altitudes ($>$ 20000 ft) to reach this branch. There is 1 ADS-B intruder.
7000011	70				X					4020			X	Branch coverage: Encounter to cover branch in DetermineVerticalProximateFactor. Ownship is at high altitudes $>$ 9000 ft to reach this branch with high negative dz_own_ave.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000012	70				X								X	Branch coverage: Encounter to cover branches in UpdateCriticalIntervalProtectionCState and CriticalIntervalProtectionCost. Ownship is at extremely high altitudes and there is an ADS-B intruder.
7000013	70				X								X	Branch coverage: Encounter to cover branches in DetermineOwnNonComplianceFactor. Ownship is at high altitudes and there is an ADS-B intruder.
7000014	70				X					4010 4020			X	Branch coverage: Encounter to cover branches in CriticalIntervalProtectionCost. Ownship's z_rel and dz_min are less than specified constants. The intruder is below ownship and all actions are penalized except increase climb.
7000015	70	X			X					1000 5201 4010 6020 5001	1 3			Branch coverage: Encounter to cover branches in UpdateIndivAdjustCounts. Reaches block of code that ensures there is no inappropriate COC annunciation when there is an intruder which was excluded from coordination and the global RA determination.
7000016	70				X								X	Branch coverage: Encounter to cover branches in HorizontalTurnRateSensitivityCostFactor. C_direct_sensitivity_factor is set to 1.0 due to ownship's vertical and turn rates.
7000017	70				X								X	Branch coverage: Encounter to cover branches in DetermineStateDependentSpeedFactor. Encounter with single ADS-B intruder that reaches a branch based on input speed and table discretization information.
7000018	70						X						X	Branch coverage: Encounter to cover branches in HorizontalTRMUpdatePrep. Setting intruder invalid by putting ownship's altitude $>$ 15500 ft and making the V2V intruder NAR.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000019	70	X			X								X	Branch coverage: Encounter to cover branches in HorizontalTRMUpdatePrep. Setting ownship to TRM_VELMODE_NONE by creating invalid WGS84_OBS messages.
7000020	70				X								X	Branch coverage: Encounter to cover branches in SelectHorizontalAdvisoryWindRelative. ADS-B intruder is determined to be out-of-scope causing the prioritized_intruders list to be empty.
7000021	70				X								X	Branch coverage: Encounter to cover branches in AddADSBTrackToReport. Ownship is in standby mode due to AGL and velocity. ADS-B intruder track is added to report, causing intruder.valid to be set to false.
7000022	70					X			X				X	Branch coverage: Encounter to cover branches in AdvanceVerticalTrack. The first 2 UAT messages come within .011 seconds of each other for the same track.
7000023	70				X				X				X	Branch coverage: Encounter to cover branches in ReceiveStateVectorPositionReport. During the call to AdvanceVerticalTrack, the type of altitude switches to or from HAE.
7000024	70				X				X				X	Branch coverage: Encounter to cover branches in ReceiveStateVectorPositionReport. The time step between position report messages for the same track is less than min_obs_toa_step = 0.01.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000025	70				X	X							X	Branch coverage: Encounter to cover branches in ReceiveStateVectorPositionReport, ReceiveStateVectorVelocityReport, and ReceivePresAltObservation. ADSB and UAT tracks are both non-ICAO. UAT track is reported, followed by the non-UAT, ADSB track. Latitude and vel_ns_kts are both reported as NaN for one cycle. PRES_ALT_OBS reports an outlier for 1 cycle.
7000026	70					X			X				X	Branch coverage: UAT intruder encounter to reach branches in AdvanceVerticalTrack.
7000027	70				X				X				X	Branch coverage: Encounter with ADS-B intruder to cover branch in ReceiveStateVectorPositionReport.
7000028	70								X				X	Branch coverage: Encounter to reach branch in ReceiveHeadingObservation. At t=12, own.heading_state = OWN_HEADING_DEGRADED while heading_degraded is set to false .
7000029	70					X							X	Branch coverage: Encounter to reach branch in ReceiveModeStatusReport. At t=7.994, rebroadcast is set to true in the MODE_STATUS_REPORT message.
7000030	70	X			X					1000 5001 6002	1 2 4		X	Branch coverage: ORNCT intruder tracks go stale after exceeding max_coasts. The track also exceeds max_coasts_ornct_high_priority and is not in a field-of-view coast.
7000031	70				X				X				X	Branch coverage: Encounter with multiple ADS-B intruders to reach branch in AdvancePassiveTrackPosition.
7000032	70				X								X	Branch coverage: Encounter with co-altitude, then descending, ADS-B intruder to reach branch in AdvancePassiveTrackVelocity.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000033	70					X			X				X	Branch coverage: UAT encounter with an intruder directly above ownship and one intruder closing from below ownship. MODE_STATUS_REPORT messages report a NACv of 4 to reach branch in ConvertNACvToSigmaHVA.
7000034	70	X								1000 5201 4010 4210	1 2		X	Branch coverage: Encounter with ORNCT intruder. Ownship's heading state is degraded while processing ORNCT intruder. Reaches branch in SetDisplayDataORNCT.
7000035	70	X											X	Branch coverage: Ownship is in STANDBY mode when ORNCT track is first added to the report.
7000036	70	X			X									Branch coverage: Reaches branches in DetermineORNCTDeletion. There are many ORNCT and ADS-B intruders. The ORNCT track to be deleted is associated with a non-empty target.
7000037	70	X					X						X	Branch coverage: Ownship has no altitude readings and correlation is happening between the V2V track and ORNCT.
7000038	70		X											Branch coverage: AGT track and Ownship moving north, AGT track suddenly jumps and moves over 10000 feet leading to decorrelation.
7000039	70		X											Branch coverage: Ownship has no altitude readings after t=5, correlation is happening between the AGT track and Ownship. AGT is non-ICAO.
7000040	70		X		X					4010 4110	3	X		Branch coverage: Multiple ADSB tracks that are non-icao start correlating, but later the tracks distance apart.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000041	70		X											Branch coverage: Ownship has no altitude readings after t=5 and correlation is happening between the V2V track and ORNCT. AGT track is ICAO.
7000042	70					X							X	Branch coverage: Covers branches in ReceiveStateVectorUATReport when position and velocity data are NaN and when the altitude source switches from non-HAE to HAE.
7000043	70		X										X	Branch coverage: Covers branches in AddAGTTrackToReport, AddTracksToReport, SelectOptimalAGTTrack. AGT has non_coop=false, is_passive=true, externally_validated=false. Ownship's opflg switches to false at t=9.994.
7000044	70						X		X	6020 5001 4010 5201 4020 4410	2 3 4	X	X	Branch coverage: Covers branches in ShouldReverse. Intruder is master and sending vertical coordination messges to ownship. Intruder is flying directly above ownship. Ownship and intruder begin descending.
7000045	70						X		X	5001 5002 6002	2 4		X	Branch coverage: Covers branches in ShouldReverse. Intruder is master and sending vertical coordination messges to ownship. Intruder is flying directly below ownship. Ownship and intruder begin ascending.
7000046	70						X		X	5001 5002 4410 4210 5201	2 4		X	Branch coverage: Covers branches in ShouldReverse. Intruder is master and sending vertical coordination messges to ownship. Intruder is flying directly below ownship. Ownship and intruder begin ascending. Intruder switches VRC from 2 to 1 around t=14 in order to reach a branch.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000047	70						X		X	5001 5002 6002	2 4		X	Branch coverage: Covers branches in ShouldReverse. Intruder is master and sending vertical coordination messges to ownship. Intruder is flying directly below ownship. Ownship and intruder begin ascending. Around t=14, the intruder has a high climb rate to trigger a branch.
7000049	70		X										X	Branch coverage: Covers branch in IndividualCostEstimation. Ownship and AGT intruder are at very low altitudes, below the allowable unequiped lower threshold according to the parameters file.
7000050	70						X	X	X	1000 5001 4010 4210 6002 6422	1 3 4	X	X	Branch coverage: Covers branch in CompatibilityCost. Intruder is responsive, ownship's sense is different than intruder's sense.
7000051	70						X		X	1000 5002 4210 6002 6422 4010	1 3 4	X	X	Branch coverage: Covers branch in CompatibilityCost. Intruder is responsive, ownship's sense is different than intruder's sense, and ownship is accelerating with a positive vertical rate, in opposition of the active VRC.
7000052	70						X		X	5201 4010 6020 5401	2 3 4		X	Branch coverage: Covers branches in BadTransitionCost. Intruder is master and sending vertical coordination messges to ownship. Intruder is flying directly above, then directly above, and then directly below ownship.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000053	70						X		X	1000 5001 4010 4210 6002 6422	1 3 4	X	X	Branch coverage: Covers branches in UpdateCoordinationDelayCState. Ownship is in surveillance-only mode for the first 10 seconds before switching into RA mode. At t=20, the intruder sends a vertical coordination message.
7000054	70						X			1000 5001 4010 4210 6002 6422	1 3 4	X	X	Branch coverage: Covers branch in ArbitrateConflictingSenses. There are 2 V2V intruders that cause a multithreat scenario and call ArbitrateConflictingSenses. Around the time this call is made, a NAR V2V intruder is added and then disappears.
7000055	70						X						X	Branch Coverage: Covers branches in ConvertNACpToSigmaHEPU. Incrementing NACp value for V2V intruder.
7000056	70	X					X			5001 4210 6020 4010	2 4			Branch Coverage: Covers branches in CorrelatePosition. Correlation done between ORNCT and V2V tracks.
7000057	70		X											Branch Coverage: Covers branches in CorrelateIndividualTracksToOwnship. Correlation between AGT tracks.
7000058	70		X											Branch Coverage: Covers branches in CorrelateIndividualTracksToOwnship. Correlation done between three AGT tracks.
7000059	70						X		X				X	Branch Coverage: Covers branches in ReceiveV2VCoordination. Varying the Vertical and Horizontal Coordination messages.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000060	70				X			X		5001	2		X	Branch coverage: Reaches branches in RemoveStaletracks and UpdatePassiveTrackerVelocity. Point Obstacle added and then marked as to_be_deleted. Nacv set to 4 for intruder after t=5.
7000061	70						X		X	1000 4010	1 3		X	Branch coverage: Reaches branches in HAEAltAtToa. Out-of-order WGS84_OBS message is sent to trigger a branch reachable when toa < own.wgs84_toa_vert. At report_time=15.996, toa of WGS84_OBS is 19.996.
7000062	70						X		X				X	Branch coverage: Covers branches in SetIntruderDegradedFlags and ReceivePresAltObservation. Horizontal Coordination message sent with incorrect parity value. Out of order pressure altitude message to hit a branch.
7000063	70				X			X					X	Branch coverage: Covers branches in AddPointObstacleTrackToReport. Ownship begins in TRM_ALTMODE_NONE due to NaN HAE and pressure altitudes. It then regains altitude reporting and switches into STANDBY mode due to setting opflg=false in OWNSHIP_DISCRETES.
7000064	70		X										X	Branch coverage: Covers branches in DeleteAGTTrack. Track status set to 4 marking track as to_be_deleted.
7000065	70						X		X	5001 4210 4010 5201	2 3 4		X	Branch coverage: Covers branches in DeleteIntent. Horizontal coordination switches from HRC of 6 to HRC of 2 in next cycle with proper parity. Because these indices return the same value when referenced in hor_intent from the parameter file, we are able to reach these branches.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
7000066	70						X						X	Branch coverage: Covers branches in ResetHTRM. Ownship is in STANDBY mode due to opflg being set to false. The encounter also triggers wind-relative mode with valid TRUE_AIRSPEED_OBS messages and prefer_wind_relative set to true in OWNSHIP_DISCRETES.
7000067	70		X							1000 6020 4020			X	Branch coverage: Covers branches in SelectHorizontalAdvisoryWindRelative. Set prefer_wind_relative in OWNSHIP_DISCRETES to true. Added TRUE_AIRSPEED_OBS messages around t=53.

Table 12. Test Group 80

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8300300	83				X					5001	1 2		X	CR Xs-003: ADS-B / V2V validation encounters (#1 of 2): ADS-B starts out validated, midway through the encounter becomes invalidated, and 10 seconds later is validated again.
8300301	83						X			5001 4210 4010	2		X	CR Xs-003: ADS-B / V2V validation encounters (#2 of 2): V2V starts out validated, midway through the encounter becomes invalidated, and 10 seconds later is validated again.
8301300	83			X										CR Xs-013: (#1 of 4) Preventing the use of zero-diagonal Sigma_vert in mahal calculation in CorrelatePosition. AGT tracking 25 ft above ownship. Ownship loses pressure and hae altitude from t=9.996 to t=29.998. AGT is pressure altitude reporting and loses pressure altitude from t=19.999 to t=29.999. AGT correlates with ownship until t=13. From t=13 to t=19, AGT does not correlate with ownship. The mahal calculation is never performed with the zero-diagonal Sigma_vert.
8301301	83		X											CR Xs-013: (#2 of 4) Preventing the use of zero-diagonal Sigma_vert in mahal calculation in CorrelatePosition. AGT tracking 25 ft above ownship. Ownship loses pressure and hae altitude from t=9.996 to t=29.998. AGT is hae altitude reporting and loses hae altitude from t=19.999 to t=29.999. AGT correlates with ownship until t=13. From t=13 to t=19, AGT does not correlate with ownship. The mahal calculation is never performed with the zero-diagonal Sigma_vert.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8301302	83			X							3			CR Xs-013: (#3 of 4) Preventing the use of zero-diagonal Sigma_vert in mahal calculation in CorrelatePosition. AGT completely independent from ownship track (flying in a different direction). Ownship loses pressure and hae altitude from t=9.996 to t=29.998. AGT is pressure altitude reporting and loses pressure altitude from t=19.999 to t=29.999. AGT does not correlate with ownship from t=1 to t=32. At t=33 and t=34, the AGT is close enough to ownship that correlation occurs. From t=35 to the end, the tracks are no longer correlated. The mahal calculation is never performed with the zero-diagonal Sigma_vert.
8301303	83		X								2			CR Xs-013: (#4 of 4) Preventing the use of zero-diagonal Sigma_vert in mahal calculation in CorrelatePosition. AGT completely independent from ownship track (flying in a different direction). Ownship loses pressure and hae altitude from t=9.996 to t=29.998. AGT is hae altitude reporting and loses hae altitude from t=19.999 to t=29.999. AGT does not correlate with ownship. AGT correlates with ownship at t=33 and t=34. The mahal calculation is never performed with the zero-diagonal Sigma_vert.
8301500	83						X			1000 5001 6002	1 3 4		X	CR Xs-015, removal of Altitude inhibits. Base encounter, AGL is between 2300 and 2447 ft. Vertical RAs: 5001, 6002. Horizontal RAs: 3, 4.
8301501	83						X			1000 5001 6002	1 3 4		X	CR Xs-015, removal of Altitude inhibits. AGL is set to 300 - 447 ft. Vertical RAs: 5001, 6002. Horizontal RAs: 3,4. (Same as in base encounter.)

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8301502	83						X			1000 5001 6002	1 3 4		X	CR Xs-015, removal of Altitude inhibits. AGL is set to 80 ft. Vertical RAs: 5001, 6002. Horizontal RAs: 3,4. (Same as in base encounter.)
8301503	83						X			1000 5001 6002	1 3 4		X	CR Xs-015, removal of Altitude inhibits. AGL is set to 5 ft. Vertical RAs: 5001, 6002. Horizontal RAs: 3,4. (Same as in base encounter.)
8301504	83						X			1000 5001 6002	1 3 4		X	CR Xs-015, removal of Altitude inhibits. AGL is set to 0 ft. Vertical RAs: 5001, 6002. Horizontal RAs: 3,4. (Same as in base encounter.)
8301505	83						X			1000 5001 6002	1 3 4		X	CR Xs-015, removal of Altitude inhibits. AGL is set to -10 ft. Vertical RAs: 5001, 6002. Horizontal RAs: 3,4. (Same as in base encounter.)
8302410	83		X		X					1000 5001 4210 4010 5201	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#1 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302411	83		X		X					1000 5001 4210 4010 5201	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#2 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8302412	83		X		X					1000 5001 4210 4010 5201	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#3 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302413	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#4 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302414	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#5 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302415	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#6 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302416	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#7 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302420	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#8 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8302421	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#9 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302422	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#10 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302423	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#11 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302424	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#12 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302425	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#13 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302426	83		X		X					1000 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#14 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8302430	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#15 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302431	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#16 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302432	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#17 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302433	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#18 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302434	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#19 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8302435	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#20 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302436	83		X		X					1000 5001 4210 4010 5201	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#21 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302440	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#22 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302441	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#23 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302442	83		X		X					1000 5201 4010 4210 5001	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#24 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8302443	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#25 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302444	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#26 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302445	83		X		X					1000 4010	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#27 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8302446	83		X		X					1000 5001 4210 4010 5201	1 2 4			CR Xs-024: alt inhibit, surveillance only mode and on ground mode testing (#28 of 28). None of them should change based on altitude. Use different altitudes and make sure no mode change occurs.
8303500	83							X					X	CR Xs-035: Tests that SetVerticalTRMSurvOnlyMode now updates point obstacles and ground point obstacles without running unnecessary functions in Surveillance Only mode. This encounter has point obstacles, and is_point_obs is set to true. Ownship is in surveillance only mode.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8303501	83												X	CR Xs-035: Tests that SetVerticalTRMSurvOnlyMode now updates point obstacles and ground point obstacles without running unnecessary functions in Surveillance Only mode. This encounter has ground point obstacles, and is_ground_point_obs is set to true. Ownship is in surveillance only mode.
8303502	83							X					X	CR Xs-035: Tests that SetVerticalTRMSurvOnlyMode now updates point obstacles and ground point obstacles without running unnecessary functions in Surveillance Only mode. This encounter has point obstacles and ground point obstacles, and is_point_obs and is_ground_point_obs are both set to true. Ownship is in surveillance only mode.
8303503	83							X					X	CR Xs-035: Tests that SetVerticalTRMSurvOnlyMode now updates point obstacles and ground point obstacles without running unnecessary functions in Surveillance Only mode. This encounter has point obstacles and ground point obstacles, and is_point_obs and is_ground_point_obs are both set to true. Ownship is switching between standby, surveillance only, and RA mode.
8401000	84	X								5201 4010 6002 4210 5001	2 4		X	CR Xs-010: ORNCT / AGT degraded surveillance (#1 of 13): ORNCT - Base encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8401001	84	X								5001 6002 4210 4010	2 4		X	CR Xs-010: ORNCT / AGT degraded surveillance (#2 of 13): ORNCT - ORNCT coast. From t=20 to t=24, degraded_surveillance is set to DEGRADED_SURVEILLANCE_ORNCT_COAST. From t=25 to t=31, the track drops before being re-initialized at t=32.
8401002	84	X								1000 4010 6020	2 4		X	CR Xs-010: ORNCT / AGT degraded surveillance (#3 of 13): ORNCT - ORNCT coast and FOV coast. From t=32 to t=36, degraded_surveillance is set to DEGRADED_SURVEILLANCE_ORNCT_COAST. From t=37 to t=42, degraded_surveillance is set to DEGRADED_SURVEILLANCE_ORNCT_FOV_COAST. The track drops at t=43 before being re-initialized at t=44. Vertical and horizontal RA drops at t=43 and starts again at t=46.
8401003	84	X								4010	1 2 4		X	CR Xs-010: ORNCT / AGT degraded surveillance (#4 of 13): ORNCT - NAR. From t=34 to t=40, degraded_surveillance is set to DEGRADED_SURVEILLANCE_NAR. From t=34 to t=42, the vertical RA drops. From t=34 to t=40, the horizontal RA drops. Note that ORNCT NAR is caused by ownship losing HAE and pressure altitudes.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8401004	84	X								5201 4010	1 2 4		X	CR Xs-010: ORNCT / AGT degraded surveillance (#5 of 13): ORNCT - ORNCT coast, FOV coast, and NAR. At t=34, degraded_surveillance is set to DEGRADED_SURVEILLANCE_ORNCT_COAST. From t=35 to t=38, degraded_surveillance is set to DEGRADED_SURVEILLANCE_ORNCT_COAST and DEGRADED_SURVEILLANCE_NAR. At t=39, degraded_surveillance is set to DEGRADED_SURVEILLANCE_ORNCT_COAST, DEGRADED_SURVEILLANCE_NAR, and DEGRADED_SURVEILLANCE_ORNCT_FOV_COAST. Vertical alerting drops from t=35 to t=41. Horizontal alerting drops from t=35 to t=39.
8401010	84		X							1000 5201 4010 4210 5001	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#6 of 13): AGT - Base encounter.
8401011	84		X							1000 5001 4010 4210	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#7 of 13): AGT - NAR. From t=34 to t=39, degraded_surveillance is set to DEGRADED_SURVEILLANCE_NAR. From t=34 to t=41, the vertical alerting drops.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8401012	84		X							1000 5201 4010 4210 5001	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#8 of 13): AGT - VERT COAST, HORIZ COAST. From t=34 to t=37, degraded_surveillance is set to DEGRADED_SURVEILLANCE_VERT_COAST and DEGRADED_SURVEILLANCE_HORIZ_COAST. Then the track reinitializes at t=38 (degraded_surveillance = 0). Then from t=39 to t=42, degraded_surveillance is set to DEGRADED_SURVEILLANCE_VERT_COAST and DEGRADED_SURVEILLANCE_HORIZ_COAST again. There is no interuption to alerting.
8401013	84		X							1000 4010	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#9 of 13): AGT - PASSIVE NOT VALIDATED. From t=37 to t=42, degraded_surveillance is set to DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED. From t=37 to t=44, vertical alerting drops. From t=37 to t=45, horizontal alerting drops.
8401014	84		X							1000 5001 4010 4210	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#10 of 13): AGT - NAR, VERT COAST, and HORIZ COAST. At t=35, degraded_surveillance is set to DEGRADED_SURVEILLANCE_NAR. From t=36 to t=39, degraded_surveillance is set to DEGRADED_SURVEILLANCE_NAR, DEGRADED_SURVEILLANCE_VERT_COAST, and DEGRADED_SURVEILLANCE_HORIZ_COAST. From t=35 to t=41, vertical alerting drops.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8401015	84		X							1000 4010	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#11 of 13): AGT - VERT COAST, HORIZ COAST, and PASSIVE NOT VALIDATED. From t=34 to t=37, degraded_surveillance is set to DEGRADED_SURVEILLANCE_VERT_COAST and DEGRADED_SURVEILLANCE_HORIZ_COAST. At t=38, degraded_surveillance is set to DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED. From t=39 to t=42, degraded_surveillance is set to DEGRADED_SURVEILLANCE_VERT_COAST, DEGRADED_SURVEILLANCE_HORIZ_COAST, and DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED. From t=35 to t=44, vertical alerting drops. From t=38 to t=45, horizontal alerting drops.
8401016	84		X							1000 5001 4010 4210	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#12 of 13): AGT - NAR and PASSIVE NOT VALIDATED. From t=34 to t=39, degraded_surveillance is set to DEGRADED_SURVEILLANCE_NAR and DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED. From t=34 to t=41, vertical alerting drops. From t=34 to t=42, horizontal alerting drops.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8401017	84		X							1000 5001 4010 4210	1 2 4			CR Xs-010: ORNCT / AGT degraded surveillance (#13 of 13): AGT - NAR, VERT COAST, HORIZ COAST, PASSIVE NOT VALIDATED. At t=34, degraded_surveillance is set to DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED. At t=35, degraded_surveillance is set to DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED and DEGRADED_SURVEILLANCE_NAR. From t=36 to t=39, degraded_surveillance is set to DEGRADED_SURVEILLANCE_PASSIVE_NOT_VALIDATED, DEGRADED_SURVEILLANCE_NAR, DEGRADED_SURVEILLANCE_VERT_COAST, and DEGRADED_SURVEILLANCE_HORIZ_COAST. From t=34 to t=41, vertical alerting drops. From t=34 to t=42, horizontal alerting drops.
8402100	84				X						2		X	CR Xs-021: (#1 of 3): Ensuring that the horizontal-NaN issue no longer happens with NAR intruders. ADS-B intruder goes NAR from t=9.966 to t=25. At t=12 in the STM report, the vertical fields become NaN but the horizontal fields remain populated.
8402101	84				X						2		X	CR Xs-021: (#2 of 3): Ensuring that the horizontal-NaN issue no longer happens with NAR intruders. ADS-B intruder has altitude less than NARS_THRESHOLD_MIN from t=9.966 to t=25. At t=12 in the STM report, the vertical fields become NaN but the horizontal fields remain populated.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8402102	84				X						2		X	CR Xs-021: (#3 of 3): Ensuring that the horizontal-NaN issue no longer happens with NAR intruders. ADS-B intruder has altitude higher than NARS_THRESHOLD_MAX from t=9.966 to t=25. At t=12 in the STM report, the vertical fields become NaN but the horizontal fields remain populated.
8404101	84		X		X					1000 4010 6020	1 2 4			CR Xs-041: Tests restored check for ADS-B velocity report. ADS-B velocity report not sent for the first 5 cycles.
8404200	84	X								1000 5201 4010 6020 5001	3 4		X	CR Xs-042: Vertical rate arrow is now based on absolute vertical rate rather than relative. Ownship gets a negative dt_vert, showing that when a negative dt_vert starts the vertical rate arrow changes from 0 to -1
8404600	84	X								5201 4010 5001	2		X	CR Xs-046: Confirm that intruders check for TRM_VELMODE_NONE and TRM_ALTMODE_NONE. Single ORNCT intruder is heading towards ownship. From t=7 to t=14, own WGS84 goes invalid. From t=15 to the end, own pressure altitude goes invalid. The corresponding warnings are output by HorizontalTRMUpdatePrep and VerticalTRMUpdatePrep. The horizontal RA disappears with TRM_VELMODE_NONE and the vertical RA disappears with TRM_ALTMODE_NONE.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8405200	84						X						X	CR Xs-052: Tests the quality checks on the V2V track. Due to poor quality in the sil value, the V2V track is invalidated and revalidated several times. At t=2.998, the quality dips below the threshold and the track drops at t=4. At t=15, the track is added back due to revalidation. At t=17, the track drops due to poor sil quality. At t=28, the track passes the quality check again and is added back for the rest of the encounter.
8405700	84				X			X		5001	2		X	CR Xs-057 (#1 of 4): Tests OWNSHIP_DISCRETES fields perform_poa and disable_gpoa. Point obstacles, ground point obstacle, and airborne intruder are present. Here, perform_poa = false and disable_gpoa = true.
8405701	84				X			X		5001	2	X	X	CR Xs-057 (#2 of 4): Tests OWNSHIP_DISCRETES fields perform_poa and disable_gpoa. Point obstacles, ground point obstacle, and airborne intruder are present. Here, perform_poa = true and disable_gpoa = true.
8405702	84				X			X		5001	2	X	X	CR Xs-057 (#3 of 4): Tests OWNSHIP_DISCRETES fields perform_poa and disable_gpoa. Point obstacles, ground point obstacle, and airborne intruder are present. Here, perform_poa = false and disable_gpoa = false.
8405703	84				X			X		5001	2	X	X	CR Xs-057 (#4 of 4): Tests OWNSHIP_DISCRETES fields perform_poa and disable_gpoa. Point obstacles, ground point obstacle, and airborne intruder are present. Here, perform_poa = true and disable_gpoa = false.
8406201	84						X			1000 4010 6020	1 2 4		X	CR Xs-062: V2V Coasting (#1 of 3): Base encounter.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8406202	84						X			1000 5001 4010	2		X	CR Xs-062: V2V Coasting (#2 of 3): V2V measurements end short before RA is issued. Track is coasted for 6 seconds, so still causes RA but RA ends then early. Similar 3 seconds later with new V2V track.
8406203	84						X			1000 5001 4010	2		X	CR Xs-062: V2V Coasting (#3 of 3): V2V measurements end short before RA is issued. V2V operational status message and externally validated V2V messages are continued. Track is coasted for 6 seconds, so still causes RA but RA ends then early. Similar 3 seconds later with new V2V track.
8406500	83												X	CR Xs-065: Quantization in ReceivePresAltObservation was hardcoded to 0, is now set to 1. Pres Alt values are not being passed through if the dt between messages is too small, in V3 all input messages would have been accepted
8406900	84						X			1000 5001 4210 4010	1 2 4		X	CR Xs-069: Tests that the check for NAR above 15,500 ft when determining intruder validity is removed. Head-on V2V intruder is above 15,500 ft and ownship still issues vertical alerts. At t=15, 5001 is issued. At t=33, 4210 is issued. At t=34, 4010 is issued until COC at t=44.
8407000	84				X					1000 5001 6002	1 2 4		X	CR Xs-070: Confirming that ADS-B horizontal tracker changes made in V3R1 are reverted back to the way it was coded up in V3R0. Testing that final ideal V2V parameters found at the end of V3R1 are utilized.
8407201	84	X								1000 4010 6020			X	CR Xs-072 (#1 of 8): ORNCT with position late in STM cycle right before STM is run

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8407202	84	X								1000 4010 6020	1 2 4		X	CR Xs-072 (#2 of 8): ORNCT with toa early in STM cycle right after previous STM cycle is run. Position however is correct for end of STM cycle.
8407203	84	X								1000 4010 6020	1 2 4		X	CR Xs-072 (#3 of 8): ORNCT with toa early in the following STM cycle. Position however is correct for end of previous STM cycle.
8407204	84		X							1000 4010 6020	1 2 4		X	CR Xs-072 (#4 of 8): AGT with position late in STM cycle right before STM is run
8407205	84		X							1000 4010 6020	1 2 4		X	CR Xs-072 (#5 of 8): AGT with toa early in STM cycle right after previous STM cycle is run. Position however is correct for end of STM cycle.
8407206	84		X							1000 4010 6020	1 2 4		X	CR Xs-072 (#6 of 8): AGT with toa early in the following STM cycle. Position however is correct for end of previous STM cycle.
8407207	84	X	X							1000 4010 6020	1 2 4			CR Xs-072 (#7 of 8): ORNCT with toa early in STM cycle right after previous STM cycle is run. In addition AGT at end of cycle. Tests that position for ORNCT is forward extrapolated for end of STM cycle and toa ORNCT not confused with AGT toa for calculating min step.
8407208	84	X	X							1000 4010 6020				CR Xs-072 (#8 of 8): AGT with toa early in STM cycle right after previous STM cycle is run. In addition ORNCT at end of cycle. Tests that position for AGT is forward extrapolated for end of STM cycle and toa AGT not confused with ORNCT toa for calculating min step.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8407800	84		X							5201 4010 6020 4210 5001	2			CR Xs-078 (#1 of 6): (1 AGT track was intended to represent ownship, but ownship cannot correlate with an AGT track that has a valid mode_s address.) 2 AGT tracks with the same mode_s address but different agt_ids do not associate in AssociateAGTToTarget. However, they do correlate by ID since both mode_s addresses are ICAO. In the STM report, both agt_ids are stored (4565 and 7898).
8407801	84		X							5201 4010 6020 4210 5001	2		X	CR Xs-078 (#2 of 6): [This behavior is not permitted by the MOPS, purely for testing.] (1 AGT track was intended to represent ownship, but ownship cannot correlate with an AGT track that has a valid mode_s address.) 2 AGT tracks with different mode_s addresses but the same agt_id associate together. In the STM report, only 1 icao24 value populates the array. However at t=21, this value switches from 111 to 222.
8407802	84		X											CR Xs-078 (#3 of 6): 2 AGT tracks with the same mode_s address but different agt_ids do not associate in AssociateAGTToTarget. However, they do correlate by ID since both mode_s addresses are ICAO. In the STM report, both agt_ids are stored (12345 and 23456).
8407803	84		X										X	CR Xs-078 (#4 of 6): [This behavior is not permitted by the MOPS, purely for testing.] 2 AGT tracks with different mode_s addresses but the same agt_id associate together. In the STM report, only 1 icao24 value populates the array (345).
8407804	84		X										X	CR Xs-078 (#5 of 6): Ownship associates with AGT track with agt_id 4565. At t=24.998, the agt_id changes to 1232, and a new AGT track initializes. These agt_ids do not show up in ownship's agt_id dictionary in the STM report.

Encounter	Test Group	ORNCT	AGT (HAE)	AGT (PRES)	ADS-B	UAT	V2V	Point Obstacles	Coordination	Vertical RA(Label 270)	Horizontal RA(Label 271)	Multi-Threat	Prescriptive	Description
8407805	84		X											CR Xs-078 (#6 of 6): Intruder AGT track has agt_id 5676 until t=23.99, when agt_id changes to 2343. This is treated as a new track. From t=24 to t=28, the agt_id dictionary in the STM report contains both 5676 and 2343, until the 5676 agt_id track coasts out.
8408901	84		X		X					1000 4010 6020	1 2 4			CR Xs-089: Tests fix to bug regarding the usage of input_int in SetHorizontalDroppedIntruderOutput.

Table 13. Branch Coverage

Algorithm	Branch Type	Line Number	Encounter
DetermineVerticalScore	THEN	13	Encounter4250002
DetermineVerticalScore	IMP_ELSE	13	Encounter1000001
DetermineVerticalScore	THEN	17	Encounter1000004
DetermineVerticalScore	IMP_ELSE	17	Encounter1000001
DetermineVerticalScore	THEN	20	Encounter1000001
DetermineVerticalScore	THEN	21	Encounter1000002
DetermineVerticalScore	ELSE	23	Encounter1000001
DetermineVerticalScore	ELSEIF	26	Encounter1000001
VerticalTRMUpdate	THEN	18	Encounter5100001
VerticalTRMUpdate	ELSEIF	22	Encounter1000001
VerticalTRMUpdate	ELSE	42	Encounter1000001
StateAndCostEstimation	THEN	26	Encounter1000001
StateAndCostEstimation	THEN	28	Encounter1000001
StateAndCostEstimation	ELSE	30	Encounter4260000
StateAndCostEstimation	ELSE	33	Encounter1000001
StateAndCostEstimation	THEN	45	Encounter1000001
StateAndCostEstimation	ELSE	52	Encounter1000004
StateAndCostEstimation	THEN	63	Encounter1000001
StateAndCostEstimation	ELSE	65	Encounter1000001
StateAndCostEstimation	THEN	68	Encounter1000001
StateAndCostEstimation	THEN	69	Encounter1000001
StateAndCostEstimation	ELSE	76	Encounter1000001
StateAndCostEstimation	ELSE	82	Encounter1000004
MinCostIndex	THEN	6	Encounter1000001
MinCostIndex	IMP_ELSE	6	Encounter1000001
InterpolateOneDimension	THEN	7	Encounter1000001
InterpolateOneDimension	ELSEIF	10	Encounter1000001
InterpolateOneDimension	ELSE	13	Encounter1000001
InterpolateOneDimension	THEN	18	Encounter1000001
InterpolateOneDimension	ELSE	20	Encounter1000001
InterpolateOneDimension	THEN	28	Encounter1000001
InterpolateOneDimension	THEN	30	Encounter1000001
InterpolateOneDimension	THEN	31	Encounter1000001
InterpolateOneDimension	ELSE	33	Encounter1000001
InterpolateOneDimension	IMP_ELSE	30	Encounter1000001
InterpolateOneDimension	THEN	41	Encounter1000001
InterpolateOneDimension	ELSE	43	Encounter4250002
InterpolateOneDimension	ELSE	47	Encounter1000001
ScaleVerticalCuts	THEN	9	Encounter1000001
ScaleVerticalCuts	ELSE	20	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
GetIndexFromClassification	THEN	10	Encounter1000001
GetIndexFromClassification	ELSEIF	13	Encounter1000001
GetIndexFromClassification	IMP_ELSE	13	Encounter1000001
GetIndexFromClassification	THEN	17	Encounter1000005
GetIndexFromClassification	THEN	18	Encounter1000005
GetIndexFromClassification	IMP_ELSE	17	Encounter1000001
VerticalTRMUpdatePrep	THEN	12	Encounter5100001
VerticalTRMUpdatePrep	ELSEIF	14	Encounter4210016
VerticalTRMUpdatePrep	ELSE	18	Encounter1000001
VerticalTRMUpdatePrep	THEN	23	Encounter1000001
VerticalTRMUpdatePrep	ELSEIF	25	Encounter1000011
VerticalTRMUpdatePrep	ELSE	27	Encounter1000001
VerticalTRMUpdatePrep	THEN	33	Encounter1000008
VerticalTRMUpdatePrep	ELSE	35	Encounter1000001
VerticalTRMUpdatePrep	THEN	41	Encounter1000001
VerticalTRMUpdatePrep	IMP_ELSE	41	Encounter1000011
VerticalTRMUpdatePrep	THEN	54	Encounter1000001
VerticalTRMUpdatePrep	IMP_ELSE	54	Encounter1000001
VerticalTRMUpdatePrep	THEN	63	Encounter1000001
VerticalTRMUpdatePrep	ELSE	65	Encounter1000001
SetHorizontalTRMSurvOnlyMode	THEN	11	Encounter5100001
SetHorizontalTRMSurvOnlyMode	ELSE	18	Encounter5100001
SetHorizontalTRMSurvOnlyMode	ELSE	22	Encounter6300004
SetVerticalTRMSurvOnlyMode	THEN	14	Encounter5100001
SetVerticalTRMSurvOnlyMode	ELSE	26	Encounter5100001
SetVerticalTRMSurvOnlyMode	THEN	50	Encounter5100001
SetVerticalTRMSurvOnlyMode	IMP_ELSE	50	Encounter5100001
GenerateSurvOnlyModeOutput	THEN	22	Encounter5100001
GenerateSurvOnlyModeOutput	IMP_ELSE	22	Encounter5100001
StateEstimation	THEN	17	Encounter1000001
StateEstimation	IMP_ELSE	17	Encounter1000001
StateEstimation	THEN	27	Encounter1000001
StateEstimation	ELSE	33	Encounter1000001
StateEstimation	THEN	34	Encounter1000001
StateEstimation	ELSE	36	Encounter4260000
HorizontalWorstCase	THEN	16	Encounter1000001
HorizontalWorstCase	IMP_ELSE	16	Encounter1000001
HorizontalWorstCase	THEN	20	Encounter1000001
HorizontalWorstCase	IMP_ELSE	20	Encounter1000001
HorizontalWorstCase	THEN	24	Encounter1000001
HorizontalWorstCase	IMP_ELSE	24	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
HorizontalWorstCase	THEN	29	Encounter1000001
HorizontalWorstCase	THEN	31	Encounter1000001
HorizontalWorstCase	ELSEIF	33	Encounter1000001
HorizontalWorstCase	IMP_ELSE	33	Encounter1000001
HorizontalWorstCase	THEN	36	Encounter1000001
HorizontalWorstCase	IMP_ELSE	36	Encounter1000001
HorizontalWorstCase	IMP_ELSE	29	Encounter1000001
LookupEntryDistribution	THEN	8	Encounter1000001
LookupEntryDistribution	IMP_ELSE	8	Encounter1000001
IsSlowClosure	THEN	8	Encounter1000007
IsSlowClosure	IMP_ELSE	8	Encounter1000001
IsVerticallyConverging	THEN	7	Encounter1000001
IsVerticallyConverging	IMP_ELSE	7	Encounter1000001
DetermineTauVerticalDistribution	THEN	22	Encounter1000007
DetermineTauVerticalDistribution	IMP_ELSE	22	Encounter1000007
TauEstimation	THEN	30	Encounter1000001
TauEstimation	IMP_ELSE	30	Encounter1000001
TauEstimation	THEN	35	Encounter1000007
TauEstimation	THEN	36	Encounter1000007
TauEstimation	IMP_ELSE	36	Encounter1000007
TauEstimation	THEN	42	Encounter1000007
TauEstimation	THEN	46	Encounter1000007
TauEstimation	IMP_ELSE	46	Encounter7000001
TauEstimation	IMP_ELSE	35	Encounter1000001
TauEstimation	THEN	56	Encounter1000001
TauEstimation	IMP_ELSE	56	Encounter1000001
CombineVerticalBeliefs	THEN	6	Encounter1000001
CombineVerticalBeliefs	IMP_ELSE	6	Encounter1000001
CombineVerticalBeliefs	THEN	13	Encounter1000001
CombineVerticalBeliefs	ELSE	15	Encounter1000001
WrapTo2Pi	THEN	4	Encounter1000001
WrapTo2Pi	IMP_ELSE	4	Encounter1000001
IntruderPrep	THEN	6	Encounter1000007
IntruderPrep	ELSEIF	14	Encounter1000004
IntruderPrep	ELSEIF	16	Encounter1000001
IntruderPrep	THEN	17	Encounter4110011
IntruderPrep	IMP_ELSE	17	Encounter1000001
IntruderPrep	IMP_ELSE	16	Encounter5100001
IntruderPrep	THEN	33	Encounter5100002
IntruderPrep	IMP_ELSE	33	Encounter1000001
IntruderPrep	THEN	36	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
IntruderPrep	IMP_ELSE	36	Encounter1000002
UpdateIntruderInputs	THEN	9	Encounter1000002
UpdateIntruderInputs	THEN	12	Encounter1000002
UpdateIntruderInputs	THEN	13	Encounter1000002
UpdateIntruderInputs	ELSEIF	15	Encounter7000002
UpdateIntruderInputs	IMP_ELSE	15	Encounter7000002
UpdateIntruderInputs	IMP_ELSE	12	Encounter1000002
UpdateIntruderInputs	ELSE	19	Encounter1000001
GetOwnHeight	THEN	5	Encounter1000001
GetOwnHeight	IMP_ELSE	5	Encounter1000001
IndividualCostEstimation	ELSE	17	Encounter1000001
IndividualCostEstimation	THEN	25	Encounter1000001
IndividualCostEstimation	THEN	26	Encounter1000011
IndividualCostEstimation	ELSE	28	Encounter1000001
IndividualCostEstimation	THEN	32	Encounter1000003
IndividualCostEstimation	IMP_ELSE	32	Encounter1000001
IndividualCostEstimation	THEN	35	Encounter1000001
IndividualCostEstimation	THEN	39	Encounter1000002
IndividualCostEstimation	THEN	44	Encounter1000002
IndividualCostEstimation	IMP_ELSE	44	Encounter1000002
IndividualCostEstimation	IMP_ELSE	39	Encounter1000001
IndividualCostEstimation	ELSEIF	50	Encounter1000010
IndividualCostEstimation	IMP_ELSE	50	Encounter1000001
IndividualCostEstimation	IMP_ELSE	25	Encounter7000003
GetTableMaxCutValue	THEN	7	Encounter1000001
GetTableMaxCutValue	IMP_ELSE	7	Encounter1000001
OfflineStatesScaleFactor	THEN	4	Encounter1000001
OfflineStatesScaleFactor	IMP_ELSE	4	Encounter1000001
OfflineStatesScaleFactor	THEN	7	Encounter1000001
OfflineStatesScaleFactor	IMP_ELSE	7	Encounter1000001
LookupOfflineCost	THEN	37	Encounter1000001
LookupOfflineCost	THEN	53	Encounter1000001
LookupOfflineCost	ELSE	55	Encounter1000001
OfflineCostEstimation	THEN	18	Encounter1000002
OfflineCostEstimation	ELSE	20	Encounter1000001
OnlineUncoordinatedActionCostEstimation	THEN	15	Encounter1000001
OnlineUncoordinatedActionCostEstimation	ELSE	45	Encounter1000001
GetModifiedGlobalRates	THEN	5	Encounter1000001
GetModifiedGlobalRates	ELSEIF	8	Encounter2100021
GetModifiedGlobalRates	THEN	10	Encounter2100021
GetModifiedGlobalRates	ELSEIF	13	Encounter2100021

Algorithm	Branch Type	Line Number	Encounter
GetModifiedGlobalRates	ELSE	20	Encounter1000001
ActionToRates	THEN	9	Encounter1000001
ActionToRates	THEN	10	Encounter1000009
ActionToRates	ELSE	12	Encounter1000001
ActionToRates	ELSE	16	Encounter1000001
MaintainRates	THEN	6	Encounter1000001
MaintainRates	THEN	8	Encounter1000001
MaintainRates	ELSE	11	Encounter1000001
MaintainRates	IMP_ELSE	8	Encounter1000001
MaintainRates	ELSE	15	Encounter1000001
MaintainRates	IMP_ELSE	17	Encounter1000001
IsIntruderMaster	THEN	11	Encounter1000003
IsIntruderMaster	IMP_ELSE	11	Encounter1000001
DetermineLDI	IMP_ELSE	15	Encounter1000001
DetermineLDI	IMP_ELSE	31	Encounter1000001
UpdateAltitudeInhibitCState	THEN	4	Encounter1000001
UpdateAltitudeInhibitCState	IMP_ELSE	4	Encounter1000001
UpdateAltitudeInhibitCState	THEN	10	Encounter1000001
UpdateAltitudeInhibitCState	ELSE	12	Encounter1000001
UpdateAltitudeInhibitCState	THEN	16	Encounter1000001
UpdateAltitudeInhibitCState	ELSEIF	19	Encounter1000001
UpdateAltitudeInhibitCState	ELSEIF	22	Encounter1000001
UpdateAltitudeInhibitCState	IMP_ELSE	22	Encounter7000003
AltitudeInhibitCost	THEN	14	Encounter1000001
AltitudeInhibitCost	IMP_ELSE	18	Encounter1000001
AltitudeInhibitCost	IMP_ELSE	14	Encounter1000001
IsCOC	THEN	4	Encounter1000001
IsCOC	IMP_ELSE	4	Encounter1000001
IsMTLO	THEN	7	Encounter2100021
IsMTLO	IMP_ELSE	7	Encounter1000001
IsStrengthening	THEN	6	Encounter1000001
IsStrengthening	THEN	7	Encounter1000003
IsStrengthening	ELSEIF	9	Encounter1000001
IsStrengthening	IMP_ELSE	9	Encounter1000001
IsStrengthening	IMP_ELSE	6	Encounter1000001
VerticalRateLimitDependentStrengthenCost	THEN	12	Encounter1000002
VerticalRateLimitDependentStrengthenCost	IMP_ELSE	12	Encounter1000001
IsMaintain	THEN	7	Encounter1000001
IsMaintain	IMP_ELSE	7	Encounter1000001
IsCrossing	THEN	5	Encounter1000001
IsCrossing	THEN	7	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
IsCrossing	IMP_ELSE	7	Encounter1000001
IsCrossing	IMP_ELSE	5	Encounter1000001
PointObsDescendCost	THEN	6	Encounter1000001
PointObsDescendCost	IMP_ELSE	6	Encounter1000001
PointObsPreventEarlyCOCCost	THEN	15	Encounter1000001
PointObsPreventEarlyCOCCost	THEN	19	Encounter1000001
PointObsPreventEarlyCOCCost	ELSE	21	Encounter1000001
PointObsPreventEarlyCOCCost	THEN	24	Encounter1000001
PointObsPreventEarlyCOCCost	ELSEIF	26	Encounter1000001
PointObsPreventEarlyCOCCost	IMP_ELSE	26	Encounter1000001
PointObsPreventEarlyCOCCost	THEN	29	Encounter1000001
PointObsPreventEarlyCOCCost	IMP_ELSE	29	Encounter1000001
PointObsPreventEarlyCOCCost	IMP_ELSE	15	Encounter1000001
PointObstacleTauEstimation	THEN	8	Encounter4260000
PointObstacleTauEstimation	ELSEIF	10	Encounter5100108
PointObstacleTauEstimation	ELSEIF	12	Encounter5100108
PointObstacleTauEstimation	ELSEIF	14	Encounter5200100
PointObstacleTauEstimation	IMP_ELSE	14	Encounter5100108
GroundPointObstacleTauEstimation	THEN	7	Encounter1000006
GroundPointObstacleTauEstimation	ELSEIF	9	Encounter1000001
GroundPointObstacleTauEstimation	ELSE	11	Encounter1000001
GroundPointObstacleTauEstimation	THEN	13	Encounter1000001
GroundPointObstacleTauEstimation	ELSEIF	15	Encounter1000005
GroundPointObstacleTauEstimation	ELSEIF	17	Encounter1000001
GroundPointObstacleTauEstimation	IMP_ELSE	17	Encounter1000001
IsProjectedCrossing	THEN	5	Encounter1000001
IsProjectedCrossing	IMP_ELSE	5	Encounter1000001
UpdateAdvisoryRestartCState	THEN	3	Encounter1000002
UpdateAdvisoryRestartCState	ELSE	6	Encounter1000001
AdvisoryRestartCost	THEN	6	Encounter1000001
AdvisoryRestartCost	IMP_ELSE	6	Encounter1000001
AdvisoryRestartCost	THEN	10	Encounter1000002
AdvisoryRestartCost	IMP_ELSE	10	Encounter1000001
InitializationCost	THEN	4	Encounter1000001
InitializationCost	IMP_ELSE	4	Encounter1000001
InitializationCost	THEN	8	Encounter1000001
InitializationCost	IMP_ELSE	8	Encounter1000001
RestrictCOCDueToReversal	THEN	7	Encounter1000001
RestrictCOCDueToReversal	IMP_ELSE	7	Encounter1000001
RestrictCOCDueToReversal	THEN	11	Encounter1000001
RestrictCOCDueToReversal	THEN	13	Encounter2100003

Algorithm	Branch Type	Line Number	Encounter
RestrictCOCDueToReversal	IMP_ELSE	13	Encounter1000001
RestrictCOCDueToReversal	IMP_ELSE	11	Encounter1000001
UpdateMaxReversalCState	THEN	8	Encounter1000010
UpdateMaxReversalCState	THEN	10	Encounter2100011
UpdateMaxReversalCState	IMP_ELSE	10	Encounter1000010
UpdateMaxReversalCState	ELSEIF	13	Encounter1000001
UpdateMaxReversalCState	IMP_ELSE	13	Encounter1000002
UpdateMaxReversalCState	ELSE	18	Encounter1000001
MaxReversalCost	THEN	10	Encounter1000001
MaxReversalCost	IMP_ELSE	10	Encounter1000001
MaxReversalCost	THEN	15	Encounter1000001
MaxReversalCost	IMP_ELSE	16	Encounter1000001
MaxReversalCost	IMP_ELSE	20	Encounter1000001
UpdateCompatibilityCState	THEN	14	Encounter1000001
UpdateCompatibilityCState	ELSE	16	Encounter1000002
CompatibilityCost	THEN	17	Encounter1000001
CompatibilityCost	IMP_ELSE	17	Encounter1000001
CompatibilityCost	THEN	24	Encounter1000002
CompatibilityCost	THEN	25	Encounter1000003
CompatibilityCost	ELSEIF	27	Encounter1000002
CompatibilityCost	THEN	28	Encounter7000002
CompatibilityCost	ELSEIF	31	Encounter1000002
CompatibilityCost	ELSE	33	Encounter1000002
CompatibilityCost	ELSEIF	36	Encounter1000002
CompatibilityCost	THEN	37	Encounter5200201
CompatibilityCost	ELSEIF	39	Encounter7000051
CompatibilityCost	ELSE	46	Encounter1000002
CompatibilityCost	IMP_ELSE	36	Encounter2100002
CompatibilityCost	IMP_ELSE	24	Encounter1000001
DetermineResponse	THEN	12	Encounter1000002
DetermineResponse	IMP_ELSE	12	Encounter1000001
DetermineResponse	THEN	15	Encounter1000001
DetermineResponse	ELSE	17	Encounter1000002
DetermineResponse	THEN	20	Encounter1000002
DetermineResponse	IMP_ELSE	20	Encounter1000002
DetermineResponse	THEN	25	Encounter7000044
DetermineResponse	ELSE	27	Encounter1000002
DetermineResponse	THEN	30	Encounter1000002
DetermineResponse	IMP_ELSE	30	Encounter1000003
DetermineResponse	THEN	35	Encounter1000002
DetermineResponse	IMP_ELSE	35	Encounter1000003

Algorithm	Branch Type	Line Number	Encounter
DetermineResponse	THEN	38	Encounter1000002
DetermineResponse	ELSEIF	40	Encounter5100107
DetermineResponse	ELSE	42	Encounter7000044
IntruderResponseEstimation	THEN	9	Encounter1000001
IntruderResponseEstimation	ELSE	11	Encounter7000000
IntruderResponseEstimation	THEN	15	Encounter1000001
IntruderResponseEstimation	ELSEIF	23	Encounter1000002
IntruderResponseEstimation	THEN	25	Encounter1000002
IntruderResponseEstimation	IMP_ELSE	25	Encounter1000002
IntruderResponseEstimation	ELSE	29	Encounter1000002
IntruderResponseEstimation	THEN	35	Encounter1000001
IntruderResponseEstimation	ELSEIF	37	Encounter2200013
IntruderResponseEstimation	IMP_ELSE	37	Encounter2200013
OwnResponseEstimation	THEN	7	Encounter1000001
OwnResponseEstimation	ELSEIF	15	Encounter1000002
OwnResponseEstimation	THEN	17	Encounter1000002
OwnResponseEstimation	IMP_ELSE	17	Encounter1000002
OwnResponseEstimation	ELSE	21	Encounter1000002
OwnResponseEstimation	THEN	27	Encounter1000001
OwnResponseEstimation	ELSEIF	29	Encounter1000002
OwnResponseEstimation	IMP_ELSE	29	Encounter1000002
UpdateSA01HeuristicCState	IMP_ELSE	14	Encounter1000001
SA01Heuristic	THEN	8	Encounter1000001
SA01Heuristic	IMP_ELSE	8	Encounter1000001
SA01Heuristic	IMP_ELSE	16	Encounter1000001
UpdateBadTransitionCState	THEN	5	Encounter1000002
UpdateBadTransitionCState	ELSE	8	Encounter1000001
UpdateBadTransitionCState	THEN	12	Encounter1000001
UpdateBadTransitionCState	THEN	13	Encounter1000001
UpdateBadTransitionCState	ELSE	16	Encounter1000001
UpdateBadTransitionCState	IMP_ELSE	12	Encounter1000001
BadTransitionCost	THEN	10	Encounter1000001
BadTransitionCost	IMP_ELSE	10	Encounter1000001
BadTransitionCost	THEN	20	Encounter1000001
BadTransitionCost	IMP_ELSE	20	Encounter1000001
BadTransitionCost	THEN	25	Encounter1000001
BadTransitionCost	ELSEIF	29	Encounter7000001
BadTransitionCost	ELSEIF	33	Encounter1000004
BadTransitionCost	ELSEIF	37	Encounter7000003
BadTransitionCost	IMP_ELSE	37	Encounter1000001
BadMaintainTransitionCost	THEN	10	Encounter1000009

Algorithm	Branch Type	Line Number	Encounter
BadMaintainTransitionCost	THEN	13	Encounter1000009
BadMaintainTransitionCost	ELSEIF	17	Encounter1000010
BadMaintainTransitionCost	ELSEIF	20	Encounter7000001
BadMaintainTransitionCost	IMP_ELSE	23	Encounter1000009
BadMaintainTransitionCost	ELSEIF	26	Encounter1000001
BadMaintainTransitionCost	THEN	29	Encounter1000002
BadMaintainTransitionCost	ELSEIF	33	Encounter5100107
BadMaintainTransitionCost	IMP_ELSE	33	Encounter1000001
BadMaintainTransitionCost	THEN	36	Encounter1000001
BadMaintainTransitionCost	IMP_ELSE	36	Encounter1000001
BadMaintainTransitionCost	IMP_ELSE	26	Encounter1000001
UpdateCrossingNoAlertCState	ELSE	8	Encounter1000001
CrossingNoAlertCost	THEN	9	Encounter1000001
CrossingNoAlertCost	IMP_ELSE	9	Encounter1000001
CrossingNoAlertCost	THEN	16	Encounter1000002
CrossingNoAlertCost	IMP_ELSE	16	Encounter1000001
UpdatePreventEarlyCOCCState	THEN	20	Encounter1000001
UpdatePreventEarlyCOCCState	ELSE	23	Encounter1000002
UpdatePreventEarlyCOCCState	THEN	24	Encounter1000002
UpdatePreventEarlyCOCCState	ELSE	27	Encounter1000002
UpdatePreventEarlyCOCCState	THEN	29	Encounter1000002
UpdatePreventEarlyCOCCState	ELSEIF	31	Encounter2200001
UpdatePreventEarlyCOCCState	IMP_ELSE	31	Encounter1000002
UpdatePreventEarlyCOCCState	THEN	37	Encounter1000002
UpdatePreventEarlyCOCCState	THEN	41	Encounter1000009
UpdatePreventEarlyCOCCState	IMP_ELSE	41	Encounter1000002
UpdatePreventEarlyCOCCState	THEN	49	Encounter7000008
UpdatePreventEarlyCOCCState	IMP_ELSE	49	Encounter1000002
UpdatePreventEarlyCOCCState	IMP_ELSE	37	Encounter1000001
PreventEarlyCOCCost	THEN	7	Encounter1000001
PreventEarlyCOCCost	IMP_ELSE	7	Encounter1000001
PreventEarlyCOCCost	THEN	12	Encounter1000002
PreventEarlyCOCCost	IMP_ELSE	12	Encounter1000001
ExpectedTau	THEN	6	Encounter1000001
ExpectedTau	IMP_ELSE	6	Encounter1000001
ExpectedTau	THEN	14	Encounter1000001
ExpectedTau	IMP_ELSE	14	Encounter1000001
AltitudeDependentCOCCost	THEN	11	Encounter1000001
AltitudeDependentCOCCost	ELSEIF	13	Encounter1000001
AltitudeDependentCOCCost	THEN	14	Encounter1000001
AltitudeDependentCOCCost	ELSE	16	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
AltitudeDependentCOCCost	ELSEIF	19	Encounter1000001
AltitudeDependentCOCCost	IMP_ELSE	19	Encounter1000001
DetermineVerticalRateReductionFactor	THEN	18	Encounter7000009
DetermineVerticalRateReductionFactor	IMP_ELSE	18	Encounter7000000
DetermineVerticalProximateFactor	THEN	15	Encounter7000009
DetermineVerticalProximateFactor	ELSEIF	17	Encounter7000000
DetermineVerticalProximateFactor	ELSE	19	Encounter7000000
DetermineVerticalProximateFactor	THEN	25	Encounter7000009
DetermineVerticalProximateFactor	IMP_ELSE	25	Encounter7000000
DetermineVerticalProximateFactor	THEN	29	Encounter7000010
DetermineVerticalProximateFactor	IMP_ELSE	29	Encounter7000000
DetermineAltitudeDependentCOCFactor	THEN	21	Encounter7000000
DetermineAltitudeDependentCOCFactor	THEN	32	Encounter7000000
DetermineAltitudeDependentCOCFactor	IMP_ELSE	32	Encounter7000000
DetermineAltitudeDependentCOCFactor	IMP_ELSE	21	Encounter1000001
PreventEarlyWeakeningCost	THEN	12	Encounter1000002
PreventEarlyWeakeningCost	THEN	16	Encounter2100012
PreventEarlyWeakeningCost	IMP_ELSE	16	Encounter1000002
PreventEarlyWeakeningCost	IMP_ELSE	12	Encounter1000001
CalculateThresholdRampDownFactor	THEN	4	Encounter1000001
CalculateThresholdRampDownFactor	ELSEIF	6	Encounter1000001
CalculateThresholdRampDownFactor	IMP_ELSE	6	Encounter1000001
CalculateThresholdRampUpFactor	THEN	4	Encounter1000001
CalculateThresholdRampUpFactor	ELSEIF	6	Encounter1000001
CalculateThresholdRampUpFactor	IMP_ELSE	6	Encounter1000002
UpdateCoordinatedRADeferralCState	THEN	17	Encounter1000002
UpdateCoordinatedRADeferralCState	THEN	21	Encounter2100022
UpdateCoordinatedRADeferralCState	IMP_ELSE	21	Encounter1000002
UpdateCoordinatedRADeferralCState	IMP_ELSE	17	Encounter1000001
CoordinatedRADeferralCost	THEN	9	Encounter1000001
CoordinatedRADeferralCost	IMP_ELSE	9	Encounter1000001
CoordinatedRADeferralCost	THEN	16	Encounter1000001
CoordinatedRADeferralCost	IMP_ELSE	16	Encounter1000001
UpdateSafeCrossingRADeferralCState	THEN	21	Encounter1000002
UpdateSafeCrossingRADeferralCState	ELSE	23	Encounter1000001
UpdateSafeCrossingRADeferralCState	THEN	30	Encounter1000001
UpdateSafeCrossingRADeferralCState	IMP_ELSE	30	Encounter1000001
UpdateSafeCrossingRADeferralCState	THEN	38	Encounter1000001
UpdateSafeCrossingRADeferralCState	ELSE	40	Encounter1000001
SafeCrossingRADeferralCost	THEN	6	Encounter1000001
SafeCrossingRADeferralCost	IMP_ELSE	6	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
SafeCrossingRADeferralCost	THEN	12	Encounter1000001
SafeCrossingRADeferralCost	IMP_ELSE	12	Encounter1000001
UpdateCoordinationDelayCState	THEN	5	Encounter7000053
UpdateCoordinationDelayCState	THEN	8	Encounter7000053
UpdateCoordinationDelayCState	ELSE	10	Encounter7000053
UpdateCoordinationDelayCState	ELSE	13	Encounter1000001
CoordinationDelayCost	THEN	5	Encounter1000001
CoordinationDelayCost	IMP_ELSE	5	Encounter1000001
CoordinationDelayCost	THEN	9	Encounter7000053
CoordinationDelayCost	IMP_ELSE	9	Encounter1000001
IsOutsideCriticalInterval	THEN	20	Encounter1000001
IsOutsideCriticalInterval	IMP_ELSE	20	Encounter7000044
CriticalIntervalRequiresVerticalDivergence	THEN	14	Encounter1000002
CriticalIntervalRequiresVerticalDivergence	IMP_ELSE	14	Encounter1000001
UpdateCriticalIntervalProtectionCState	THEN	24	Encounter1000001
UpdateCriticalIntervalProtectionCState	ELSE	26	Encounter7000044
UpdateCriticalIntervalProtectionCState	THEN	34	Encounter7000044
UpdateCriticalIntervalProtectionCState	THEN	40	Encounter7000044
UpdateCriticalIntervalProtectionCState	IMP_ELSE	40	Encounter7000044
UpdateCriticalIntervalProtectionCState	THEN	49	Encounter7000045
UpdateCriticalIntervalProtectionCState	IMP_ELSE	49	Encounter7000044
UpdateCriticalIntervalProtectionCState	THEN	53	Encounter1000002
UpdateCriticalIntervalProtectionCState	IMP_ELSE	53	Encounter1000001
CriticalIntervalProtectionCost	THEN	13	Encounter1000001
CriticalIntervalProtectionCost	IMP_ELSE	13	Encounter1000001
CriticalIntervalProtectionCost	THEN	20	Encounter7000045
CriticalIntervalProtectionCost	THEN	21	Encounter7000045
CriticalIntervalProtectionCost	IMP_ELSE	23	Encounter7000045
CriticalIntervalProtectionCost	ELSEIF	26	Encounter1000002
CriticalIntervalProtectionCost	THEN	27	Encounter1000002
CriticalIntervalProtectionCost	ELSEIF	29	Encounter2100002
CriticalIntervalProtectionCost	IMP_ELSE	29	Encounter1000002
CriticalIntervalProtectionCost	ELSEIF	32	Encounter7000045
CriticalIntervalProtectionCost	IMP_ELSE	32	Encounter1000001
DetermineOwnNonComplianceFactor	IMP_ELSE	20	Encounter1000001
UpdateTimeBasedNonComplianceCState	THEN	25	Encounter1000001
UpdateTimeBasedNonComplianceCState	ELSE	29	Encounter1000001
UpdateTimeBasedNonComplianceCState	THEN	35	Encounter1000001
UpdateTimeBasedNonComplianceCState	ELSE	39	Encounter1000002
UpdateTimeBasedNonComplianceCState	THEN	40	Encounter1000009
UpdateTimeBasedNonComplianceCState	IMP_ELSE	40	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
UpdateTimeBasedNonComplianceCState	THEN	43	Encounter1000002
UpdateTimeBasedNonComplianceCState	IMP_ELSE	43	Encounter1000002
UpdateTimeBasedNonComplianceCState	THEN	47	Encounter1000001
UpdateTimeBasedNonComplianceCState	IMP_ELSE	47	Encounter1000002
UpdateTimeBasedNonComplianceCState	THEN	53	Encounter7000044
UpdateTimeBasedNonComplianceCState	IMP_ELSE	53	Encounter1000001
UpdateTimeBasedNonComplianceCState	THEN	57	Encounter1000001
UpdateTimeBasedNonComplianceCState	IMP_ELSE	57	Encounter1000002
TimeBasedNonComplianceCost	THEN	11	Encounter1000001
TimeBasedNonComplianceCost	IMP_ELSE	11	Encounter1000001
TimeBasedNonComplianceCost	THEN	17	Encounter1000002
TimeBasedNonComplianceCost	THEN	18	Encounter1000002
TimeBasedNonComplianceCost	ELSEIF	20	Encounter1000002
TimeBasedNonComplianceCost	IMP_ELSE	20	Encounter1000002
TimeBasedNonComplianceCost	IMP_ELSE	17	Encounter1000001
FindCompliantRateDelta	THEN	4	Encounter1000002
FindCompliantRateDelta	ELSEIF	6	Encounter1000003
FindCompliantRateDelta	IMP_ELSE	6	Encounter1000001
DetermineHorizontalWorstCasePhiSpreadFactor	THEN	14	Encounter1000002
DetermineHorizontalWorstCasePhiSpreadFactor	THEN	15	Encounter1000002
DetermineHorizontalWorstCasePhiSpreadFactor	ELSE	17	Encounter1000002
DetermineHorizontalWorstCasePhiSpreadFactor	IMP_ELSE	14	Encounter1000001
ActionSelection	THEN	16	Encounter1000001
ActionSelection	IMP_ELSE	16	Encounter1000004
ActionSelection	THEN	20	Encounter1000001
ActionSelection	ELSE	25	Encounter1000001
AllowUnequippedMTLO	THEN	17	Encounter1000001
AllowUnequippedMTLO	THEN	20	Encounter1000002
AllowUnequippedMTLO	ELSEIF	22	Encounter1000009
AllowUnequippedMTLO	THEN	23	Encounter1000009
AllowUnequippedMTLO	ELSEIF	25	Encounter1000010
AllowUnequippedMTLO	IMP_ELSE	25	Encounter1000009
AllowUnequippedMTLO	IMP_ELSE	22	Encounter1000001
AllowUnequippedMTLO	IMP_ELSE	17	Encounter1000001
AllowUnequippedMTLO	THEN	31	Encounter1000002
AllowUnequippedMTLO	IMP_ELSE	31	Encounter1000001
DetermineMinimumCostAction	THEN	12	Encounter1000001
DetermineMinimumCostAction	IMP_ELSE	12	Encounter1000004
DetermineMultiIntruderAction	THEN	11	Encounter1000001
DetermineMultiIntruderAction	IMP_ELSE	11	Encounter1000002
DetermineMultiIntruderAction	THEN	22	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
DetermineMultiIntruderAction	THEN	33	Encounter1000001
DetermineMultiIntruderAction	IMP_ELSE	38	Encounter1000001
DetermineMultiIntruderAction	ELSE	42	Encounter2100021
DetermineMultiIntruderAction	THEN	46	Encounter2100021
DetermineMultiIntruderAction	ELSE	55	Encounter2100021
DetermineMultiIntruderAction	IMP_ELSE	46	Encounter1000001
DetermineSenses	THEN	11	Encounter1000009
DetermineSenses	THEN	12	Encounter1000009
DetermineSenses	ELSE	15	Encounter1000009
DetermineSenses	ELSE	18	Encounter1000001
DetermineConflictingSenses	THEN	6	Encounter1000001
DetermineConflictingSenses	THEN	7	Encounter1000001
DetermineConflictingSenses	ELSEIF	9	Encounter1000003
DetermineConflictingSenses	IMP_ELSE	9	Encounter1000001
DetermineConflictingSenses	IMP_ELSE	6	Encounter3211001
MultithreatCostBalancing	THEN	24	Encounter1000002
MultithreatCostBalancing	THEN	29	Encounter1000010
MultithreatCostBalancing	THEN	33	Encounter1000010
MultithreatCostBalancing	IMP_ELSE	33	Encounter1000010
MultithreatCostBalancing	THEN	37	Encounter1000010
MultithreatCostBalancing	IMP_ELSE	37	Encounter2100021
MultithreatCostBalancing	IMP_ELSE	29	Encounter1000002
MultithreatCostBalancing	IMP_ELSE	24	Encounter1000001
MultithreatCostBalancing	THEN	44	Encounter1000010
MultithreatCostBalancing	ELSEIF	48	Encounter1000010
MultithreatCostBalancing	ELSEIF	52	Encounter1000010
MultithreatCostBalancing	THEN	53	Encounter1000010
MultithreatCostBalancing	ELSEIF	55	Encounter2100021
MultithreatCostBalancing	ELSE	57	Encounter2100021
MultithreatCostBalancing	ELSE	62	Encounter1000001
SandwichPreventionCost	THEN	9	Encounter1000002
SandwichPreventionCost	IMP_ELSE	9	Encounter1000001
SandwichPreventionCost	THEN	17	Encounter1000001
SandwichPreventionCost	THEN	19	Encounter1000001
SandwichPreventionCost	ELSEIF	21	Encounter1000001
SandwichPreventionCost	IMP_ELSE	21	Encounter1000001
SandwichPreventionCost	IMP_ELSE	17	Encounter1000001
SandwichPreventionCost	THEN	26	Encounter1000001
SandwichPreventionCost	THEN	31	Encounter1000001
SandwichPreventionCost	IMP_ELSE	31	Encounter1000001
SandwichPreventionCost	IMP_ELSE	26	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
UnequippedCostFusion	THEN	13	Encounter1000001
UnequippedCostFusion	IMP_ELSE	13	Encounter1000001
UnequippedCostFusion	THEN	19	Encounter1000010
UnequippedCostFusion	IMP_ELSE	19	Encounter1000001
MTLODetermination	IMP_ELSE	11	Encounter1000001
MTLODetermination	THEN	17	Encounter1000001
IndividualSelectionCostEstimation	THEN	19	Encounter1000001
IndividualSelectionCostEstimation	ELSE	21	Encounter1000002
IndividualSelectionCostEstimation	THEN	24	Encounter1000001
IndividualSelectionCostEstimation	IMP_ELSE	24	Encounter1000002
IndividualSelectionCostEstimation	THEN	33	Encounter1000001
IndividualSelectionCostEstimation	THEN	35	Encounter1000001
IndividualSelectionCostEstimation	IMP_ELSE	35	Encounter1000002
IndividualSelectionCostEstimation	IMP_ELSE	33	Encounter1000001
ArbitrateConflictingSenses	THEN	14	Encounter1000010
ArbitrateConflictingSenses	THEN	15	Encounter1000010
ArbitrateConflictingSenses	IMP_ELSE	15	Encounter1000010
ArbitrateConflictingSenses	THEN	20	Encounter1000010
ArbitrateConflictingSenses	IMP_ELSE	20	Encounter1000010
ArbitrateConflictingSenses	IMP_ELSE	14	Encounter7000054
ArbitrateConflictingSenses	THEN	26	Encounter1000010
ArbitrateConflictingSenses	IMP_ELSE	26	Encounter1000010
ArbitrateConflictingSenses	THEN	29	Encounter1000010
ArbitrateConflictingSenses	THEN	31	Encounter1000010
ArbitrateConflictingSenses	IMP_ELSE	31	Encounter1000010
ArbitrateConflictingSenses	IMP_ELSE	29	Encounter1000010
ArbitrateConflictingSenses	THEN	36	Encounter1000010
ArbitrateMatchingSenses	THEN	6	Encounter1000001
ArbitrateMatchingSenses	IMP_ELSE	6	Encounter1000001
ArbitrateMatchingSenses	THEN	11	Encounter1000002
ArbitrateMatchingSenses	THEN	18	Encounter1000002
ArbitrateMatchingSenses	IMP_ELSE	18	Encounter1000002
ArbitrateMatchingSenses	THEN	24	Encounter1000002
ArbitrateMatchingSenses	IMP_ELSE	24	Encounter1000002
ArbitrateMatchingSenses	IMP_ELSE	11	Encounter1000001
ActionArbitration	THEN	7	Encounter1000010
ActionArbitration	ELSE	9	Encounter1000001
PersistMTLO	THEN	6	Encounter2100021
PersistMTLO	IMP_ELSE	12	Encounter2100021
PersistMTLO	IMP_ELSE	6	Encounter1000001
IsCorrective	THEN	5	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
IsCorrective	IMP_ELSE	5	Encounter1000002
IsDiverging	THEN	5	Encounter2100022
IsDiverging	IMP_ELSE	5	Encounter1000002
IsPreventive	THEN	5	Encounter1000001
IsPreventive	IMP_ELSE	5	Encounter1000001
IsDND	THEN	4	Encounter1000002
IsDND	IMP_ELSE	4	Encounter1000002
IsDNC	THEN	4	Encounter1000002
IsDNC	IMP_ELSE	4	Encounter1000002
IsReversal	THEN	4	Encounter1000001
IsReversal	IMP_ELSE	4	Encounter1000001
IsMasterForcingReversal	THEN	4	Encounter1000003
IsMasterForcingReversal	IMP_ELSE	4	Encounter1000001
IsWithinRateBounds	THEN	6	Encounter1000002
IsWithinRateBounds	ELSEIF	8	Encounter1000002
IsWithinRateBounds	IMP_ELSE	8	Encounter1000002
VRCToSense	THEN	4	Encounter1000001
VRCToSense	ELSEIF	6	Encounter1000002
VRCToSense	ELSE	8	Encounter1000002
RatesToAction	ELSEIF	12	Encounter2100021
RatesToAction	IMP_ELSE	12	Encounter2100021
RatesToAction	THEN	20	Encounter2100021
RatesToSense	THEN	4	Encounter1000001
RatesToSense	ELSEIF	6	Encounter1000001
RatesToSense	ELSE	8	Encounter1000001
CoordinationSelection	THEN	8	Encounter1000001
CoordinationSelection	ELSEIF	10	Encounter1000009
CoordinationSelection	ELSEIF	14	Encounter1000009
CoordinationSelection	ELSE	18	Encounter1000001
CoordinationSelection	THEN	24	Encounter1000001
CoordinationSelection	ELSE	26	Encounter1000001
EncodeVRC	THEN	7	Encounter1000001
EncodeVRC	ELSEIF	9	Encounter1000002
EncodeVRC	ELSEIF	11	Encounter1000003
EncodeVRC	ELSE	13	Encounter1000002
EncodeCVC	THEN	7	Encounter1000001
EncodeCVC	ELSEIF	9	Encounter1000002
EncodeCVC	ELSEIF	11	Encounter2100001
EncodeCVC	ELSEIF	13	Encounter1000002
EncodeCVC	ELSEIF	15	Encounter1000002
EncodeCVC	ELSEIF	17	Encounter2100003

Algorithm	Branch Type	Line Number	Encounter
EncodeCVC	ELSEIF	19	Encounter2100012
EncodeCVC	ELSE	21	Encounter1000002
DetermineCode	THEN	4	Encounter1000001
DetermineCode	ELSE	6	Encounter1000001
DetermineCrossing	THEN	12	Encounter1000002
DetermineCrossing	IMP_ELSE	12	Encounter1000001
DetermineCrossing	THEN	17	Encounter1000001
DetermineCrossing	THEN	18	Encounter1000001
DetermineCrossing	THEN	19	Encounter4110060
DetermineCrossing	ELSE	21	Encounter1000001
DetermineCrossing	ELSE	24	Encounter1000002
DetermineCrossing	ELSE	27	Encounter1000002
DetermineCrossing	THEN	31	Encounter1000002
DetermineCrossing	ELSEIF	33	Encounter6100000
DetermineCrossing	IMP_ELSE	33	Encounter1000001
DetermineCrossing	IMP_ELSE	17	Encounter1000001
DetermineGroundAlert	THEN	7	Encounter1000001
DetermineGroundAlert	THEN	9	Encounter1000001
DetermineGroundAlert	IMP_ELSE	9	Encounter1000002
DetermineGroundAlert	IMP_ELSE	7	Encounter1000001
DetermineGroundAlert	THEN	14	Encounter1000001
DetermineGroundAlert	THEN	15	Encounter1000001
DetermineGroundAlert	ELSE	17	Encounter1000010
DetermineGroundAlert	IMP_ELSE	14	Encounter1000001
DetermineDisplayData	THEN	7	Encounter1000001
DetermineDisplayData	IMP_ELSE	7	Encounter1000001
DetermineDisplayData	THEN	15	Encounter1000001
DetermineDisplayData	IMP_ELSE	15	Encounter1000001
DetermineDisplayData	THEN	19	Encounter1000001
DetermineDisplayData	ELSEIF	21	Encounter1000003
DetermineDisplayData	IMP_ELSE	21	Encounter1000001
GetMatchingLabel270Rule	THEN	32	Encounter1000001
GetMatchingLabel270Rule	THEN	37	Encounter1000001
GetMatchingLabel270Rule	ELSEIF	39	Encounter1000003
GetMatchingLabel270Rule	ELSEIF	41	Encounter1000003
GetMatchingLabel270Rule	ELSE	43	Encounter1000001
GetMatchingLabel270Rule	IMP_ELSE	32	Encounter1000001
Consistent	THEN	14	Encounter1000001
Consistent	IMP_ELSE	14	Encounter1000001
GenerateVTRMOutput	THEN	9	Encounter5100001
GenerateVTRMOutput	ELSEIF	13	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
GenerateVTRMOutput	ELSE	18	Encounter1000001
GenerateVTRMOutput	THEN	26	Encounter1000008
GenerateVTRMOutput	IMP_ELSE	26	Encounter1000001
TRMIntruderStateUpdate	THEN	12	Encounter1000001
TRMIntruderStateUpdate	THEN	17	Encounter1000004
TRMIntruderStateUpdate	ELSE	23	Encounter1000001
TRMIntruderStateUpdate	IMP_ELSE	12	Encounter1000001
UpdateIndivAdjustThreatInfo	THEN	5	Encounter1000001
UpdateIndivAdjustThreatInfo	THEN	7	Encounter1000001
UpdateIndivAdjustThreatInfo	ELSE	9	Encounter1000003
UpdateIndivAdjustThreatInfo	THEN	12	Encounter1000001
UpdateIndivAdjustThreatInfo	ELSEIF	15	Encounter1000001
UpdateIndivAdjustThreatInfo	ELSE	17	Encounter1000010
UpdateIndivAdjustThreatInfo	ELSE	20	Encounter1000001
UpdateIndivAdjustCounts	THEN	5	Encounter1000004
UpdateIndivAdjustCounts	IMP_ELSE	5	Encounter1000001
UpdateIndivAdjustCounts	THEN	8	Encounter1000001
UpdateIndivAdjustCounts	THEN	9	Encounter1000001
UpdateIndivAdjustCounts	THEN	11	Encounter4310001
UpdateIndivAdjustCounts	IMP_ELSE	11	Encounter1000001
UpdateIndivAdjustCounts	IMP_ELSE	9	Encounter1000001
UpdateIndivAdjustCounts	IMP_ELSE	8	Encounter1000001
GenerateRAModeOutput	THEN	8	Encounter1000001
GenerateRAModeOutput	ELSEIF	10	Encounter1000001
GenerateRAModeOutput	ELSE	12	Encounter1000001
GenerateRAModeOutput	THEN	17	Encounter4310001
GenerateRAModeOutput	IMP_ELSE	17	Encounter1000001
GenerateRAModeOutput	THEN	26	Encounter1000001
GenerateRAModeOutput	IMP_ELSE	26	Encounter1000001
DroppedIntrudersAdjustment	THEN	18	Encounter1000001
DroppedIntrudersAdjustment	THEN	20	Encounter1000008
DroppedIntrudersAdjustment	ELSE	23	Encounter1000001
DroppedIntrudersAdjustment	ELSE	29	Encounter1000001
DetermineVerticalRAData	THEN	8	Encounter1000001
DetermineVerticalRAData	ELSEIF	17	Encounter1000001
DetermineVerticalRAData	ELSE	20	Encounter1000001
SetRABroadcastData	THEN	15	Encounter1000001
SetRABroadcastData	THEN	18	Encounter1000001
SetRABroadcastData	IMP_ELSE	18	Encounter1000001
SetRABroadcastData	ELSE	21	Encounter1000001
SetRABroadcastData	THEN	25	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
SetRABroadcastData	IMP_ELSE	25	Encounter1000001
EncodeTIDAltitude	THEN	5	Encounter4140010
EncodeTIDAltitude	ELSE	11	Encounter3110000
EncodeTIDAltitude	IMP_ELSE	15	Encounter3110000
EncodeTIDRange	ELSEIF	8	Encounter3110000
EncodeTIDRange	ELSE	12	Encounter3110000
EncodeTIDRange	IMP_ELSE	16	Encounter3110000
EncodeTIDBearing	THEN	4	Encounter3110000
EncodeTIDBearing	THEN	6	Encounter3130000
EncodeTIDBearing	IMP_ELSE	6	Encounter3110000
EncodeTIDBearing	IMP_ELSE	10	Encounter3110000
DetermineTTIAndTID	THEN	6	Encounter1000002
DetermineTTIAndTID	THEN	7	Encounter1000002
DetermineTTIAndTID	ELSEIF	11	Encounter3212001
DetermineTTIAndTID	ELSE	15	Encounter3110000
HorizontalRateToAction	THEN	9	Encounter1000001
HorizontalRateToAction	IMP_ELSE	9	Encounter1000002
HorizontalRateToAdvisory	THEN	5	Encounter1000002
HorizontalRateToAdvisory	ELSEIF	7	Encounter2100003
HorizontalRateToAdvisory	IMP_ELSE	7	Encounter1000002
HorizontalRateToSense	THEN	5	Encounter1000001
HorizontalRateToSense	ELSEIF	7	Encounter1000001
HorizontalRateToSense	ELSEIF	9	Encounter1000001
HorizontalRateToSense	ELSE	11	Encounter1000001
HRCToAdvisory	THEN	4	Encounter1000001
HRCToAdvisory	ELSEIF	6	Encounter1000002
HRCToAdvisory	ELSEIF	8	Encounter2100003
HRCToAdvisory	ELSEIF	10	Encounter1000002
HRCToAdvisory	ELSEIF	12	Encounter2100003
IsHorizontalCOC	THEN	4	Encounter1000001
IsHorizontalCOC	IMP_ELSE	4	Encounter1000001
IsHorizontalReversal	THEN	5	Encounter1000002
IsHorizontalReversal	IMP_ELSE	5	Encounter1000001
IsHorizontalWeakening	THEN	5	Encounter1000002
IsHorizontalWeakening	THEN	6	Encounter1000002
IsHorizontalWeakening	ELSE	8	Encounter1000002
IsHorizontalWeakening	IMP_ELSE	11	Encounter1000002
IsHorizontalWeakening	IMP_ELSE	5	Encounter1000001
GetPerformanceBasedParams	THEN	15	Encounter1000001
GetPerformanceBasedParams	IMP_ELSE	15	Encounter1000001
HorizontalInitialCACost	THEN	6	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
HorizontalInitialCACost	IMP_ELSE	6	Encounter1000001
HorizontalReversalCost	THEN	7	Encounter1000002
HorizontalReversalCost	IMP_ELSE	7	Encounter1000001
HorizontalDesensitivityCost	THEN	5	Encounter1000001
HorizontalDesensitivityCost	ELSEIF	7	Encounter1000002
HorizontalDesensitivityCost	IMP_ELSE	7	Encounter1000001
HorizontalTurnRateSensitivityCostFactor	THEN	5	Encounter2100011
HorizontalTurnRateSensitivityCostFactor	THEN	6	Encounter2100011
HorizontalTurnRateSensitivityCostFactor	IMP_ELSE	6	Encounter3110000
HorizontalTurnRateSensitivityCostFactor	THEN	9	Encounter2100011
HorizontalTurnRateSensitivityCostFactor	IMP_ELSE	9	Encounter2100011
HorizontalTurnRateSensitivityCostFactor	IMP_ELSE	5	Encounter1000001
HorizontalWeakeningCostFactor	THEN	5	Encounter1000002
HorizontalWeakeningCostFactor	IMP_ELSE	5	Encounter1000001
HorizontalCoordinationCost	THEN	6	Encounter1000001
HorizontalCoordinationCost	THEN	8	Encounter1000002
HorizontalCoordinationCost	THEN	9	Encounter1000002
HorizontalCoordinationCost	THEN	11	Encounter2100012
HorizontalCoordinationCost	IMP_ELSE	11	Encounter1000002
HorizontalCoordinationCost	THEN	14	Encounter2100003
HorizontalCoordinationCost	IMP_ELSE	14	Encounter1000002
HorizontalCoordinationCost	IMP_ELSE	9	Encounter1000002
HorizontalCoordinationCost	IMP_ELSE	8	Encounter1000001
HorizontalCoordinationCost	IMP_ELSE	6	Encounter1000001
HorizontalMultiThreatReversalCost	THEN	9	Encounter1000001
HorizontalMultiThreatReversalCost	THEN	11	Encounter2100021
HorizontalMultiThreatReversalCost	IMP_ELSE	11	Encounter1000001
HorizontalMultiThreatReversalCost	IMP_ELSE	9	Encounter1000001
HorizontalMultiThreatReversalCost	THEN	17	Encounter1000001
HorizontalMultiThreatReversalCost	IMP_ELSE	17	Encounter1000002
HorizontalMultiThreatReversalCost	THEN	21	Encounter2100021
HorizontalMultiThreatReversalCost	IMP_ELSE	21	Encounter1000001
HorizontalMaintainHeadingIncentive	THEN	7	Encounter1000002
HorizontalMaintainHeadingIncentive	IMP_ELSE	7	Encounter1000001
HorizontalInitialMaintainCost	THEN	6	Encounter1000001
HorizontalInitialMaintainCost	IMP_ELSE	6	Encounter1000001
HorizontalTurnFromMaintainIncentive	THEN	7	Encounter1000002
HorizontalTurnFromMaintainIncentive	IMP_ELSE	7	Encounter1000001
HorizontalMultiThreatMaintainHeadingCost	THEN	9	Encounter1000001
HorizontalMultiThreatMaintainHeadingCost	THEN	11	Encounter2100021
HorizontalMultiThreatMaintainHeadingCost	IMP_ELSE	11	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
HorizontalMultiThreatMaintainHeadingCost	IMP_ELSE	9	Encounter1000001
HorizontalMultiThreatMaintainHeadingCost	THEN	17	Encounter1000001
HorizontalMultiThreatMaintainHeadingCost	IMP_ELSE	17	Encounter1000002
HorizontalMultiThreatMaintainHeadingCost	THEN	21	Encounter2100021
HorizontalMultiThreatMaintainHeadingCost	IMP_ELSE	21	Encounter1000001
HorizontalMultiThreatTurnFromMaintainCost	THEN	9	Encounter1000001
HorizontalMultiThreatTurnFromMaintainCost	THEN	11	Encounter2100021
HorizontalMultiThreatTurnFromMaintainCost	IMP_ELSE	11	Encounter1000001
HorizontalMultiThreatTurnFromMaintainCost	IMP_ELSE	9	Encounter1000001
HorizontalMultiThreatTurnFromMaintainCost	THEN	17	Encounter1000001
HorizontalMultiThreatTurnFromMaintainCost	IMP_ELSE	17	Encounter1000002
HorizontalMultiThreatTurnFromMaintainCost	THEN	21	Encounter5200005
HorizontalMultiThreatTurnFromMaintainCost	ELSEIF	24	Encounter1000002
HorizontalMultiThreatTurnFromMaintainCost	IMP_ELSE	24	Encounter1000001
HorizontalCoordinatedReversalCost	THEN	6	Encounter1000001
HorizontalCoordinatedReversalCost	THEN	9	Encounter1000001
HorizontalCoordinatedReversalCost	THEN	11	Encounter1000001
HorizontalCoordinatedReversalCost	THEN	13	Encounter1000002
HorizontalCoordinatedReversalCost	THEN	16	Encounter1000002
HorizontalCoordinatedReversalCost	IMP_ELSE	16	Encounter1000002
HorizontalCoordinatedReversalCost	IMP_ELSE	13	Encounter1000001
HorizontalCoordinatedReversalCost	IMP_ELSE	11	Encounter1000003
HorizontalCoordinatedReversalCost	IMP_ELSE	9	Encounter2100021
HorizontalCoordinatedReversalCost	IMP_ELSE	6	Encounter1000001
UpdateNumHorizontalReversals	THEN	4	Encounter4110060
UpdateNumHorizontalReversals	IMP_ELSE	4	Encounter1000001
UpdateNumHorizontalReversals	THEN	7	Encounter1000001
UpdateNumHorizontalReversals	IMP_ELSE	7	Encounter1000002
ApplyStateDependentOnlineCosts	THEN	13	Encounter1000001
ApplyStateDependentOnlineCosts	IMP_ELSE	13	Encounter1000001
CalculatePolicyState	THEN	26	Encounter1000001
CalculatePolicyState	IMP_ELSE	26	Encounter1000001
CalculatePolicyState	THEN	30	Encounter1000001
CalculatePolicyState	ELSE	32	Encounter1000001
CalculatePolicyState	THEN	34	Encounter1000001
CalculatePolicyState	ELSE	36	Encounter1000001
CreatePolicyStateBeliefs	THEN	16	Encounter1000001
CreatePolicyStateBeliefs	IMP_ELSE	16	Encounter1000001
DetermineStateDependentSpeedFactor	THEN	9	Encounter1000004
DetermineStateDependentSpeedFactor	ELSEIF	11	Encounter4280201
DetermineStateDependentSpeedFactor	ELSEIF	13	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
DetermineStateDependentSpeedFactor	ELSE	15	Encounter1000001
GetHorizontalPolicyCosts	THEN	10	Encounter1000001
GetHorizontalPolicyCosts	THEN	17	Encounter1000001
GetHorizontalPolicyCosts	ELSE	19	Encounter1000002
GetHorizontalPolicyCosts	ELSE	22	Encounter1000001
GetPolicyTableCosts	THEN	61	Encounter1000001
GetPolicyTableCosts	IMP_ELSE	61	Encounter1000001
GetPolicyTableCosts	THEN	65	Encounter1000002
GetPolicyTableCosts	IMP_ELSE	65	Encounter1000001
ModifyPolicySpeedBinsIndividual	THEN	10	Encounter1000001
ModifyPolicySpeedBinsIndividual	THEN	12	Encounter1000001
ModifyPolicySpeedBinsIndividual	IMP_ELSE	12	Encounter1000001
ModifyPolicySpeedBinsIndividual	THEN	17	Encounter1000001
ModifyPolicySpeedBinsIndividual	IMP_ELSE	17	Encounter1000009
ModifyPolicySpeedBinsIndividual	IMP_ELSE	10	Encounter1000001
StateDependentCrossingCostFactors	THEN	19	Encounter1000002
StateDependentCrossingCostFactors	THEN	21	Encounter1000002
StateDependentCrossingCostFactors	ELSEIF	23	Encounter1000002
StateDependentCrossingCostFactors	IMP_ELSE	23	Encounter1000002
StateDependentCrossingCostFactors	IMP_ELSE	19	Encounter1000001
StateDependentCrossingCostFactors	THEN	28	Encounter1000002
StateDependentCrossingCostFactors	THEN	30	Encounter1000002
StateDependentCrossingCostFactors	ELSEIF	32	Encounter1000002
StateDependentCrossingCostFactors	IMP_ELSE	32	Encounter1000002
StateDependentCrossingCostFactors	IMP_ELSE	28	Encounter1000001
CalculateCrossingCostPsiFactor	THEN	7	Encounter1000004
CalculateCrossingCostPsiFactor	IMP_ELSE	7	Encounter1000001
CalculateCrossingCostPsiFactor	THEN	10	Encounter1000002
CalculateCrossingCostPsiFactor	IMP_ELSE	10	Encounter1000001
CalculateCrossingCostPsiFactor	THEN	13	Encounter1000001
CalculateCrossingCostPsiFactor	ELSE	15	Encounter1000001
CalculateCrossingCostPsiFactor	IMP_ELSE	20	Encounter1000001
CalculateCrossingCostSpeedFactor	THEN	8	Encounter1000005
CalculateCrossingCostSpeedFactor	IMP_ELSE	8	Encounter1000001
CalculateCrossingCostThetaFactor	THEN	11	Encounter1000002
CalculateCrossingCostThetaFactor	IMP_ELSE	11	Encounter1000001
CalculateCrossingCostThetaFactor	THEN	15	Encounter1000002
CalculateCrossingCostThetaFactor	IMP_ELSE	15	Encounter1000001
CalculateCrossingCostThetaFactor	THEN	19	Encounter1000001
CalculateCrossingCostThetaFactor	IMP_ELSE	23	Encounter1000002
CalculateCrossingCostVerticalTauFactor	THEN	6	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
CalculateCrossingCostVerticalTauFactor	THEN	8	Encounter1000002
CalculateCrossingCostVerticalTauFactor	IMP_ELSE	8	Encounter1000001
UpdatePolicySpeedBinIndividual	IMP_ELSE	14	Encounter1000001
GetCutpointIndexVector	THEN	4	Encounter1000001
GetCutpointIndexVector	IMP_ELSE	4	Encounter1000001
FindMaximumWeightAndIndex	THEN	8	Encounter1000001
FindMaximumWeightAndIndex	IMP_ELSE	8	Encounter1000001
ConvertIntruderInputsToEnuBeliefs	THEN	8	Encounter1000007
ConvertIntruderInputsToEnuBeliefs	IMP_ELSE	8	Encounter1000001
ConvertIntruderInputsToEnuBeliefs	THEN	14	Encounter4260000
ConvertIntruderInputsToEnuBeliefs	ELSE	16	Encounter1000001
EnuBeliefEstimation	THEN	18	Encounter1000001
EnuBeliefEstimation	ELSE	20	Encounter1000001
EnuBeliefEstimation	THEN	32	Encounter1000001
EnuBeliefEstimation	IMP_ELSE	32	Encounter1000001
HorizontalStateEstimation	THEN	6	Encounter1000001
HorizontalStateEstimation	ELSEIF	10	Encounter1000007
HorizontalStateEstimation	ELSEIF	14	Encounter6100003
IsIntruderEquipped	THEN	4	Encounter1000002
IsIntruderEquipped	IMP_ELSE	4	Encounter1000001
IsIntruderInScope	THEN	14	Encounter1000001
IsIntruderInScope	IMP_ELSE	14	Encounter1000001
PrioritizeAndFilterIntruders	THEN	14	Encounter1000004
PrioritizeAndFilterIntruders	ELSE	16	Encounter1000001
PrioritizeAndFilterIntruders	THEN	17	Encounter4110011
PrioritizeAndFilterIntruders	IMP_ELSE	17	Encounter1000001
PrioritizeAndFilterIntruders	THEN	23	Encounter1000001
PrioritizeAndFilterIntruders	IMP_ELSE	23	Encounter1000001
PrioritizeAndFilterIntruders	THEN	31	Encounter1000001
PrioritizeAndFilterIntruders	THEN	36	Encounter1000001
PrioritizeAndFilterIntruders	IMP_ELSE	36	Encounter1000004
PrioritizeAndFilterIntruders	IMP_ELSE	31	Encounter1000001
HorizontalTRMUpdate	THEN	16	Encounter5100001
HorizontalTRMUpdate	ELSEIF	18	Encounter1000001
HorizontalTRMUpdate	THEN	20	Encounter1000001
HorizontalTRMUpdate	THEN	25	Encounter1000001
HorizontalTRMUpdate	ELSE	28	Encounter1000007
HorizontalTRMUpdate	ELSE	34	Encounter6100003
HorizontalTRMUpdate	ELSE	41	Encounter1000001
HorizontalTRMUpdatePrep	THEN	9	Encounter5100001
HorizontalTRMUpdatePrep	ELSEIF	11	Encounter6100001

Algorithm	Branch Type	Line Number	Encounter
HorizontalTRMUpdatePrep	ELSE	15	Encounter1000001
HorizontalTRMUpdatePrep	THEN	23	Encounter1000008
HorizontalTRMUpdatePrep	ELSE	26	Encounter1000001
HorizontalTRMUpdatePrep	THEN	34	Encounter5100002
HorizontalTRMUpdatePrep	IMP_ELSE	34	Encounter5100001
HorizontalTRMUpdatePrep	THEN	47	Encounter1000001
HorizontalTRMUpdatePrep	IMP_ELSE	47	Encounter1000001
ResetHorizontalTRMOwnshipState	THEN	4	Encounter1000001
ResetHorizontalTRMOwnshipState	IMP_ELSE	4	Encounter1000001
ResetHTRMState	THEN	4	Encounter1000001
ResetHTRMState	ELSEIF	8	Encounter4260002
ResetHTRMState	ELSEIF	12	Encounter6300005
ResetHTRMState	THEN	21	Encounter1000001
ResetHTRMState	IMP_ELSE	21	Encounter5100001
ResetHTRMState	THEN	25	Encounter5100001
ResetHTRMState	IMP_ELSE	25	Encounter1000001
FindHighestThreatIndex	THEN	5	Encounter1000001
FindHighestThreatIndex	IMP_ELSE	5	Encounter1000001
HorizontalAlertLookahead	THEN	28	Encounter1000002
HorizontalAlertLookahead	IMP_ELSE	28	Encounter1000001
HorizontalAlertLookahead	THEN	34	Encounter1000001
HorizontalAlertLookahead	ELSEIF	37	Encounter1000002
HorizontalAlertLookahead	THEN	39	Encounter1000002
HorizontalAlertLookahead	IMP_ELSE	39	Encounter4110060
HorizontalAlertLookahead	THEN	43	Encounter1000002
HorizontalAlertLookahead	IMP_ELSE	43	Encounter1000002
HorizontalAlertLookahead	ELSE	48	Encounter3110000
HorizontalAlertLookahead	THEN	53	Encounter1000001
HorizontalAlertLookahead	ELSE	55	Encounter1000002
HorizontalCostFusion	THEN	17	Encounter2100021
HorizontalCostFusion	IMP_ELSE	17	Encounter1000001
HorizontalCostFusion	THEN	20	Encounter1000001
HorizontalCostFusion	IMP_ELSE	20	Encounter1000001
HorizontalCostFusion	THEN	23	Encounter1000002
HorizontalCostFusion	IMP_ELSE	23	Encounter1000001
HorizontalCostFusion	THEN	33	Encounter1000002
HorizontalCostFusion	IMP_ELSE	33	Encounter1000001
HorizontalCostFusion	THEN	36	Encounter1000001
HorizontalCostFusion	THEN	38	Encounter1000002
HorizontalCostFusion	IMP_ELSE	38	Encounter1000001
HorizontalCostFusion	IMP_ELSE	36	Encounter2100021

Algorithm	Branch Type	Line Number	Encounter
SelectHorizontalAdvisory	THEN	8	Encounter1000001
SelectHorizontalAdvisory	THEN	11	Encounter1000002
SelectHorizontalAdvisory	IMP_ELSE	11	Encounter1000001
SelectHorizontalAdvisory	ELSE	14	Encounter1000001
SelectHorizontalAdvisory	THEN	17	Encounter1000002
SelectHorizontalAdvisory	THEN	21	Encounter1000002
SelectHorizontalAdvisory	ELSE	23	Encounter2100021
SelectHorizontalAdvisory	ELSE	28	Encounter1000001
SelectHorizontalAdvisoryWindRelative	THEN	9	Encounter4260000
SelectHorizontalAdvisoryWindRelative	THEN	12	Encounter6300005
SelectHorizontalAdvisoryWindRelative	IMP_ELSE	12	Encounter4260000
SelectHorizontalAdvisoryWindRelative	ELSE	15	Encounter1000007
SelectHorizontalAdvisoryWindRelative	THEN	18	Encounter4260000
SelectHorizontalAdvisoryWindRelative	THEN	22	Encounter4260000
SelectHorizontalAdvisoryWindRelative	ELSE	24	Encounter6100004
SelectHorizontalAdvisoryWindRelative	ELSE	29	Encounter1000007
SelectHorizontalAlertLookaheadAction	THEN	18	Encounter1000001
SelectHorizontalAlertLookaheadAction	IMP_ELSE	18	Encounter1000001
SelectHorizontalAlertLookaheadAction	THEN	21	Encounter1000002
SelectHorizontalAlertLookaheadAction	THEN	24	Encounter1000002
SelectHorizontalAlertLookaheadAction	THEN	25	Encounter1000002
SelectHorizontalAlertLookaheadAction	IMP_ELSE	25	Encounter1000002
SelectHorizontalAlertLookaheadAction	THEN	28	Encounter1000002
SelectHorizontalAlertLookaheadAction	THEN	29	Encounter1000002
SelectHorizontalAlertLookaheadAction	IMP_ELSE	29	Encounter1000002
SelectHorizontalAlertLookaheadAction	IMP_ELSE	24	Encounter1000002
SelectHorizontalAlertLookaheadAction	IMP_ELSE	21	Encounter1000001
SelectHorizontalAlertLookaheadAction	THEN	36	Encounter1000002
SelectHorizontalAlertLookaheadAction	IMP_ELSE	36	Encounter1000001
Consistent271	THEN	4	Encounter1000002
Consistent271	IMP_ELSE	4	Encounter1000001
DetermineDisplayTrackAngleAndTurnRate	THEN	6	Encounter1000002
DetermineDisplayTrackAngleAndTurnRate	IMP_ELSE	6	Encounter1000001
DetermineLabel271	THEN	12	Encounter1000001
DetermineLabel271	IMP_ELSE	12	Encounter1000002
DetermineHorizontalDesensitivity	THEN	5	Encounter1000001
DetermineHorizontalDesensitivity	ELSEIF	7	Encounter1000001
DetermineHorizontalDesensitivity	IMP_ELSE	7	Encounter3110000
UpdateHTRMIntruderState	THEN	3	Encounter1000008
UpdateHTRMIntruderState	IMP_ELSE	3	Encounter1000001
UpdateHTRMIntruderState	THEN	10	Encounter1000008

Algorithm	Branch Type	Line Number	Encounter
UpdateHTRMIntruderState	IMP_ELSE	10	Encounter1000001
SetHorizontalIntruderOutput	THEN	7	Encounter5100001
SetHorizontalIntruderOutput	IMP_ELSE	7	Encounter1000001
SetHorizontalIntruderOutput	THEN	21	Encounter5100001
SetHorizontalIntruderOutput	IMP_ELSE	21	Encounter1000001
EncodeHRC	THEN	7	Encounter1000009
EncodeHRC	ELSEIF	9	Encounter1000002
EncodeHRC	THEN	10	Encounter2100012
EncodeHRC	ELSE	12	Encounter1000002
EncodeHRC	ELSEIF	15	Encounter2100003
EncodeHRC	THEN	16	Encounter2100011
EncodeHRC	ELSE	18	Encounter2100003
EncodeHRC	IMP_ELSE	15	Encounter1000002
EncodeCHC	THEN	10	Encounter1000001
EncodeCHC	ELSEIF	12	Encounter1000002
EncodeCHC	ELSEIF	14	Encounter1000002
EncodeCHC	ELSEIF	16	Encounter2100003
EncodeCHC	ELSEIF	18	Encounter1000002
EncodeCHC	ELSEIF	20	Encounter4280207
EncodeCHC	ELSEIF	22	Encounter6200011
EncodeCHC	ELSE	24	Encounter1000002
GetHorizontalCoordinationOffsetFactor	THEN	5	Encounter1000002
GetHorizontalCoordinationOffsetFactor	THEN	6	Encounter1000002
GetHorizontalCoordinationOffsetFactor	ELSE	8	Encounter1000002
GetHorizontalCoordinationOffsetFactor	ELSEIF	11	Encounter1000009
GetHorizontalCoordinationOffsetFactor	THEN	12	Encounter2100012
GetHorizontalCoordinationOffsetFactor	ELSE	14	Encounter1000009
GetHorizontalCoordinationOffsetFactor	ELSE	17	Encounter1000002
GetHorizontalCoordinationOffsetFactor	THEN	18	Encounter1000002
GetHorizontalCoordinationOffsetFactor	ELSE	20	Encounter1000009
GetHorizontalCoordinationPolicy	THEN	23	Encounter1000002
GetHorizontalCoordinationPolicy	ELSE	25	Encounter1000002
GetHorizontalCoordinationPolicy	THEN	31	Encounter1000009
GetHorizontalCoordinationPolicy	IMP_ELSE	31	Encounter1000002
HorizontalCoordinationSelection	THEN	8	Encounter1000002
HorizontalCoordinationSelection	THEN	13	Encounter1000009
HorizontalCoordinationSelection	ELSE	15	Encounter1000002
HorizontalCoordinationSelection	IMP_ELSE	8	Encounter1000001
ApplyHorizontalCoordinationHysteresis	THEN	4	Encounter1000002
ApplyHorizontalCoordinationHysteresis	THEN	7	Encounter1000002
ApplyHorizontalCoordinationHysteresis	THEN	11	Encounter2100011

Algorithm	Branch Type	Line Number	Encounter
ApplyHorizontalCoordinationHysteresis	THEN	12	Encounter2100011
ApplyHorizontalCoordinationHysteresis	ELSEIF	16	Encounter1000002
ApplyHorizontalCoordinationHysteresis	THEN	17	Encounter1000002
ApplyHorizontalCoordinationHysteresis	IMP_ELSE	17	Encounter6200011
ApplyHorizontalCoordinationHysteresis	IMP_ELSE	16	Encounter1000002
ApplyHorizontalCoordinationHysteresis	IMP_ELSE	7	Encounter4280207
ApplyHorizontalCoordinationHysteresis	IMP_ELSE	4	Encounter1000002
DetermineHorizontalCode	THEN	5	Encounter1000002
DetermineHorizontalCode	IMP_ELSE	5	Encounter1000001
DetermineHorizontalScore	THEN	11	Encounter1000001
DetermineHorizontalScore	IMP_ELSE	11	Encounter1000001
DetermineHorizontalScore	THEN	15	Encounter1000004
DetermineHorizontalScore	IMP_ELSE	15	Encounter1000001
DetermineHorizontalScore	THEN	18	Encounter1000002
DetermineHorizontalScore	THEN	19	Encounter1000002
DetermineHorizontalScore	ELSE	21	Encounter1000009
DetermineHorizontalScore	ELSEIF	24	Encounter1000001
UpdateTargetTrackAngle	THEN	8	Encounter1000002
UpdateTargetTrackAngle	THEN	9	Encounter2100011
UpdateTargetTrackAngle	IMP_ELSE	9	Encounter1000002
UpdateTargetTrackAngle	THEN	12	Encounter1000002
UpdateTargetTrackAngle	ELSEIF	14	Encounter1000009
UpdateTargetTrackAngle	ELSE	16	Encounter1000002
UpdateTargetTrackAngle	THEN	23	Encounter1000002
UpdateTargetTrackAngle	THEN	24	Encounter1000002
UpdateTargetTrackAngle	ELSEIF	26	Encounter1000009
UpdateTargetTrackAngle	THEN	27	Encounter1000009
UpdateTargetTrackAngle	IMP_ELSE	27	Encounter2100003
UpdateTargetTrackAngle	ELSE	30	Encounter1000002
UpdateTargetTrackAngle	IMP_ELSE	23	Encounter1000002
SetAHRAReportData	THEN	3	Encounter1000001
SetAHRAReportData	ELSE	5	Encounter1000001
SetAHRAReportData	THEN	8	Encounter1000002
SetAHRAReportData	IMP_ELSE	8	Encounter1000001
SetAHRAReportData	THEN	12	Encounter1000001
SetAHRAReportData	ELSE	14	Encounter1000002
SetAHRAReportData	THEN	17	Encounter1000002
SetAHRAReportData	ELSE	19	Encounter2100021
SetAHRAReportData	THEN	24	Encounter1000009
SetAHRAReportData	ELSEIF	26	Encounter1000002
SetAHRAReportData	ELSEIF	28	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
SetAHRAReportData	ELSE	30	Encounter1000001
SetAHRAReportData	THEN	33	Encounter1000001
SetAHRAReportData	ELSE	35	Encounter1000002
UpdateRaPubState	THEN	5	Encounter1000002
UpdateRaPubState	THEN	7	Encounter1000002
UpdateRaPubState	IMP_ELSE	7	Encounter1000002
UpdateRaPubState	IMP_ELSE	5	Encounter1000001
UpdateRaPubState	THEN	14	Encounter1000002
UpdateRaPubState	THEN	16	Encounter3110000
UpdateRaPubState	IMP_ELSE	16	Encounter1000002
UpdateRaPubState	IMP_ELSE	14	Encounter1000001
UpdateRaPubState	THEN	23	Encounter1000002
UpdateRaPubState	IMP_ELSE	23	Encounter1000002
DetermineDimensionalRAState	THEN	8	Encounter1000001
DetermineDimensionalRAState	IMP_ELSE	8	Encounter1000001
DetermineDimensionalRAState	THEN	11	Encounter1000002
DetermineDimensionalRAState	IMP_ELSE	11	Encounter1000001
DetermineDimensionalRAState	THEN	14	Encounter1000001
DetermineDimensionalRAState	IMP_ELSE	14	Encounter1000001
DetermineDimensionalRAState	THEN	17	Encounter1000002
DetermineDimensionalRAState	IMP_ELSE	17	Encounter1000001
SetReportRATerminationBits	THEN	7	Encounter2100001
SetReportRATerminationBits	ELSEIF	12	Encounter1000002
SetReportRATerminationBits	ELSEIF	17	Encounter1000009
SetReportRATerminationBits	IMP_ELSE	17	Encounter1000001
SetReportRATerminationBits	THEN	23	Encounter1000001
SetReportRATerminationBits	ELSEIF	33	Encounter1000001
SetReportRATerminationBits	IMP_ELSE	33	Encounter1000001
SetReportRATerminationBits	THEN	39	Encounter1000002
SetReportRATerminationBits	ELSEIF	49	Encounter1000002
SetReportRATerminationBits	IMP_ELSE	49	Encounter1000001
SetPrevVerticalTerminationData	THEN	9	Encounter1000002
SetPrevVerticalTerminationData	IMP_ELSE	9	Encounter1000002
SetPrevHorizontalTerminationData	THEN	8	Encounter1000009
SetPrevHorizontalTerminationData	IMP_ELSE	8	Encounter1000009
SetThreatIdentityData	THEN	5	Encounter1000002
SetThreatIdentityData	ELSE	12	Encounter2100021
SetRaPublishedState	THEN	6	Encounter1000002
SetRaPublishedState	ELSEIF	10	Encounter2100021
SetRaPublishedState	ELSEIF	13	Encounter2100021
SetRaPublishedState	ELSE	15	Encounter2100021

Algorithm	Branch Type	Line Number	Encounter
SetBlendedIntruderThreatData	THEN	11	Encounter1000002
SetBlendedIntruderThreatData	IMP_ELSE	11	Encounter1000002
SetBlendedIntruderThreatData	THEN	19	Encounter2100021
SetBlendedIntruderThreatData	THEN	21	Encounter2100021
SetBlendedIntruderThreatData	IMP_ELSE	21	Encounter2100021
SetBlendedIntruderThreatData	ELSEIF	28	Encounter1000002
SetBlendedIntruderThreatData	ELSEIF	32	Encounter2100021
SetBlendedIntruderThreatData	THEN	36	Encounter2100021
SetBlendedIntruderThreatData	THEN	41	Encounter2100021
SetBlendedIntruderThreatData	ELSE	46	Encounter2100021
SetBlendedIntruderThreatData	IMP_ELSE	36	Encounter2100021
SetBlendedIntruderThreatData	THEN	54	Encounter4420001
SetBlendedIntruderThreatData	IMP_ELSE	54	Encounter2100021
SetBlendedIntruderThreatData	IMP_ELSE	32	Encounter1000001
SetReportDataOnRATerm	THEN	5	Encounter1000002
SetReportDataOnRATerm	ELSEIF	7	Encounter1000009
SetReportDataOnRATerm	ELSEIF	9	Encounter1000001
SetReportDataOnRATerm	IMP_ELSE	9	Encounter1000001
GenerateHTRMOutput	THEN	14	Encounter1000002
GenerateHTRMOutput	IMP_ELSE	14	Encounter1000001
GenerateHTRMOutput	THEN	36	Encounter1000002
GenerateHTRMOutput	ELSE	38	Encounter1000001
SetBlendedCoordinationData	THEN	8	Encounter1000001
GenerateBlendedsXuTRMOutput	THEN	12	Encounter1000001
GenerateBlendedsXuTRMOutput	ELSE	14	Encounter1000001
DetermineMultidimensionalMultithreat	THEN	7	Encounter1000002
DetermineMultidimensionalMultithreat	IMP_ELSE	7	Encounter1000001
DetermineMultidimensionalMultithreat	THEN	11	Encounter1000002
DetermineMultidimensionalMultithreat	THEN	13	Encounter1000002
DetermineMultidimensionalMultithreat	IMP_ELSE	13	Encounter1000002
DetermineMultidimensionalMultithreat	THEN	16	Encounter3213007
DetermineMultidimensionalMultithreat	IMP_ELSE	16	Encounter1000002
DetermineMultidimensionalMultithreat	THEN	20	Encounter3213007
DetermineMultidimensionalMultithreat	IMP_ELSE	20	Encounter1000002
DetermineMultidimensionalMultithreat	IMP_ELSE	11	Encounter1000001
DetermineMultidimensionalMultithreat	THEN	24	Encounter4140010
DetermineMultidimensionalMultithreat	IMP_ELSE	24	Encounter1000001
AddADSBTrackToDB	THEN	4	Encounter1000005
AddADSBTrackToReport	THEN	7	Encounter4220000
AddADSBTrackToReport	ELSE	9	Encounter1000005
AddADSBTrackToReport	THEN	22	Encounter7000063

Algorithm	Branch Type	Line Number	Encounter
AddADSBTrackToReport	IMP_ELSE	22	Encounter1000005
AddADSBTrackToReport	THEN	28	Encounter1000005
AddADSBTrackToReport	IMP_ELSE	28	Encounter4210016
AddADSBTrackToReport	THEN	33	Encounter1000005
AddADSBTrackToReport	IMP_ELSE	33	Encounter1000005
AddAltBiasAndSample	THEN	3	Encounter1000001
AddAltBiasAndSample	THEN	4	Encounter1000001
AddAltBiasAndSample	THEN	6	Encounter4210002
AddAltBiasAndSample	ELSEIF	8	Encounter3211000
AddAltBiasAndSample	IMP_ELSE	8	Encounter1000001
AddAltBiasAndSample	ELSE	11	Encounter1000002
AddAltBiasAndSample	THEN	13	Encounter1000005
AddAltBiasAndSample	ELSE	15	Encounter1000002
AddAltBiasAndSample	THEN	21	Encounter1000001
AddAltBiasAndSample	ELSEIF	23	Encounter3211000
AddAltBiasAndSample	ELSE	25	Encounter1000002
AddAltBiasAndSample	THEN	32	Encounter1000001
AddAltBiasAndSample	THEN	33	Encounter1000001
AddAltBiasAndSample	ELSE	35	Encounter1000002
AddAltBiasAndSample	THEN	39	Encounter1000001
AddAltBiasAndSample	ELSE	41	Encounter1000002
AddAltBiasAndSample	ELSE	47	Encounter1000007
GetPresAEMSigma	THEN	6	Encounter1000001
GetPresAEMSigma	IMP_ELSE	6	Encounter1000001
AddV2VTrackToReport	THEN	18	Encounter5100001
AddV2VTrackToReport	IMP_ELSE	18	Encounter1000002
AddV2VTrackToReport	THEN	25	Encounter1000002
AddV2VTrackToReport	IMP_ELSE	25	Encounter4230020
AddV2VTrackToReport	THEN	30	Encounter1000004
AddV2VTrackToReport	IMP_ELSE	30	Encounter1000002
AssociateV2VToTarget	THEN	5	Encounter1000002
AssociateV2VToTarget	THEN	7	Encounter1000002
AssociateV2VToTarget	IMP_ELSE	7	Encounter1000004
AssociateV2VToTarget	IMP_ELSE	5	Encounter2200000
AssociateV2VToV2VAndAGTTargets	THEN	6	Encounter1000002
AssociateV2VToV2VAndAGTTargets	THEN	8	Encounter1000002
AssociateV2VToV2VAndAGTTargets	IMP_ELSE	8	Encounter2100021
AssociateV2VToV2VAndAGTTargets	IMP_ELSE	6	Encounter1000005
AssociateV2VToV2VAndAGTTargets	IMP_ELSE	14	Encounter7000000
ReceiveStateVectorV2VReport	IMP_ELSE	5	Encounter1000002
ReceiveStateVectorV2VReport	THEN	8	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
ReceiveStateVectorV2VReport	ELSE	12	Encounter3211000
ReceiveStateVectorV2VReport	THEN	22	Encounter3211000
ReceiveStateVectorV2VReport	ELSE	25	Encounter1000002
ReceiveStateVectorV2VReport	THEN	31	Encounter1000002
ReceiveStateVectorV2VReport	ELSE	38	Encounter1000002
ReceiveStateVectorV2VReport	THEN	42	Encounter1000002
ReceiveStateVectorV2VReport	THEN	47	Encounter1000007
ReceiveStateVectorV2VReport	IMP_ELSE	47	Encounter1000002
ReceiveStateVectorV2VReport	IMP_ELSE	42	Encounter1000004
AddPointObstacleTrackToReport	IMP_ELSE	13	Encounter4260000
AddPointObstacleTrackToReport	THEN	17	Encounter4260000
AssociatePointObstacleReportToTarget	THEN	5	Encounter4410001
AssociatePointObstacleReportToTarget	THEN	7	Encounter4410001
AssociatePointObstacleReportToTarget	IMP_ELSE	7	Encounter5100108
AssociatePointObstacleReportToTarget	IMP_ELSE	5	Encounter4410001
ReceivePointObstacleReport	THEN	3	Encounter4260000
ReceivePointObstacleReport	THEN	5	Encounter4260000
ReceivePointObstacleReport	ELSE	12	Encounter4410001
ReceivePointObstacleReport	IMP_ELSE	3	Encounter4260002
GetRelativeAltitude	THEN	5	Encounter1000002
GetRelativeAltitude	ELSEIF	7	Encounter1000001
GetRelativeAltitude	ELSE	9	Encounter4210009
AddHorBias	ELSE	12	Encounter1000002
AdvanceVerticalTrack	THEN	8	Encounter3211000
AdvanceVerticalTrack	THEN	9	Encounter3211000
AdvanceVerticalTrack	ELSE	12	Encounter4230002
AdvanceVerticalTrack	ELSE	16	Encounter1000002
AdvanceVerticalTrack	THEN	17	Encounter1000005
AdvanceVerticalTrack	ELSE	20	Encounter1000002
AdvanceVerticalTrack	THEN	30	Encounter4120010
AdvanceVerticalTrack	ELSEIF	32	Encounter1000002
AdvanceVerticalTrack	ELSE	34	Encounter1000002
AdvanceVerticalTrack	THEN	39	Encounter1000002
AdvanceVerticalTrack	THEN	41	Encounter1000004
AdvanceVerticalTrack	THEN	42	Encounter1000004
AdvanceVerticalTrack	ELSE	44	Encounter1000006
AdvanceVerticalTrack	ELSE	49	Encounter1000002
AdvanceVerticalTrack	THEN	50	Encounter1000002
AdvanceVerticalTrack	IMP_ELSE	50	Encounter1000002
AdvanceVerticalTrack	THEN	57	Encounter1000002
AdvanceVerticalTrack	IMP_ELSE	57	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
AdvanceVerticalTrack	IMP_ELSE	39	Encounter7000022
AssociateADSBReportToTarget	THEN	4	Encounter1000005
AssociateADSBReportToTarget	THEN	5	Encounter1000005
AssociateADSBReportToTarget	IMP_ELSE	5	Encounter1000005
AssociateADSBReportToTarget	IMP_ELSE	4	Encounter3211000
AssociateICAOtoTarget	THEN	4	Encounter1000005
AssociateICAOtoTarget	IMP_ELSE	5	Encounter1000005
AssociateICAOtoTarget	IMP_ELSE	4	Encounter3211000
AssociateORNCTIdToTarget	THEN	5	Encounter3212000
AssociateORNCTIdToTarget	IMP_ELSE	5	Encounter3212000
PresAltAtToa	THEN	4	Encounter4350002
PresAltAtToa	ELSE	9	Encounter1000001
HAEAltAtToa	THEN	4	Encounter1000001
HAEAltAtToa	THEN	6	Encounter7000061
HAEAltAtToa	ELSEIF	10	Encounter1000001
HAEAltAtToa	ELSE	12	Encounter1000001
DeleteIntent	THEN	6	Encounter1000002
DeleteIntent	THEN	11	Encounter5200201
DeleteIntent	IMP_ELSE	11	Encounter1000002
DeleteIntent	THEN	15	Encounter1000002
DeleteIntent	THEN	16	Encounter1000002
DeleteIntent	ELSE	18	Encounter1000002
DeleteIntent	IMP_ELSE	15	Encounter5200201
DeleteIntent	IMP_ELSE	6	Encounter2100004
FilterTracksForTRM	THEN	14	Encounter1000001
FilterTracksForTRM	THEN	18	Encounter1000002
FilterTracksForTRM	IMP_ELSE	20	Encounter1000001
FilterTracksForTRM	THEN	24	Encounter4260000
FilterTracksForTRM	IMP_ELSE	24	Encounter1000001
FilterTracksForTRM	THEN	36	Encounter1000001
FilterTracksForTRM	IMP_ELSE	36	Encounter1000001
FilterTracksForTRM	THEN	45	Encounter4260000
FilterTracksForTRM	IMP_ELSE	45	Encounter1000001
SlantRange	THEN	3	Encounter1000007
SlantRange	ELSE	5	Encounter1000001
HeadingAtToa	THEN	5	Encounter1000001
HeadingAtToa	ELSEIF	10	Encounter1000001
HeadingAtToa	ELSE	14	Encounter7000018
HeadingAtToa	IMP_ELSE	5	Encounter1000001
InitializeHeadingTracker	THEN	5	Encounter1000001
InitializeHeadingTracker	ELSE	7	Encounter7000018

Algorithm	Branch Type	Line Number	Encounter
InitializePassiveTrackFile	THEN	4	Encounter1000002
InitializePassiveTrackFile	THEN	6	Encounter3211000
InitializePassiveTrackFile	ELSE	8	Encounter1000002
InitializePassiveTrackFile	ELSE	11	Encounter1000005
InitializePassiveTrackFile	THEN	13	Encounter4210002
InitializePassiveTrackFile	ELSE	15	Encounter1000005
InitializePassiveTrackFile	THEN	20	Encounter1000002
InitializePassiveTrackFile	IMP_ELSE	20	Encounter1000005
InitializePassiveTrackFile	THEN	30	Encounter4120010
InitializePassiveTrackFile	IMP_ELSE	30	Encounter1000002
InitializeVerticalTracker	THEN	3	Encounter3211000
InitializeVerticalTracker	THEN	4	Encounter3211000
InitializeVerticalTracker	ELSE	7	Encounter4230002
InitializeVerticalTracker	ELSE	11	Encounter1000002
InitializeVerticalTracker	THEN	12	Encounter1000005
InitializeVerticalTracker	ELSE	15	Encounter1000002
InitializeVerticalTracker	THEN	20	Encounter1000007
InitializeVerticalTracker	ELSE	26	Encounter1000002
InitializeOwnVerticalTracker	THEN	3	Encounter1000001
InitializeOwnVerticalTracker	ELSE	6	Encounter1000001
InitializeOwnVerticalTracker	THEN	12	Encounter1000001
InitializeOwnVerticalTracker	ELSE	14	Encounter1000001
IsOutlier	THEN	3	Encounter1000001
IsOutlier	ELSE	6	Encounter1000001
mahal	THEN	5	Encounter1000001
mahal	IMP_ELSE	5	Encounter1000001
mahal	THEN	9	Encounter1000001
mahal	IMP_ELSE	9	Encounter1000001
mahal	THEN	12	Encounter1000001
mahal	IMP_ELSE	13	Encounter1000001
mahal	ELSE	18	Encounter1000001
OwnWgs84Timeout	THEN	4	Encounter4350002
OwnWgs84Timeout	IMP_ELSE	4	Encounter1000001
nanmin	THEN	3	Encounter4210016
nanmin	ELSE	7	Encounter1000001
nanmax	ELSEIF	5	Encounter1000001
nanmax	ELSE	7	Encounter1000001
nanminimum	THEN	3	Encounter4110011
nanminimum	ELSE	5	Encounter1000001
nanmaximum	THEN	3	Encounter4110011
nanmaximum	ELSE	5	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
PredictVerticalTracker	THEN	3	Encounter1000001
PredictVerticalTracker	THEN	4	Encounter1000001
PredictVerticalTracker	ELSE	6	Encounter1000001
PredictVerticalTracker	THEN	7	Encounter3211000
PredictVerticalTracker	ELSE	9	Encounter1000001
PredictVerticalTracker	ELSE	13	Encounter1000001
PredictVerticalTracker	THEN	14	Encounter1000001
PredictVerticalTracker	ELSE	16	Encounter1000002
PredictVerticalTracker	THEN	17	Encounter1000005
PredictVerticalTracker	ELSE	19	Encounter1000002
PredictedAltitude	THEN	15	Encounter1000005
ReceiveStateVectorPositionReport	THEN	4	Encounter1000005
ReceiveStateVectorPositionReport	THEN	7	Encounter1000005
ReceiveStateVectorPositionReport	ELSE	11	Encounter1000005
ReceiveStateVectorPositionReport	THEN	14	Encounter1000005
ReceiveStateVectorPositionReport	THEN	16	Encounter1000005
ReceiveStateVectorPositionReport	THEN	20	Encounter7000023
ReceiveStateVectorPositionReport	IMP_ELSE	20	Encounter1000005
ReceiveStateVectorPositionReport	IMP_ELSE	16	Encounter7000024
ReceiveStateVectorPositionReport	IMP_ELSE	14	Encounter7000025
ReceiveStateVectorPositionReport	IMP_ELSE	4	Encounter7000025
ReceiveStateVectorVelocityReport	THEN	4	Encounter1000005
ReceiveStateVectorVelocityReport	THEN	6	Encounter1000005
ReceiveStateVectorVelocityReport	THEN	9	Encounter1000005
ReceiveStateVectorVelocityReport	IMP_ELSE	9	Encounter7000025
ReceiveStateVectorVelocityReport	IMP_ELSE	6	Encounter7000013
ReceiveStateVectorVelocityReport	IMP_ELSE	4	Encounter7000025
ReceivePresAltObservation	IMP_ELSE	8	Encounter1000001
ReceivePresAltObservation	THEN	21	Encounter1000001
ReceivePresAltObservation	THEN	23	Encounter4210012
ReceivePresAltObservation	ELSEIF	25	Encounter1000001
ReceivePresAltObservation	ELSE	27	Encounter1000001
ReceivePresAltObservation	THEN	32	Encounter1000001
ReceivePresAltObservation	THEN	33	Encounter1000005
ReceivePresAltObservation	ELSEIF	35	Encounter1000001
ReceivePresAltObservation	ELSE	40	Encounter4210012
ReceivePresAltObservation	ELSE	44	Encounter1000001
ReceivePresAltObservation	THEN	45	Encounter1000001
ReceivePresAltObservation	IMP_ELSE	45	Encounter1000001
ReceivePresAltObservation	IMP_ELSE	21	Encounter7000062
ReceiveADSBOperationalStatusMessage	THEN	4	Encounter2200101

Algorithm	Branch Type	Line Number	Encounter
ReceiveV2VOperationalStatusMessage	THEN	5	Encounter1000002
ReceiveHeadingObservation	THEN	11	Encounter1000001
ReceiveHeadingObservation	THEN	13	Encounter1000001
ReceiveHeadingObservation	THEN	15	Encounter7000028
ReceiveHeadingObservation	ELSEIF	17	Encounter7000028
ReceiveHeadingObservation	IMP_ELSE	17	Encounter1000001
ReceiveHeadingObservation	THEN	20	Encounter1000001
ReceiveHeadingObservation	THEN	21	Encounter1000001
ReceiveHeadingObservation	THEN	23	Encounter1000007
ReceiveHeadingObservation	ELSE	25	Encounter1000001
ReceiveHeadingObservation	IMP_ELSE	21	Encounter7000018
ReceiveHeadingObservation	IMP_ELSE	20	Encounter7000028
ReceiveHeadingObservation	IMP_ELSE	13	Encounter7000018
ReceiveHeadingObservation	THEN	34	Encounter7000018
ReceiveHeadingObservation	IMP_ELSE	34	Encounter1000001
ReceiveHeadingObservation	IMP_ELSE	11	Encounter1000001
ReceiveHeadingObservation	THEN	38	Encounter1000001
ReceiveHeadingObservation	THEN	42	Encounter1000001
ReceiveHeadingObservation	THEN	43	Encounter7000028
ReceiveHeadingObservation	ELSE	45	Encounter1000001
ReceiveHeadingObservation	ELSE	50	Encounter7000018
ReceiveHeadingObservation	IMP_ELSE	38	Encounter1000001
ReceiveModeStatusReport	THEN	3	Encounter1000005
ReceiveModeStatusReport	THEN	5	Encounter1000005
ReceiveModeStatusReport	THEN	8	Encounter1000005
ReceiveModeStatusReport	IMP_ELSE	8	Encounter7000000
ReceiveModeStatusReport	IMP_ELSE	3	Encounter7000029
ReceiveWgs84Observation	THEN	11	Encounter1000001
ReceiveWgs84Observation	THEN	15	Encounter1000001
ReceiveWgs84Observation	THEN	16	Encounter3212000
ReceiveWgs84Observation	THEN	18	Encounter3212000
ReceiveWgs84Observation	IMP_ELSE	18	Encounter3212000
ReceiveWgs84Observation	IMP_ELSE	16	Encounter1000001
ReceiveWgs84Observation	IMP_ELSE	15	Encounter1000005
ReceiveWgs84Observation	THEN	38	Encounter1000001
ReceiveWgs84Observation	ELSE	44	Encounter1000001
ReceiveWgs84Observation	THEN	47	Encounter1000001
ReceiveWgs84Observation	THEN	48	Encounter1000001
ReceiveWgs84Observation	ELSEIF	50	Encounter2100001
ReceiveWgs84Observation	ELSE	56	Encounter1000005
ReceiveWgs84Observation	ELSE	61	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
ReceiveWgs84Observation	IMP_ELSE	11	Encounter4350001
RemoveStaleADSBTrack	THEN	7	Encounter7000025
RemoveStaleADSBTrack	THEN	8	Encounter7000029
RemoveStaleADSBTrack	IMP_ELSE	8	Encounter7000025
RemoveStaleADSBTrack	ELSE	13	Encounter1000005
RemoveStaleADSBTrack	THEN	14	Encounter7000029
RemoveStaleADSBTrack	IMP_ELSE	14	Encounter1000005
RemoveStaleORNCTTrack	THEN	8	Encounter7000030
RemoveStaleORNCTTrack	THEN	13	Encounter7000030
RemoveStaleORNCTTrack	THEN	14	Encounter7000030
RemoveStaleORNCTTrack	IMP_ELSE	13	Encounter7000030
RemoveStaleORNCTTrack	IMP_ELSE	8	Encounter3212000
SetOwnV2VOperationalStatusMessage	THEN	3	Encounter1000001
SetOwnV2VOperationalStatusMessage	ELSEIF	5	Encounter5100001
SetOwnV2VOperationalStatusMessage	ELSEIF	7	Encounter1000001
SetCoordination	THEN	7	Encounter1000002
SetCoordination	THEN	8	Encounter1000002
SetCoordination	THEN	9	Encounter1000002
SetCoordination	ELSEIF	13	Encounter2100001
SetCoordination	ELSEIF	17	Encounter2200014
SetCoordination	ELSEIF	22	Encounter2200022
SetCoordination	ELSEIF	27	Encounter2200001
SetCoordination	ELSE	31	Encounter1000001
SetCoordination	THEN	32	Encounter2200102
SetCoordination	THEN	33	Encounter2200102
SetCoordination	ELSE	37	Encounter2200102
SetCoordination	ELSE	42	Encounter1000001
SetCoordination	THEN	43	Encounter2200107
SetCoordination	ELSEIF	47	Encounter1000001
SetIntruderDegradedFlags	THEN	7	Encounter3212000
SetIntruderDegradedFlags	THEN	8	Encounter3212000
SetIntruderDegradedFlags	IMP_ELSE	8	Encounter3212000
SetIntruderDegradedFlags	THEN	11	Encounter7000030
SetIntruderDegradedFlags	IMP_ELSE	11	Encounter3212000
SetIntruderDegradedFlags	THEN	14	Encounter4250008
SetIntruderDegradedFlags	IMP_ELSE	14	Encounter3212000
SetIntruderDegradedFlags	ELSE	17	Encounter1000001
SetIntruderDegradedFlags	THEN	20	Encounter1000007
SetIntruderDegradedFlags	IMP_ELSE	20	Encounter1000001
SetIntruderDegradedFlags	THEN	23	Encounter1000004
SetIntruderDegradedFlags	IMP_ELSE	23	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
SetIntruderDegradedFlags	THEN	26	Encounter1000004
SetIntruderDegradedFlags	IMP_ELSE	26	Encounter1000001
SetIntruderDegradedFlags	THEN	29	Encounter1000004
SetIntruderDegradedFlags	IMP_ELSE	29	Encounter1000001
SetIntruderDegradedFlags	THEN	33	Encounter2100014
SetIntruderDegradedFlags	IMP_ELSE	33	Encounter1000001
SetIntruderDegradedFlags	THEN	37	Encounter7000062
SetIntruderDegradedFlags	IMP_ELSE	37	Encounter1000001
SetClassification	THEN	5	Encounter1000005
SetClassification	IMP_ELSE	5	Encounter1000001
SetClassification	THEN	8	Encounter1000002
SetClassification	IMP_ELSE	8	Encounter1000001
SetClassification	THEN	11	Encounter3212000
SetClassification	IMP_ELSE	11	Encounter1000001
SetClassification	THEN	14	Encounter1000001
SetClassification	IMP_ELSE	14	Encounter1000002
SetClassification	THEN	20	Encounter4280215
SetClassification	ELSEIF	22	Encounter3211000
SetClassification	ELSEIF	24	Encounter1000005
SetClassification	ELSEIF	26	Encounter1000001
VerticalRateArrowUpdate	THEN	6	Encounter1000001
VerticalRateArrowUpdate	IMP_ELSE	6	Encounter1000001
VerticalRateArrowUpdate	THEN	10	Encounter3212000
VerticalRateArrowUpdate	THEN	13	Encounter3212000
VerticalRateArrowUpdate	ELSEIF	15	Encounter4250006
VerticalRateArrowUpdate	IMP_ELSE	15	Encounter4250008
VerticalRateArrowUpdate	IMP_ELSE	10	Encounter1000001
VerticalRateArrowUpdate	THEN	19	Encounter1000001
VerticalRateArrowUpdate	ELSEIF	21	Encounter1000002
VerticalRateArrowUpdate	ELSE	23	Encounter1000001
VerticalRateArrowUpdate	THEN	30	Encounter1000001
VerticalRateArrowUpdate	ELSEIF	32	Encounter1000002
VerticalRateArrowUpdate	ELSE	34	Encounter1000001
VerticalRateArrowUpdate	THEN	38	Encounter1000001
VerticalRateArrowUpdate	ELSEIF	40	Encounter1000004
VerticalRateArrowUpdate	ELSEIF	42	Encounter1000001
VerticalRateArrowUpdate	IMP_ELSE	42	Encounter1000001
SetDisplayDataPassive	THEN	12	Encounter1000001
SetDisplayDataPassive	ELSE	17	Encounter7000018
SetDisplayDataPassive	THEN	26	Encounter1000001
SetDisplayDataPassive	ELSE	30	Encounter1000007

Algorithm	Branch Type	Line Number	Encounter
SetDisplayDataPassive	THEN	35	Encounter1000001
SetDisplayDataPassive	ELSE	37	Encounter1000007
TrackExists	THEN	3	Encounter1000001
TrackExists	ELSE	5	Encounter1000001
uchol	THEN	8	Encounter1000001
uchol	IMP_ELSE	8	Encounter1000001
uchol	THEN	13	Encounter1000001
uchol	IMP_ELSE	13	Encounter1000001
UpdatePassiveQualityHistory	THEN	3	Encounter1000005
UpdatePassiveQualityHistory	ELSEIF	12	Encounter1000002
UpdatePassiveQualityHistory	THEN	21	Encounter1000002
UpdatePassiveQualityHistory	IMP_ELSE	21	Encounter1000002
RemoveStaleV2VTrack	THEN	7	Encounter7000054
RemoveStaleV2VTrack	THEN	8	Encounter7000054
RemoveStaleV2VTrack	ELSE	13	Encounter1000002
RemoveStaleV2VTrack	IMP_ELSE	14	Encounter1000002
UpdateHorizontalVelocityMode	THEN	9	Encounter1000001
UpdateHorizontalVelocityMode	THEN	11	Encounter6100004
UpdateHorizontalVelocityMode	ELSEIF	13	Encounter1000007
UpdateHorizontalVelocityMode	THEN	14	Encounter6100006
UpdateHorizontalVelocityMode	ELSE	16	Encounter1000007
UpdateHorizontalVelocityMode	ELSEIF	19	Encounter1000001
UpdateHorizontalVelocityMode	ELSEIF	21	Encounter1000004
UpdateHorizontalVelocityMode	IMP_ELSE	21	Encounter1000004
UpdateHorizontalVelocityMode	ELSEIF	24	Encounter6100003
UpdateHorizontalVelocityMode	ELSE	26	Encounter1000001
GetGroundSpeed	THEN	4	Encounter1000001
UpdateAdvisoryMode	THEN	3	Encounter1000001
UpdateAdvisoryMode	ELSEIF	5	Encounter5100001
UpdateAdvisoryMode	ELSE	7	Encounter1000001
UpdateHistory	THEN	6	Encounter1000001
UpdateHistory	IMP_ELSE	6	Encounter1000001
UpdateIntruderVRC	THEN	5	Encounter1000001
UpdateIntruderVRC	IMP_ELSE	6	Encounter1000001
UpdateIntruderVRC	THEN	10	Encounter1000002
UpdateIntruderVRC	IMP_ELSE	10	Encounter1000001
UpdateIntruderVRC	IMP_ELSE	5	Encounter1000001
UpdateVerticalTracker	THEN	4	Encounter1000001
UpdateVerticalTracker	THEN	5	Encounter1000001
UpdateVerticalTracker	ELSE	7	Encounter3211000
UpdateVerticalTracker	THEN	8	Encounter3211000

Algorithm	Branch Type	Line Number	Encounter
UpdateVerticalTracker	ELSE	10	Encounter4230002
UpdateVerticalTracker	ELSE	14	Encounter1000001
UpdateVerticalTracker	THEN	15	Encounter1000001
UpdateVerticalTracker	ELSE	17	Encounter1000002
UpdateVerticalTracker	THEN	18	Encounter1000005
UpdateVerticalTracker	ELSE	20	Encounter1000002
UpdateVerticalTracker	THEN	26	Encounter1000001
UpdateVerticalTracker	ELSE	28	Encounter1000001
UpdateVerticalTracker	THEN	33	Encounter1000005
UpdateVerticalTracker	IMP_ELSE	33	Encounter1000001
UpdateVerticalTracker	THEN	38	Encounter1000005
UpdateVerticalTracker	IMP_ELSE	38	Encounter1000001
ConvertECEFToWGS84	THEN	18	Encounter1000001
ConvertECEFToWGS84	THEN	35	Encounter1000002
ConvertECEFToWGS84	IMP_ELSE	35	Encounter1000001
GetAltForHorTrkObservation	THEN	3	Encounter1000002
GetAltForHorTrkObservation	ELSE	6	Encounter1000007
UpdateAltInHorGeoPos	THEN	3	Encounter7000006
UpdateAltInHorGeoPos	ELSE	6	Encounter1000007
ConvertWGS84ToECEF	THEN	8	Encounter1000001
ConvertWGS84ToECEF	IMP_ELSE	8	Encounter1000001
PropagateOwnshipToToa	THEN	5	Encounter1000001
PropagateOwnshipToToa	THEN	6	Encounter1000001
PropagateOwnshipToToa	ELSE	8	Encounter4350001
PropagateOwnshipToToa	ELSE	14	Encounter1000001
PropagateOwnshipToToa	THEN	19	Encounter1000001
PropagateOwnshipToToa	THEN	22	Encounter1000001
PropagateOwnshipToToa	ELSE	26	Encounter1000005
PropagateOwnshipToToa	ELSE	34	Encounter1000001
PropagateGeodeticToToa	THEN	5	Encounter1000001
PropagateGeodeticToToa	THEN	8	Encounter1000001
PropagateGeodeticToToa	ELSE	10	Encounter1000001
PropagateGeodeticToToa	ELSE	15	Encounter1000004
ReCenterHorizontalTrackLocation	THEN	4	Encounter1000001
ReCenterHorizontalTrackLocation	ELSE	7	Encounter1000007
AlignAltitudeTypesUsedInGeodeticPositions	THEN	7	Encounter1000001
AlignAltitudeTypesUsedInGeodeticPositions	THEN	8	Encounter1000001
AlignAltitudeTypesUsedInGeodeticPositions	THEN	11	Encounter4210006
AlignAltitudeTypesUsedInGeodeticPositions	IMP_ELSE	11	Encounter1000001
AlignAltitudeTypesUsedInGeodeticPositions	ELSEIF	15	Encounter4210009
AlignAltitudeTypesUsedInGeodeticPositions	ELSE	20	Encounter4210017

Algorithm	Branch Type	Line Number	Encounter
AlignAltitudeTypesUsedInGeodeticPositions	ELSE	26	Encounter1000002
AlignAltitudeTypesUsedInGeodeticPositions	THEN	27	Encounter1000002
AlignAltitudeTypesUsedInGeodeticPositions	THEN	30	Encounter1000007
AlignAltitudeTypesUsedInGeodeticPositions	IMP_ELSE	30	Encounter1000002
AlignAltitudeTypesUsedInGeodeticPositions	ELSEIF	34	Encounter4210012
AlignAltitudeTypesUsedInGeodeticPositions	ELSE	39	Encounter4210016
AdvancePassiveTrackPosition	THEN	11	Encounter1000005
AdvancePassiveTrackPosition	THEN	14	Encounter1000009
AdvancePassiveTrackPosition	THEN	15	Encounter1000009
AdvancePassiveTrackPosition	ELSE	17	Encounter1000009
AdvancePassiveTrackPosition	ELSE	27	Encounter1000005
AdvancePassiveTrackPosition	THEN	28	Encounter7000031
AdvancePassiveTrackPosition	IMP_ELSE	28	Encounter1000005
AdvancePassiveTrackVelocity	THEN	10	Encounter1000005
AdvancePassiveTrackVelocity	THEN	13	Encounter1000005
AdvancePassiveTrackVelocity	ELSEIF	16	Encounter7000032
AdvancePassiveTrackVelocity	IMP_ELSE	16	Encounter7000032
AdvancePassiveTrackVelocity	IMP_ELSE	10	Encounter1000005
AdvancePassiveTrackVelocity	THEN	22	Encounter1000005
AdvancePassiveTrackVelocity	THEN	24	Encounter1000005
AdvancePassiveTrackVelocity	IMP_ELSE	24	Encounter1000005
AdvancePassiveTrackVelocity	IMP_ELSE	22	Encounter1000009
ConvertNACpToSigmaHEPU	THEN	4	Encounter7000055
ConvertNACpToSigmaHEPU	ELSEIF	6	Encounter7000055
ConvertNACpToSigmaHEPU	ELSEIF	8	Encounter7000055
ConvertNACpToSigmaHEPU	ELSEIF	10	Encounter7000055
ConvertNACpToSigmaHEPU	ELSEIF	12	Encounter7000055
ConvertNACpToSigmaHEPU	ELSEIF	14	Encounter4340003
ConvertNACpToSigmaHEPU	ELSEIF	16	Encounter7000001
ConvertNACpToSigmaHEPU	ELSEIF	18	Encounter3211000
ConvertNACpToSigmaHEPU	ELSEIF	20	Encounter7000055
ConvertNACpToSigmaHEPU	ELSEIF	22	Encounter1000001
ConvertNACpToSigmaHEPU	ELSEIF	24	Encounter7000055
ConvertNACpToSigmaHEPU	IMP_ELSE	24	Encounter7000055
ConvertNACvToSigmaHVA	THEN	4	Encounter1000002
ConvertNACvToSigmaHVA	ELSEIF	6	Encounter1000001
ConvertNACvToSigmaHVA	ELSEIF	8	Encounter7000033
ConvertNACvToSigmaHVA	IMP_ELSE	8	Encounter1000005
ConvertGVAToSigmaVEPU	ELSEIF	6	Encounter4210002
InitializePassiveTrackerPosition	THEN	3	Encounter1000002
InitializePassiveTrackerPosition	ELSE	7	Encounter1000005

Algorithm	Branch Type	Line Number	Encounter
PredictPassiveTracker	THEN	3	Encounter1000002
PredictPassiveTracker	ELSE	5	Encounter1000001
SurroundingPoints	ELSE	8	Encounter4350002
SurroundingPoints	THEN	10	Encounter4350002
SurroundingPoints	IMP_ELSE	10	Encounter4350002
PassiveQualityCheck	THEN	5	Encounter1000002
PassiveQualityCheck	THEN	6	Encounter1000005
PassiveQualityCheck	ELSEIF	12	Encounter1000002
PassiveQualityCheck	THEN	20	Encounter1000002
PassiveQualityCheck	ELSEIF	22	Encounter1000005
PassiveQualityCheck	IMP_ELSE	22	Encounter1000002
PassiveQualityCheck	THEN	25	Encounter1000002
PassiveQualityCheck	ELSEIF	27	Encounter4320002
PassiveQualityCheck	ELSEIF	29	Encounter1000002
PassiveQualityCheck	IMP_ELSE	29	Encounter1000002
PassiveQualityCheck	IMP_ELSE	5	Encounter1000001
AddDecorrelatedTrackToTarget	THEN	3	Encounter3231000
AddDecorrelatedTrackToTarget	ELSEIF	5	Encounter3212000
AddDecorrelatedTrackToTarget	ELSEIF	7	Encounter3130000
AddTracksToReport	THEN	4	Encounter1000001
AddTracksToReport	THEN	10	Encounter1000001
AddTracksToReport	THEN	11	Encounter1000005
AddTracksToReport	ELSEIF	13	Encounter1000002
AddTracksToReport	ELSEIF	15	Encounter3212000
AddTracksToReport	ELSEIF	17	Encounter1000001
AddTracksToReport	ELSEIF	20	Encounter4260000
AddTracksToReport	IMP_ELSE	20	Encounter1000002
AddTracksToReport	IMP_ELSE	4	Encounter7000025
TrackSourceSelection	THEN	14	Encounter1000009
TrackSourceSelection	ELSEIF	16	Encounter1000002
TrackSourceSelection	ELSEIF	18	Encounter1000001
TrackSourceSelection	ELSEIF	20	Encounter4120030
TrackSourceSelection	ELSEIF	22	Encounter4130041
TrackSourceSelection	ELSEIF	24	Encounter4140010
TrackSourceSelection	ELSEIF	26	Encounter1000005
TrackSourceSelection	ELSEIF	28	Encounter1000004
TrackSourceSelection	ELSEIF	30	Encounter4110011
TrackSourceSelection	ELSEIF	32	Encounter4120080
TrackSourceSelection	ELSEIF	34	Encounter1000007
TrackSourceSelection	ELSEIF	36	Encounter4140011
TrackSourceSelection	IMP_ELSE	36	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
OptimalPreTrackedTracks	THEN	7	Encounter3212000
OptimalPreTrackedTracks	IMP_ELSE	7	Encounter1000001
OptimalPreTrackedTracks	THEN	10	Encounter1000001
OptimalPreTrackedTracks	THEN	13	Encounter1000001
OptimalPreTrackedTracks	THEN	14	Encounter1000001
OptimalPreTrackedTracks	ELSE	16	Encounter4140010
OptimalPreTrackedTracks	ELSE	19	Encounter4110011
OptimalPreTrackedTracks	THEN	20	Encounter4110011
OptimalPreTrackedTracks	ELSE	22	Encounter4140011
OptimalPreTrackedTracks	IMP_ELSE	10	Encounter4350001
OptimalQualityTrack	THEN	13	Encounter1000001
OptimalQualityTrack	ELSE	15	Encounter4140010
OptimalQualityTrack	THEN	19	Encounter1000001
OptimalQualityTrack	IMP_ELSE	19	Encounter1000001
PredictPreTrackedTrack	THEN	4	Encounter1000001
PredictPreTrackedTrack	ELSE	12	Encounter3212000
SelectFinalTrack	THEN	5	Encounter1000001
SelectFinalTrack	THEN	10	Encounter1000001
SelectFinalTrack	IMP_ELSE	10	Encounter3213017
SelectFinalTrack	THEN	13	Encounter1000001
SelectFinalTrack	IMP_ELSE	13	Encounter1000001
SelectFinalTrack	THEN	24	Encounter1000001
SelectFinalTrack	ELSE	27	Encounter1000001
SelectFinalTrack	ELSE	31	Encounter1000002
OptimalTrackHistoryUpdate	THEN	5	Encounter1000001
OptimalTrackHistoryUpdate	THEN	6	Encounter1000001
OptimalTrackHistoryUpdate	IMP_ELSE	6	Encounter1000001
OptimalTrackHistoryUpdate	ELSE	10	Encounter1000002
GetPreviouslySelectedTrack	THEN	5	Encounter1000001
GetPreviouslySelectedTrack	THEN	6	Encounter1000005
GetPreviouslySelectedTrack	THEN	7	Encounter1000005
GetPreviouslySelectedTrack	ELSEIF	10	Encounter1000002
GetPreviouslySelectedTrack	THEN	11	Encounter1000002
GetPreviouslySelectedTrack	ELSEIF	14	Encounter3212000
GetPreviouslySelectedTrack	THEN	15	Encounter3212000
GetPreviouslySelectedTrack	ELSEIF	18	Encounter1000001
GetPreviouslySelectedTrack	THEN	20	Encounter1000001
GetPreviouslySelectedTrack	IMP_ELSE	20	Encounter3213017
GetPreviouslySelectedTrack	IMP_ELSE	5	Encounter1000001
CoordinationTimeoutCheck	THEN	8	Encounter1000002
CoordinationTimeoutCheck	THEN	9	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
CoordinationTimeoutCheck	IMP_ELSE	9	Encounter2100004
CoordinationTimeoutCheck	IMP_ELSE	8	Encounter1000001
CoordinationTimeoutCheck	THEN	16	Encounter1000002
CoordinationTimeoutCheck	THEN	17	Encounter1000002
CoordinationTimeoutCheck	IMP_ELSE	17	Encounter2100004
CoordinationTimeoutCheck	IMP_ELSE	16	Encounter1000001
CorrelatePosition	THEN	13	Encounter1000001
CorrelatePosition	THEN	14	Encounter1000001
CorrelatePosition	THEN	17	Encounter3211000
CorrelatePosition	ELSE	20	Encounter1000001
CorrelatePosition	IMP_ELSE	14	Encounter7000000
CorrelatePosition	IMP_ELSE	13	Encounter3110000
CorrelatePosition	THEN	26	Encounter7000036
CorrelatePosition	ELSEIF	28	Encounter7000008
CorrelatePosition	ELSEIF	30	Encounter7000037
CorrelatePosition	IMP_ELSE	30	Encounter1000001
CorrelatePosition	THEN	40	Encounter7000039
CorrelatePosition	ELSEIF	42	Encounter4120010
CorrelatePosition	THEN	45	Encounter4120010
CorrelatePosition	ELSE	47	Encounter4120010
CorrelatePosition	ELSE	50	Encounter1000001
CorrelatePosition	THEN	53	Encounter1000001
CorrelatePosition	THEN	55	Encounter4120010
CorrelatePosition	IMP_ELSE	55	Encounter1000001
CorrelatePosition	ELSE	59	Encounter7000041
CorrelatePosition	THEN	62	Encounter3110000
CorrelatePosition	ELSE	64	Encounter1000001
GenerateLeadTrackList	THEN	7	Encounter1000005
GenerateLeadTrackList	ELSEIF	9	Encounter1000002
GenerateLeadTrackList	ELSEIF	11	Encounter3212000
GenerateLeadTrackList	ELSEIF	13	Encounter1000001
GenerateLeadTrackList	IMP_ELSE	13	Encounter4260000
GenerateLeadTrackList	THEN	16	Encounter1000001
GenerateLeadTrackList	IMP_ELSE	16	Encounter4260000
ExtrapolateTrack	THEN	4	Encounter1000005
ExtrapolateTrack	ELSEIF	6	Encounter1000002
ExtrapolateTrack	ELSEIF	8	Encounter3212000
ExtrapolateTrack	ELSEIF	10	Encounter1000001
GetAircraftTracks	THEN	4	Encounter1000005
GetAircraftTracks	IMP_ELSE	4	Encounter1000001
GetAircraftTracks	THEN	7	Encounter1000002

Algorithm	Branch Type	Line Number	Encounter
GetAircraftTracks	IMP_ELSE	7	Encounter1000001
GetAircraftTracks	THEN	10	Encounter3212000
GetAircraftTracks	IMP_ELSE	10	Encounter1000001
MergeTargets	THEN	6	Encounter3213011
MergeTargets	IMP_ELSE	6	Encounter2200000
MergeTargets	THEN	10	Encounter3211000
MergeTargets	IMP_ELSE	10	Encounter2200000
MergeTargets	THEN	16	Encounter2200000
MergeTargets	IMP_ELSE	16	Encounter3211000
MergeTargets	THEN	21	Encounter3213006
MergeTargets	ELSEIF	23	Encounter3213017
MergeTargets	IMP_ELSE	23	Encounter2200000
MergeTargets	IMP_ELSE	27	Encounter2200000
MergeTargets	IMP_ELSE	32	Encounter2200000
MergeTargets	THEN	41	Encounter3211000
MergeTargets	IMP_ELSE	41	Encounter2200000
MergeTargets	THEN	46	Encounter3212003
MergeTargets	ELSEIF	48	Encounter2200000
MergeTargets	IMP_ELSE	48	Encounter3212000
RemoveDecorrelatedTrackFromTarget	THEN	3	Encounter3212000
RemoveDecorrelatedTrackFromTarget	IMP_ELSE	3	Encounter3231001
RemoveDecorrelatedTrackFromTarget	THEN	7	Encounter3231000
RemoveDecorrelatedTrackFromTarget	IMP_ELSE	7	Encounter3212000
RemoveDecorrelatedTrackFromTarget	THEN	11	Encounter3212000
RemoveDecorrelatedTrackFromTarget	IMP_ELSE	11	Encounter3233000
RemoveDecorrelatedTrackFromTarget	THEN	16	Encounter3233000
RemoveDecorrelatedTrackFromTarget	IMP_ELSE	16	Encounter3233017
RemoveStaleTracks	THEN	6	Encounter1000005
RemoveStaleTracks	IMP_ELSE	6	Encounter1000001
RemoveStaleTracks	IMP_ELSE	9	Encounter1000001
RemoveStaleTracks	THEN	12	Encounter3212000
RemoveStaleTracks	IMP_ELSE	12	Encounter1000001
RemoveStaleTracks	THEN	17	Encounter8407805
RemoveStaleTracks	IMP_ELSE	18	Encounter8407805
RemoveStaleTracks	IMP_ELSE	17	Encounter1000001
RemoveStaleTracks	THEN	25	Encounter1000002
RemoveStaleTracks	IMP_ELSE	25	Encounter1000001
RemoveStaleTracks	THEN	28	Encounter4260000
RemoveStaleTracks	THEN	29	Encounter7000060
RemoveStaleTracks	IMP_ELSE	29	Encounter4260000
RemoveStaleTracks	IMP_ELSE	28	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
RemoveStaleTracks	THEN	34	Encounter7000025
RemoveStaleTracks	ELSE	37	Encounter1000001
ReceiveV2VCoordination	THEN	9	Encounter1000002
ReceiveV2VCoordination	THEN	15	Encounter1000002
ReceiveV2VCoordination	THEN	16	Encounter2100014
ReceiveV2VCoordination	ELSEIF	18	Encounter7000059
ReceiveV2VCoordination	ELSE	20	Encounter1000002
ReceiveV2VCoordination	THEN	22	Encounter2100004
ReceiveV2VCoordination	IMP_ELSE	22	Encounter1000002
ReceiveV2VCoordination	THEN	26	Encounter1000002
ReceiveV2VCoordination	THEN	30	Encounter7000002
ReceiveV2VCoordination	IMP_ELSE	30	Encounter1000002
ReceiveV2VCoordination	IMP_ELSE	26	Encounter1000002
ReceiveV2VCoordination	IMP_ELSE	15	Encounter1000002
ReceiveV2VCoordination	THEN	38	Encounter1000002
ReceiveV2VCoordination	THEN	44	Encounter1000002
ReceiveV2VCoordination	THEN	46	Encounter2100004
ReceiveV2VCoordination	IMP_ELSE	46	Encounter1000002
ReceiveV2VCoordination	THEN	50	Encounter1000002
ReceiveV2VCoordination	THEN	54	Encounter5200201
ReceiveV2VCoordination	IMP_ELSE	54	Encounter1000002
ReceiveV2VCoordination	IMP_ELSE	50	Encounter2100002
ReceiveV2VCoordination	IMP_ELSE	44	Encounter2100014
ReceiveV2VCoordination	IMP_ELSE	38	Encounter1000002
SetDisplayDataORNCT	THEN	12	Encounter3212000
SetDisplayDataORNCT	ELSE	17	Encounter7000034
SetDisplayDataORNCT	THEN	23	Encounter3212000
SetDisplayDataORNCT	ELSE	26	Encounter4250008
SetDisplayDataORNCT	THEN	31	Encounter3212000
SetDisplayDataORNCT	ELSE	33	Encounter4250008
AddORNCTTrackToReport	IMP_ELSE	15	Encounter3212000
AddORNCTTrackToReport	THEN	22	Encounter3212000
AddORNCTTrackToReport	IMP_ELSE	22	Encounter3212002
AddORNCTTrackToReport	THEN	25	Encounter3212000
AddORNCTTrackToReport	IMP_ELSE	25	Encounter4250008
ReceiveOwnRelNonCoopTrack	ELSEIF	12	Encounter3212000
ReceiveOwnRelNonCoopTrack	THEN	20	Encounter3212000
ReceiveOwnRelNonCoopTrack	THEN	33	Encounter3212000
ReceiveOwnRelNonCoopTrack	ELSE	35	Encounter3212000
ReceiveOwnRelNonCoopTrack	ELSE	38	Encounter3212000
UpdateORNCTTrack	THEN	21	Encounter3212000

Algorithm	Branch Type	Line Number	Encounter
UpdateORNCTTrack	ELSE	24	Encounter3212000
UpdateORNCTTrack	THEN	25	Encounter3212000
UpdateORNCTTrack	THEN	27	Encounter3212000
UpdateORNCTTrack	IMP_ELSE	27	Encounter3212000
UpdateORNCTTrack	IMP_ELSE	25	Encounter8401004
SetOwnshipVelocity	THEN	6	Encounter1000001
SetOwnshipVelocity	ELSE	11	Encounter1000001
ReceiveAbsoluteGeodeticTrack	IMP_ELSE	4	Encounter1000001
ReceiveAbsoluteGeodeticTrack	THEN	9	Encounter1000001
ReceiveAbsoluteGeodeticTrack	ELSEIF	12	Encounter3110000
ReceiveAbsoluteGeodeticTrack	IMP_ELSE	12	Encounter1000001
ReceiveAbsoluteGeodeticTrack	THEN	17	Encounter1000001
ReceiveAbsoluteGeodeticTrack	THEN	19	Encounter1000001
ReceiveAbsoluteGeodeticTrack	ELSEIF	21	Encounter1000001
AssembleAGTTrack	THEN	5	Encounter1000001
AssembleAGTTrack	ELSE	7	Encounter3213000
AssembleAGTTrack	THEN	18	Encounter1000001
AssembleAGTTrack	IMP_ELSE	18	Encounter3110000
InitializeAGTTrack	THEN	23	Encounter4140010
InitializeAGTTrack	IMP_ELSE	23	Encounter1000001
InitializeAGTTrack	THEN	29	Encounter3120000
InitializeAGTTrack	ELSE	31	Encounter1000001
UpdateAGTTrack	THEN	7	Encounter1000001
UpdateAGTTrack	THEN	10	Encounter4140010
UpdateAGTTrack	IMP_ELSE	10	Encounter1000001
UpdateAGTTrack	THEN	21	Encounter1000001
UpdateAGTTrack	ELSE	28	Encounter4140010
UpdateAGTTrack	IMP_ELSE	7	Encounter8401014
UpdateAGTTrack	THEN	37	Encounter3110000
UpdateAGTTrack	THEN	38	Encounter3110000
UpdateAGTTrack	IMP_ELSE	41	Encounter3110000
UpdateAGTTrack	IMP_ELSE	37	Encounter1000001
UpdateAGTTrack	THEN	46	Encounter1000001
UpdateAGTTrack	THEN	56	Encounter1000001
UpdateAGTTrack	THEN	67	Encounter4140010
UpdateAGTTrack	IMP_ELSE	67	Encounter1000001
UpdateAGTTrack	THEN	73	Encounter1000001
UpdateAGTTrack	IMP_ELSE	73	Encounter4140010
UpdateAGTTrack	IMP_ELSE	46	Encounter3110000
AssociateAGTToTarget	THEN	7	Encounter1000001
AssociateAGTToTarget	IMP_ELSE	7	Encounter3213017

Algorithm	Branch Type	Line Number	Encounter
AssociateAGTToOwnship	THEN	5	Encounter3110000
AssociateAGTToOwnship	IMP_ELSE	5	Encounter7000057
AddAGTTrackToReport	THEN	16	Encounter7000043
AddAGTTrackToReport	IMP_ELSE	16	Encounter1000001
AddAGTTrackToReport	THEN	22	Encounter3110000
AddAGTTrackToReport	IMP_ELSE	22	Encounter1000001
AddAGTTrackToReport	THEN	28	Encounter1000001
AddAGTTrackToReport	IMP_ELSE	28	Encounter4240020
ExtrapolateAGTTrack	THEN	31	Encounter1000001
ExtrapolateAGTTrack	IMP_ELSE	31	Encounter1000001
DecorrelateTargetsFromOwnship	THEN	15	Encounter3110000
DecorrelateTargetsFromOwnship	ELSE	17	Encounter3120000
DecorrelateTargetsFromOwnship	THEN	21	Encounter3110000
DecorrelateTargetsFromOwnship	IMP_ELSE	21	Encounter3110000
DecorrelateTargetsFromOwnship	THEN	25	Encounter3130000
DecorrelateTargetsFromOwnship	THEN	29	Encounter3130000
DecorrelateTargetsFromOwnship	IMP_ELSE	29	Encounter3130000
DecorrelateTargetsFromOwnship	IMP_ELSE	25	Encounter3110000
DecorrelateTargetsFromOwnship	THEN	33	Encounter3130000
DecorrelateTargetsFromOwnship	IMP_ELSE	33	Encounter3110000
CorrelateToOwnshipByType	THEN	4	Encounter3120000
CorrelateToOwnshipByType	ELSEIF	6	Encounter1000001
CorrelateToOwnshipByType	THEN	7	Encounter3110000
CorrelateToOwnshipByType	ELSE	9	Encounter1000001
CorrelateToOwnshipByType	ELSE	12	Encounter3140001
CorrelateTargetsToOwnship	THEN	6	Encounter1000001
CorrelateTargetsToOwnship	IMP_ELSE	6	Encounter1000002
CorrelateTargetsToOwnship	IMP_ELSE	14	Encounter1000001
CorrelateTargetsToOwnship	THEN	18	Encounter1000001
CorrelateTargetsToOwnship	IMP_ELSE	18	Encounter1000001
CorrelateIndividualTracksToOwnship	THEN	10	Encounter1000001
CorrelateIndividualTracksToOwnship	ELSE	12	Encounter1000001
CorrelateIndividualTracksToOwnship	THEN	16	Encounter3110000
CorrelateIndividualTracksToOwnship	THEN	20	Encounter3110000
CorrelateIndividualTracksToOwnship	IMP_ELSE	16	Encounter1000001
CorrelateIndividualTracksToOwnship	THEN	24	Encounter3110000
CorrelateIndividualTracksToOwnship	THEN	28	Encounter3110000
CorrelateIndividualTracksToOwnship	IMP_ELSE	28	Encounter7000058
CorrelateIndividualTracksToOwnship	THEN	33	Encounter3110000
CorrelateIndividualTracksToOwnship	IMP_ELSE	33	Encounter7000058
CorrelateIndividualTracksToOwnship	IMP_ELSE	24	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
ExtrapolateOwnshipTrack	THEN	12	Encounter1000001
ExtrapolateOwnshipTrack	ELSEIF	17	Encounter4210012
ExtrapolateOwnshipTrack	ELSE	22	Encounter1000001
RemoveDecorrelatedTrackFromOwnship	THEN	4	Encounter3130000
ReceiveExternallyValidatedADSB	THEN	4	Encounter1000009
ReceiveExternallyValidatedADSB	IMP_ELSE	4	Encounter3211001
ReceiveExternallyValidatedV2V	THEN	4	Encounter1000002
ReceiveExternallyValidatedV2V	IMP_ELSE	4	Encounter5200201
SetAvailableSources	THEN	4	Encounter1000005
SetAvailableSources	THEN	5	Encounter4220000
SetAvailableSources	ELSE	7	Encounter1000005
SetAvailableSources	IMP_ELSE	4	Encounter1000001
SetAvailableSources	THEN	11	Encounter3212000
SetAvailableSources	IMP_ELSE	11	Encounter1000001
SetAvailableSources	THEN	14	Encounter1000001
SetAvailableSources	IMP_ELSE	14	Encounter1000002
SetAvailableSources	THEN	17	Encounter1000002
SetAvailableSources	IMP_ELSE	17	Encounter1000001
SetAvailableSources	THEN	20	Encounter4260000
SetAvailableSources	IMP_ELSE	20	Encounter1000001
SetIntruderIDDirectory	THEN	3	Encounter1000005
SetIntruderIDDirectory	THEN	4	Encounter1000005
SetIntruderIDDirectory	ELSE	6	Encounter3211000
SetIntruderIDDirectory	IMP_ELSE	3	Encounter1000001
SetIntruderIDDirectory	THEN	10	Encounter1000002
SetIntruderIDDirectory	THEN	12	Encounter1000002
SetIntruderIDDirectory	THEN	13	Encounter1000002
SetIntruderIDDirectory	ELSEIF	15	Encounter3211001
SetIntruderIDDirectory	IMP_ELSE	15	Encounter2200000
SetIntruderIDDirectory	IMP_ELSE	12	Encounter3212002
SetIntruderIDDirectory	IMP_ELSE	10	Encounter1000001
SetIntruderIDDirectory	THEN	20	Encounter3212000
SetIntruderIDDirectory	IMP_ELSE	20	Encounter1000001
SetIntruderIDDirectory	THEN	25	Encounter3140001
SetIntruderIDDirectory	IMP_ELSE	25	Encounter1000001
SetIntruderIDDirectory	THEN	28	Encounter3213002
SetIntruderIDDirectory	THEN	29	Encounter3213002
SetIntruderIDDirectory	ELSEIF	31	Encounter3213003
SetIntruderIDDirectory	IMP_ELSE	31	Encounter3221001
SetIntruderIDDirectory	IMP_ELSE	28	Encounter1000001
SetIntruderIDDirectory	THEN	36	Encounter4260000

Algorithm	Branch Type	Line Number	Encounter
SetIntruderIDDirectory	IMP_ELSE	36	Encounter1000001
ReceiveStateVectorUatReport	THEN	5	Encounter7000042
ReceiveStateVectorUatReport	IMP_ELSE	5	Encounter4220000
ReceiveStateVectorUatReport	THEN	10	Encounter4220000
ReceiveStateVectorUatReport	ELSE	17	Encounter4220000
ReceiveStateVectorUatReport	THEN	20	Encounter4220000
ReceiveStateVectorUatReport	THEN	22	Encounter4220000
ReceiveStateVectorUatReport	THEN	26	Encounter7000042
ReceiveStateVectorUatReport	IMP_ELSE	26	Encounter4220000
ReceiveStateVectorUatReport	IMP_ELSE	20	Encounter7000000
AdvancePassiveTrackState	THEN	4	Encounter1000002
AdvancePassiveTrackState	ELSE	9	Encounter4220000
AdvancePassiveTrackState	THEN	18	Encounter1000002
AdvancePassiveTrackState	THEN	22	Encounter1000002
AdvancePassiveTrackState	IMP_ELSE	22	Encounter4220000
AdvancePassiveTrackState	THEN	25	Encounter1000004
AdvancePassiveTrackState	THEN	26	Encounter1000004
AdvancePassiveTrackState	ELSE	28	Encounter1000004
AdvancePassiveTrackState	ELSE	40	Encounter1000002
AdvancePassiveTrackState	THEN	42	Encounter4340002
AdvancePassiveTrackState	IMP_ELSE	42	Encounter1000002
UpdatePassiveTrackerState	THEN	3	Encounter1000002
UpdatePassiveTrackerState	ELSE	6	Encounter4220000
UpdateIntruderHRC	THEN	5	Encounter1000001
UpdateIntruderHRC	IMP_ELSE	6	Encounter1000001
UpdateIntruderHRC	THEN	10	Encounter1000002
UpdateIntruderHRC	IMP_ELSE	10	Encounter1000001
UpdateIntruderHRC	IMP_ELSE	5	Encounter1000004
ValidCovarianceMatrix	THEN	3	Encounter4280306
ValidCovarianceMatrix	ELSE	5	Encounter1000001
SetOwnshipData	THEN	19	Encounter1000001
SetOwnshipData	ELSEIF	21	Encounter1000005
SetOwnshipData	IMP_ELSE	21	Encounter1000001
SetOwnshipData	THEN	24	Encounter1000001
SetOwnshipData	ELSEIF	26	Encounter4350002
SetOwnshipData	IMP_ELSE	26	Encounter1000001
SetOwnshipData	THEN	29	Encounter1000001
SetOwnshipData	ELSEIF	31	Encounter1000007
SetOwnshipData	IMP_ELSE	31	Encounter1000001
SetOwnshipData	THEN	34	Encounter1000001
SetOwnshipData	IMP_ELSE	34	Encounter1000001

Algorithm	Branch Type	Line Number	Encounter
SetOwnshipData	THEN	39	Encounter1000001
SetOwnshipData	IMP_ELSE	39	Encounter1000001
SetOwnshipData	THEN	43	Encounter4210001
SetOwnshipData	ELSEIF	46	Encounter1000001
SetOwnshipData	ELSE	49	Encounter1000001
Predict1DKalmanFilter	THEN	4	Encounter1000001
Predict1DKalmanFilter	ELSE	6	Encounter1000001
Predict1DKalmanFilter	THEN	12	Encounter1000001
Predict2DKalmanFilter	THEN	4	Encounter1000001
Predict2DKalmanFilter	ELSE	6	Encounter1000001
StmHousekeepingPrioritization	THEN	13	Encounter1000001
StmHousekeepingPrioritization	ELSE	15	Encounter1000002
StmHousekeepingPrioritization	THEN	18	Encounter1000001
StmHousekeepingPrioritization	ELSE	20	Encounter1000002
CorrelateByType	THEN	5	Encounter3211000
CorrelateByType	IMP_ELSE	9	Encounter3211000
CorrelateByType	THEN	12	Encounter3211000
CorrelateByType	THEN	13	Encounter3211000
CorrelateByType	ELSE	15	Encounter3211000
CorrelateByType	IMP_ELSE	5	Encounter1000004
CorrelationTrackType	THEN	5	Encounter1000005
CorrelationTrackType	THEN	6	Encounter1000005
CorrelationTrackType	ELSE	8	Encounter3211000
CorrelationTrackType	ELSEIF	11	Encounter1000002
CorrelationTrackType	THEN	12	Encounter3212002
CorrelationTrackType	ELSEIF	14	Encounter1000002
CorrelationTrackType	ELSE	16	Encounter3211001
CorrelationTrackType	ELSEIF	19	Encounter3212000
CorrelationTrackType	ELSEIF	21	Encounter1000001
CorrelationTrackType	THEN	22	Encounter1000001
CorrelationTrackType	THEN	23	Encounter1000001
CorrelationTrackType	ELSEIF	25	Encounter3213002
CorrelationTrackType	ELSE	27	Encounter3213003
CorrelationTrackType	ELSE	30	Encounter3120000
CorrelationTrackType	THEN	31	Encounter3120000
CorrelationTrackType	ELSEIF	33	Encounter3213004
CorrelationTrackType	ELSE	35	Encounter3213005
GetCorrelationMechanism	THEN	7	Encounter1000004
CorrelateID	THEN	5	Encounter1000004
CorrelateID	ELSE	12	Encounter2200000
CorrelateID	THEN	14	Encounter2200000

Algorithm	Branch Type	Line Number	Encounter
CorrelateID	IMP_ELSE	14	Encounter2200000
CorrelateID	THEN	24	Encounter2200000
CorrelateID	ELSEIF	27	Encounter3223000
CorrelateID	ELSEIF	30	Encounter3224000
CorrelateID	ELSEIF	33	Encounter1000004
CorrelateID	IMP_ELSE	33	Encounter3211000
CorrelateID	THEN	37	Encounter1000004
CorrelateID	THEN	38	Encounter2200000
CorrelateID	ELSE	40	Encounter1000004
CorrelateID	IMP_ELSE	37	Encounter3211000
CorrelateID	THEN	44	Encounter1000004
CorrelateID	IMP_ELSE	44	Encounter2200000
CorrelateIndividualTracks	THEN	9	Encounter1000004
CorrelateIndividualTracks	IMP_ELSE	9	Encounter1000004
CorrelateIndividualTracks	THEN	16	Encounter3211000
CorrelateIndividualTracks	THEN	18	Encounter3211000
CorrelateIndividualTracks	IMP_ELSE	16	Encounter1000004
CorrelateIndividualTracks	THEN	22	Encounter2200000
CorrelateIndividualTracks	THEN	23	Encounter2200000
CorrelateIndividualTracks	ELSE	26	Encounter3211000
CorrelateIndividualTracks	IMP_ELSE	22	Encounter1000004
DecorrelateTargets	THEN	16	Encounter1000001
DecorrelateTargets	ELSE	18	Encounter1000001
DecorrelateTargets	THEN	27	Encounter3212000
DecorrelateTargets	IMP_ELSE	27	Encounter1000001
DecorrelateTargets	THEN	30	Encounter3212000
DecorrelateTargets	IMP_ELSE	30	Encounter1000001
DecorrelateTargets	THEN	33	Encounter1000001
DecorrelateTargets	IMP_ELSE	33	Encounter2200000
DecorrelateTargets	THEN	37	Encounter3212000
DecorrelateTargets	THEN	39	Encounter3212000
DecorrelateTargets	IMP_ELSE	37	Encounter2200000
DecorrelateTargets	THEN	43	Encounter3212000
DecorrelateTargets	IMP_ELSE	43	Encounter2200000
CorrelateTargets	THEN	9	Encounter1000001
CorrelateTargets	THEN	12	Encounter1000004
CorrelateTargets	THEN	15	Encounter1000006
CorrelateTargets	ELSEIF	18	Encounter1000004
CorrelateTargets	THEN	20	Encounter2200000
CorrelateTargets	IMP_ELSE	20	Encounter1000004
CorrelateTargets	IMP_ELSE	18	Encounter1000006

Algorithm	Branch Type	Line Number	Encounter
CorrelateTargets	IMP_ELSE	12	Encounter1000001
CorrelateTargets	IMP_ELSE	9	Encounter3211000
CorrelationHistoryCheck	THEN	7	Encounter3110000
CorrelationHistoryCheck	IMP_ELSE	7	Encounter4110010
CorrelationHistoryUpdate	THEN	6	Encounter3130000
CorrelationHistoryUpdate	THEN	10	Encounter3130000
CorrelationHistoryUpdate	IMP_ELSE	10	Encounter4130041
CorrelationHistoryUpdate	IMP_ELSE	6	Encounter3110000
CalculateOwnshipVelocityOffsets	THEN	10	Encounter3212000
CalculateOwnshipVelocityOffsets	ELSEIF	12	Encounter8401004
CalculateOwnshipVelocityOffsets	ELSE	14	Encounter8401004